BIOLOGY GRADE 10: PLANT GASEOUS EXCHANGE AND RESPIRATION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.3 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 2.0: Anatomy and Physiology of Plants
Sub-Strand	Sub-Strand 2.3: Plant Gaseous Exchange and Respiration
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Sites of gaseous exchange in plants (stomata, cuticle, lenticels, pneumatophores) – Structure and adaptations of stomata – Mechanisms of stomatal opening and closing (potassium ion theory, starch–sugar hypothesis) – Gaseous exchange in leaves, stems, and roots – Adaptations of plants in different environments (xerophytes, hydrophytes, mesophytes) for gaseous exchange – Aerobic respiration — process, reactants, products, and energy yield – Anaerobic respiration — process, reactants, and products (ethanol and lactic acid pathways) – Differences between aerobic and anaerobic respiration – Economic and ecological importance of anaerobic respiration (fermentation, biogas production, brewing) – Factors affecting the rate of respiration in plants – Relationship between photosynthesis, gaseous exchange, and respiration – Project: Investigating anaerobic respiration in yeast using locally available materials (e.g., maize flour, sugar-cane juice)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify and describe the sites of gaseous exchange in plants and explain their structural adaptations, b) explain the mechanisms of stomatal opening and closing and relate them to environmental conditions found in Kenyan ecosystems, c) describe the processes of aerobic and anaerobic respiration in plants, including reactants, products, and energy transformations, d) distinguish between aerobic and anaerobic respiration using experimental evidence and data analysis, e) evaluate the economic and ecological importance of anaerobic respiration in real-world Kenyan contexts such as fermentation, biogas production, and the food industry, f) design and carry out a simple investigation on anaerobic respiration using locally available materials, g) appreciate the significance of gaseous exchange and respiration in maintaining life processes in plants and their relevance to human livelihoods.
Core Competencies	– Communication and Collaboration: Learners collaboratively analyse animations of stomatal mechanisms, share findings from investigations, and present project results on anaerobic respiration to peers, developing scientific argumentation and teamwork skills. – Critical Thinking and Problem Solving: Learners design investigations to distinguish aerobic from anaerobic respiration,

	interpret experimental data, and evaluate competing theories of stomatal opening and closing. – Creativity and Imagination: Learners devise project-based investigations using locally available Kenyan materials (e.g., maize flour, sugar-cane juice, sorghum) to demonstrate anaerobic respiration, and construct progressively refined models of gaseous exchange. – Digital Literacy: Learners interact with online animations and simulations of gaseous exchange and stomatal mechanisms, developing the ability to extract and critically evaluate digital scientific content. – Self-Efficacy: Learners develop confidence in conducting biological investigations and explaining complex physiological processes through hands-on experimentation and iterative model building.
Core Values	– Responsibility: Learners handle biological specimens and laboratory materials carefully and responsibly, and search for information on respiration and gaseous exchange from safe, credible internet sources. – Respect: Learners appreciate diverse scientific explanations and peer contributions during collaborative discussions on stomatal theories and anaerobic respiration data. – Unity: Learners work in mixed-ability groups during investigations and project work, recognising that collaborative effort produces stronger scientific explanations. – Social Justice: Learners connect the economic importance of anaerobic respiration (biogas, fermentation) to equitable energy and food security solutions relevant to Kenyan communities, including rural farmers in the Rift Valley and Coast regions.
Science & Engineering Practices	– SEP 1 – Asking Questions and Defining Problems: Learners generate and refine investigable questions about why plants produce bubbles in warm water and how stomata regulate gas movement, anchoring inquiry throughout the unit. – SEP 2 – Developing and Using Models: Learners iteratively construct and revise annotated diagrams and physical models of a stoma, a leaf cross-section, and a plant respiratory pathway to represent and explain gaseous exchange mechanisms. – SEP 3 – Planning and Carrying Out Investigations: Learners design controlled experiments to compare aerobic and anaerobic respiration rates in yeast and to verify sites of gaseous exchange (e.g., petroleum jelly blocking stomata experiment). – SEP 4 – Analysing and Interpreting Data: Learners record, tabulate, and interpret quantitative data from respiration experiments (e.g., CO_2 production rates, bubble counts) and draw evidence-based conclusions. – SEP 6 – Constructing Explanations: Learners use the potassium ion theory and starch–sugar hypothesis to construct written and oral explanations of stomatal behaviour under varying light and water conditions. – SEP 7 – Engaging in Argument from Evidence: Learners evaluate experimental results and competing theories of stomatal opening, defending their interpretations with data during structured class debates. – SEP 8 – Obtaining, Evaluating, and Communicating Information: Learners read scientific texts, watch animations, and present research on the economic importance of anaerobic respiration, evaluating source reliability and clarity.
Pertinent & Contemporary Issues (PCIs)	– Environmental Conservation: Learners examine how plant gaseous exchange and respiration contribute to carbon cycling and ecosystem balance in Kenyan environments such as the Kakamega Rainforest and Maasai Mara savannah, reinforcing the value of conserving plant biodiversity. – Health Education: Learners connect anaerobic respiration in human muscle tissue (lactic acid production during athletics — referencing Kenyan long-distance runners from Eldoret/Iten) to understanding fatigue and recovery, promoting awareness of healthy physical activity. – Financial Literacy and Entrepreneurship: Learners explore how anaerobic respiration underpins profitable industries in Kenya — including traditional fermentation of uji (fermented porridge), busaa brewing, biogas generation for rural homes, and commercial bread-making using yeast — linking science to income-generating opportunities. – Safety and Security: Learners observe safe laboratory procedures when conducting respiration experiments involving yeast, limewater, and warm water baths, and access digital resources from secure, vetted websites. – Climate Change: Learners analyse how changes in temperature and CO_2 levels — driven by climate change — affect rates of plant respiration and gaseous exchange, particularly for Kenyan staple crops such as maize and beans in semi-arid regions.

Career Connections	– Botanist: This sub-strand develops foundational understanding of plant physiology — specifically gaseous exchange and respiration — directly relevant to careers in botanical research and conservation in institutions such as the National Museums of Kenya. – Agricultural Scientist / Agronomist: Understanding stomatal adaptations and respiration rates informs crop management strategies for Kenyan farmers growing maize, tea (Kericho), and horticultural produce, supporting careers in agricultural research at KALRO (Kenya Agricultural and Livestock Research Organisation). – Biotechnologist: Knowledge of anaerobic respiration and fermentation pathways supports careers in biotechnology, including biofuel production, food fermentation technology, and pharmaceutical manufacturing in Kenya. – Environmental Scientist: Understanding the role of plant gaseous exchange in carbon cycling prepares learners for careers in environmental monitoring, climate research, and ecosystem management with organisations such as NEMA (National Environment Management Authority). – Food Scientist / Food Technologist: The economic importance of anaerobic respiration (fermentation of uji, togwa, and dairy products) connects directly to careers in Kenya's growing food processing and quality assurance industry. – Medical/Clinical Scientist: Understanding aerobic and anaerobic respiration at the cellular level provides foundational knowledge for careers in medicine, physiology, and sports science — relevant given Kenya's prominence in athletics.
Focus for Lessons	This unit anchors learning in the observable phenomenon of bubbles produced by leaves submerged in warm water, driving students to investigate how plants exchange gases and release energy through respiration. Learners progress from identifying physical sites of gaseous exchange — stomata, lenticels, cuticle, and pneumatophores — to understanding the molecular mechanisms of stomatal regulation, and ultimately to comparing aerobic and anaerobic respiration through hands-on experimentation. Throughout the unit, Kenyan contexts such as biogas production in rural homesteads, fermentation of traditional foods like uji and busaa, and the physiology of elite Kenyan runners from Iten ground abstract biological concepts in meaningful, real-world applications.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do plants "breathe," and what happens inside their cells when they release the energy stored in food? KICD KEY INQUIRY QUESTIONS: 1. How do plants exchange gases with their environment? 2. How do plants obtain energy from organic compounds through respiration?
Anchoring Phenomenon	ANCHORING PHENOMENON: A student at Mwalimu Njoroge's biology class in Nairobi brings freshly cut sukuma wiki (kale) leaves from the school garden and submerges them in a beaker of warm water. Within seconds, tiny bubbles begin streaming from the surface of the leaves — not randomly, but from specific points on the leaf surface. When the same leaves are submerged in cold water, far fewer bubbles appear. The students are puzzled: Where exactly are these bubbles coming from? What gas is inside them? Why does warming the water increase bubble production? Why do bubbles appear to come from specific spots rather than the entire leaf surface? And if plants are releasing gas, does this mean they are "breathing" like animals do? This puzzling observation launches the unit-long investigation into the sites, mechanisms, and processes of plant gaseous exchange and respiration.
Storyline Thread	Lesson 1 — ANCHORING PHENOMENON & PREDICT PHASE: Students observe sukuma wiki leaves submerged in warm vs. cold water and record where bubbles emerge. They draw initial annotated diagrams predicting the sites of gas exchange and populate the Driving Question Board (DQB) with their "What we notice / What we wonder" observations. The teacher introduces the driving question and unit storyline. Lesson 2 — SITES OF GASEOUS EXCHANGE (Reading + Observe): Students read the relevant section of the student textbook and observe a teacher-guided animation of gaseous exchange across leaf surfaces. They identify and label the sites — stomata,

	cuticle, lenticels, and pneumatophores — and connect each site back to the bubble locations observed in Lesson 1. Groups begin constructing a leaf cross-section model. Lesson 3 — STOMATAL MECHANISMS (Animation + Collaborative Inquiry): Students investigate the mechanisms of stomatal opening and closing using teacher-curated animations. They evaluate the potassium ion theory and the starch–sugar hypothesis through structured group discussion, constructing written explanations and debating which theory better explains the wilting maize seedling phenomenon. Model diagrams are revised to include guard cell mechanisms. Lesson 4 — AEROBIC & ANAEROBIC RESPIRATION (Research + Experiment + Data Analysis): Students read about aerobic and anaerobic respiration and design a simple investigation comparing CO_2 production by yeast in aerobic vs. anaerobic conditions using limewater indicators. They collect, tabulate, and interpret data, constructing word and symbol equations for both pathways. Lesson 5 — IMPORTANCE OF ANAEROBIC RESPIRATION (Application + Real-World Contexts): Students examine the economic and ecological importance of anaerobic respiration in Kenyan contexts — fermentation of uji and busaa, biogas generation, bread-making with yeast, and ethanol production. They connect the mandazi dough supporting phenomenon to their experimental findings and begin planning their unit project. Lesson 6 — FACTORS AFFECTING RESPIRATION RATE (Investigation + Data Interpretation): Students investigate how temperature, glucose concentration, and oxygen availability affect respiration rates in yeast, using locally available materials (sugar-cane juice, maize flour). They graph results and explain trends using their understanding of enzyme activity and cellular respiration. Lesson 7 — ADAPTATIONS FOR GASEOUS EXCHANGE IN DIFFERENT ENVIRONMENTS (Comparative Analysis): Students compare adaptations for gaseous exchange in xerophytes (e.g., cactus, acacia), hydrophytes (e.g., water hyacinth in Lake Victoria), and mesophytes (e.g., sukuma wiki), relating structural features to environmental conditions across Kenya's diverse ecosystems. Lesson 8 — PHOTOSYNTHESIS, GASEOUS EXCHANGE & RESPIRATION LINKAGES (Synthesis + Model Building): Students map the relationships between photosynthesis, gaseous exchange, and respiration on a systems diagram, tracking the flow of CO_2, O_2, water, and glucose. They revise their cumulative unit models and update the DQB to reflect growing understanding. Lesson 9 — PROJECT LAUNCH & PLANNING (Guided Inquiry Project Design): Students are introduced to the unit project: designing and conducting an investigation into anaerobic respiration using locally available Kenyan materials. Groups choose their focus (yeast fermentation, biogas simulation, or food fermentation), formulate hypotheses, and plan procedures, materials lists, and data collection methods. Lesson 10 — PROJECT EXECUTION (Hands-On Investigation): Student groups carry out their planned investigations, collecting and recording data. The teacher circulates with structured prompts, checking that groups are controlling variables and recording observations accurately. Groups begin preparing evidence-based explanations for their findings. Lesson 11 — PROJECT PRESENTATIONS & PEER REVIEW (Communication + Argumentation): Groups present their project findings to the class, connecting results back to the unit driving question and the anchoring phenomenon. Peers provide structured feedback using a co-created rubric. The class debates how their findings confirm or challenge their original Lesson 1 predictions on

	the DQB. Lesson 12 — CONSOLIDATION, MODEL FINALISATION & UNIT REVIEW: Students finalise and annotate their cumulative unit models of plant gaseous exchange and respiration, incorporating all evidence gathered across the unit. The class revisits and resolves all questions on the DQB, reflecting on how their understanding has changed since Lesson 1. The teacher conducts a structured end-of-unit review connecting all phenomena, concepts, and real-world applications.
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — The Wilting Pot Plant: A potted maize seedling left in the hot Nairobi midday sun wilts dramatically, but recovers in the evening. Students observe that the wilting coincides with stomata closing to reduce water loss — directly connecting stomatal mechanism to a visible, familiar plant behaviour. – SUPPORTING PHENOMENON 2 — The Rising Dough: A mama mboga (vegetable vendor) near Gikomba Market mixes maize flour, warm water, and a pinch of sugar with yeast to make mandazi dough. The dough visibly expands and becomes bubbly overnight. Students are puzzled about what is causing the dough to rise and where the bubbles inside come from — anchoring the investigation into anaerobic respiration and CO_2 production by yeast. – SUPPORTING PHENOMENON 3 — The Biogas Flame: A farmer in Kirinyaga County has a biogas digester fed with cow manure and plant waste. The gas produced burns with a steady blue flame used for cooking ugali. Students question what biological process produces this flammable gas from organic plant and animal waste, connecting anaerobic respiration to real-world energy production. – SUPPORTING PHENOMENON 4 — The Athlete's Burning Muscles: A Kenyan 800-metre runner from Iten collapses on the track immediately after a race, with cramping leg muscles. Students question why muscles "burn" and cramp during intense exercise even though the athlete has been breathing throughout the race — bridging plant and animal anaerobic respiration and demonstrating the universality of these processes.

LESSON 1 (80 minutes): Anchoring Phenomenon & Predict Phase: Where Do the Bubbles Come From?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit anchoring phenomenon and elicit students' prior knowledge and curiosity about plant gaseous exchange and respiration.
Knowledge	Students will know that gases move in and out of plant leaves, that bubbles emerging from submerged sukuma wiki leaves are evidence of gas release, and that the location and rate of bubble production varies depending on water temperature and specific sites on the leaf surface.
Skills	Students will observe and record the emergence of bubbles from submerged sukuma wiki leaves, annotate diagrams to predict the sites of gas exchange, and formulate investigable questions to populate the Driving Question Board (DQB).
Attitudes	Students will appreciate that careful observation of everyday Kenyan food plants can reveal complex biological processes, and will develop curiosity and openness to revising their initial ideas as evidence accumulates.
Key Inquiry Question	How do plants exchange gases with their environment?
Purpose in Storyline	This lesson launches the unit storyline by anchoring student curiosity in a puzzling, observable phenomenon — bubbles streaming from sukuma wiki leaves submerged in warm vs. cold water. Students make initial predictions about where and why bubbles emerge, draw annotated diagrams, and populate the Driving Question Board with 'What we notice / What we wonder' observations. Their predictions and questions will be revisited, tested, and refined across all six lessons in the unit.
Safety Notes	Ensure beakers are placed on stable flat surfaces to avoid spills of warm water. Warn students that warm water (not boiling) is used — target temperature 40–45°C — and that they should not touch the beaker base immediately after heating. Freshly cut sukuma wiki leaves should be handled with care; students with known plant allergies should inform the teacher. No sharp instruments are used in this lesson.

B. LESSON OVERVIEW
In this opening lesson, students are introduced to the unit anchoring phenomenon: freshly cut sukuma wiki (kale) leaves submerged in warm water produce streams of tiny bubbles from specific points on the leaf surface, while the same leaves in cold water produce far fewer bubbles. Students first predict — without instruction — where the bubbles come from, what gas they contain, and why temperature affects bubble production. They then observe the phenomenon directly using beakers, warm and cold water, and sukuma wiki leaves from the school garden or a local market. Students record observations, draw annotated diagrams of bubble sites, and share initial explanations with peers. The lesson closes with the teacher introducing the unit driving question and guiding students to populate the Driving Question Board (DQB) with their noticings and wonderings. No formal vocabulary is introduced yet — this lesson is entirely about launching curiosity, surfacing prior knowledge, and opening productive questions that the rest of the unit will answer.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Before touching any materials, students individually examine a	Distribute one sukuma wiki leaf and one hand lens per student.	Think-Pair-Share with public prediction wall: By externalising	Teacher circulates during individual writing and looks for: (a)

📹 **VIDEO:** [Elements and atoms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	single dry sukuma wiki leaf placed on their desk. They study both sides of the leaf with the naked eye and, if available, a hand lens. Each student records in their science notebook: (1) a labeled diagram of the leaf showing any surface features they can see, (2) a written prediction answering: 'If this leaf were submerged in warm water, where do you think bubbles would come from, and what gas would be inside them?' Students then share predictions with a partner (Think-Pair-Share) before three to four pairs share with the whole class. The teacher records all predictions — without confirming or correcting — on the board under the heading 'Our Predictions.'	Say: 'Before we do anything else, I want you to become a detective. Look very carefully at this leaf — both sides. You have 3 minutes of silent observation. Draw what you see and label anything that catches your eye.' [Allow 3 minutes of silent observation.] Then say: 'Now, in your notebook, answer this question: If I submerged this leaf in warm water, where exactly would bubbles come from, and what gas do you think would be inside those bubbles?' [Allow 4 minutes of individual writing. Wait for pens to stop moving.] After writing, say: 'Turn to your partner and share your prediction. You each have 90 seconds.' [Allow 3 minutes for paired discussion.] Cold-call at least 4 pairs — spread across the room — using: 'Amina, what did you and your partner predict?' Record every prediction verbatim on the board. Do NOT affirm or correct. Say: 'These are all interesting ideas. We are going to test them right now.'	individual predictions before observation, this strategy activates prior knowledge, creates cognitive investment in the outcome, and gives the teacher a baseline assessment of students' mental models of plant structure. Recording predictions publicly (on the board) creates accountability and allows students to compare their initial ideas to evidence later in the lesson.	whether students identify the underside of the leaf as different from the upper side, (b) whether any students spontaneously mention 'pores,' 'holes,' or 'stomata,' (c) whether students predict oxygen or carbon dioxide or 'air' as the gas. Teacher notes the 3–4 most common prediction types to use as anchors during the Explain phase. Observable indicator: every student has a written prediction and a labeled diagram in their notebook before the observation begins.
Observe Phase 📹 **VIDEO:** [Elements and atoms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of four. Each group receives: two 500 mL beakers (one filled with warm water at ~42°C, one with cold water at ~15°C or room temperature), two freshly cut sukuma wiki leaves of similar size, and a printed observation recording sheet with an outline diagram of a leaf. Students simultaneously submerge one leaf in warm water and one in cold water and observe for 5 minutes. They mark on their leaf outline diagrams exactly where bubbles appear, count approximate bubble streams, and note differences between warm and cold	Before distributing materials, say: 'Your job as scientists right now is not to explain — your job is to observe as precisely as possible. Record exactly where bubbles appear, not why.' Demonstrate the setup with one beaker: hold a leaf at the stem end and lower it gently into warm water so students see the technique. Say: 'Notice I am submerging the whole leaf including the edges. Now watch — do not explain yet, just watch and record.' [Allow groups 5 minutes of uninterrupted observation. Circulate silently, crouching to student level.] At the 3-minute mark, prompt quietly to any group	Structured Observation with Annotated Diagram: By constraining students to observe before explaining, this strategy builds disciplinary habits of separating data from interpretation (NGSS SEP 3: Planning and Carrying Out Investigations; SEP 4: Analyzing and Interpreting Data). The annotated diagram forces spatial precision — students must commit to a location on the leaf rather than giving vague descriptions. Comparing warm vs. cold water introduces the variable of temperature as an evidence anchor that will connect to enzyme activity and stomatal opening in	Teacher collects one annotated diagram per group at the end of the observation phase. Look for: (a) students marking the underside of the leaf as the primary bubble source, (b) students identifying discrete point-sources rather than the whole surface, (c) students noting the warm vs. cold difference quantitatively (more vs. fewer streams). Observable indicator: each group's diagram has at least 3 labeled observation notes and a comparison between warm and cold water conditions.

	conditions. Groups also note: which surface of the leaf (upper or lower) produces more bubbles; whether bubbles come from the edges, veins, or flat surface; and how quickly bubble production starts. After 5 minutes, groups rotate their beakers so every student observes both conditions. Students update their individual notebooks with annotated diagrams and written observations.	not yet annotating: 'Where exactly are the bubbles coming from? Can you mark that spot on your diagram?' After 5 minutes, call attention: 'Groups, pause. Rotate your beakers — the person who observed warm water now observes cold. You have 2 more minutes.' [Allow 2 minutes.] After rotation, ask the whole class: 'Raise your hand if you saw more bubbles in the warm water than the cold.' [Pause 10 seconds. Count hands aloud: 'I see 24 hands — almost everyone.'] Then cold-call 3 students: 'Juma, tell me one specific thing you noticed about where the bubbles came from — not why, just where.' Repeat for two more students. Record their observations on the board under 'What We Observed.'	later lessons.	
Explain Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to their original individual predictions on the board. As a class, they compare 'What We Predicted' with 'What We Observed.' Students discuss in groups: Which predictions were supported by the evidence? Which were surprised by the evidence? Each student then writes a 'revised initial explanation' in their notebook — 2 to 3 sentences answering: 'Based on what I observed, I now think the bubbles come from _______ because _______.' Groups share revised explanations. The teacher introduces — for the first time — the unit driving question: 'How do plants breathe, and what happens inside their cells when they release the energy stored in food?' The teacher explicitly names that this question will take the whole unit to answer, and that today's observation is only the first clue.	Point to the prediction board: 'Let us be honest scientists. Look at what you predicted before you observed. Thumbs up if your prediction was close to what you saw; thumbs sideways if you were partly right; thumbs down if you were surprised.' [Wait 5 seconds. Scan the room.] Say: 'Being surprised is the best thing that can happen in science — it means you have learned something new.' Cold-call one student with thumbs down: 'Fatuma, what surprised you, and what does your new observation make you think?' [Wait 10 seconds for student to respond.] After 3 student shares, say: 'Now in your notebooks, write a revised explanation — 2 to 3 sentences. Start with: I now think the bubbles come from... because...' [Allow 4 minutes of individual writing.] Introduce the driving question by writing it on the	Predict-Observe-Explain (POE) Cycle with Claim-Evidence-Reasoning scaffold: Students explicitly compare predictions to observations and construct a revised explanation using the sentence frame 'I now think... because...' This mirrors the scientific practice of revising models based on evidence (NGSS SEP 6: Constructing Explanations). The reading card functions as a minimal anchor text — enough to introduce the concept of internal leaf structure without front-loading vocabulary, preserving productive struggle.	Teacher reads 6–8 revised explanations over students' shoulders during writing time. Look for: (a) students citing a specific location (e.g., 'underside of the leaf,' 'tiny holes') rather than vague answers, (b) students connecting bubble production to something inside the leaf rather than the water itself, (c) students using evidence from their observation ('I saw more bubbles in warm water, so...') in their reasoning. Observable indicator: revised explanation contains a specific claim AND a piece of observational evidence.

	Students then read a short 150-word ARES reading card (provided on the platform) describing the sukuma wiki leaf cross-section as an introductory primer — no technical vocabulary is given yet, only the idea that leaves have internal air spaces and surface openings. Students annotate the reading card with connections to their observations.	board in full. Say: 'This is the question our entire unit will answer. We cannot answer it today — and that is exactly the point. Today you found one clue. Over the next five lessons, you will find more clues until you can answer this fully.' Distribute the ARES reading card. Say: 'Read this silently and circle any part that connects to what you just observed. You have 4 minutes.' [Allow 4 minutes.] Cold-call 2 students: 'Otieno, what did you circle and why?' After shares, say: 'Hold onto your questions — we are about to put them somewhere important.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each student receives three sticky notes of different colors: yellow for 'What I noticed,' blue for 'What I wonder,' and pink for 'What I think I already know.' Students individually write one idea per sticky note (at least one of each color) based on the lesson's phenomenon. Students then bring their sticky notes to a large class DQB chart paper posted on the wall (or a designated section of the board). The teacher facilitates a 5-minute whole-class clustering of the sticky notes into emerging theme groups (e.g., 'Questions about where gas exchange happens,' 'Questions about what the gas is,' 'Questions about temperature effects,' 'Questions about whether plants breathe like we do'). The DQB remains posted in the classroom for the duration of the unit and will be revisited and updated at the start and end of each subsequent lesson.	Distribute sticky notes. Say: 'Yellow note: write one thing you noticed during today's observation. Blue note: write one question you are still wondering about — make it a real question you genuinely want answered. Pink note: write one thing you already think you know about how plants get energy or exchange gases.' [Allow 4 minutes of silent individual writing.] Circulate and prompt any student who is stuck: 'What was the most surprising thing you saw today?' or 'What would you need to know to explain the bubbles fully?' After 4 minutes, say: 'Bring your notes to the board. As you stick them up, read your blue wonder question aloud.' [Allow students to approach the board in a controlled flow — row by row or group by group.] After all notes are posted, step back and say: 'I am going to group these questions. Help me — where do you think these two questions belong together?' Facilitate clustering for 5 minutes, naming each cluster with a strip of	Driving Question Board (DQB) as a Living Knowledge-Tracking Tool: The DQB externalises the class's collective curiosity and makes the arc of learning visible over the unit. Clustering questions by theme gives students agency in shaping the unit narrative and mirrors how scientists organise research problems. The three sticky note colors distinguish observation (data), inquiry (questions), and prior knowledge (mental models), building metacognitive awareness of different types of knowing (NGSS SEP 1: Asking Questions and Defining Problems).	Teacher photographs the DQB at the end of the lesson for reference. Assess: (a) whether students' wonder questions are scientifically productive and investigable (e.g., 'Do all leaves produce the same number of bubbles?' vs. 'Why is biology hard?'), (b) whether any students spontaneously use terms like 'stomata,' 'respiration,' or 'photosynthesis' on their pink prior knowledge notes — these students may be ready for extension, (c) whether the class collectively generates questions covering the main unit concepts (gas exchange sites, gas identity, temperature effects, cellular respiration). Observable indicator: every student has contributed at least three sticky notes to the DQB.

		paper. End by pointing to the driving question written on the board: 'Every question you wrote today lives somewhere inside this big question. By the end of this unit, we will come back to this board and answer most of them. Let us see how far we get.'		
Model Building Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students draw their first 'Initial Model' of a sukuma wiki leaf cross-section in their science notebooks. This is a free-hand sketch — no template provided — showing their current best thinking about: (a) what is inside the leaf, (b) where gases enter and exit, and (c) how temperature might affect the process. Students label their model with at least three annotations. They write at the bottom: 'This is my model on [date]. I think it shows ___. I am not sure about ___.' Models are not graded — they are a baseline that students will revise in every subsequent lesson as they gain new evidence. Students share their model with one partner and identify one similarity and one difference between their two models.	Say: 'Scientists do not just write notes — they build models to show how they think something works. Your model does not have to be correct. It has to show your honest best thinking right now.' Write on the board: 'Your model must show: (1) The inside of the leaf. (2) Where gases go in and out. (3) Why warm water made more bubbles.' [Allow 6 minutes of individual model drawing. Play soft background music if available to signal focused work time.] At 3 minutes, circulate and prompt: 'Where exactly on your diagram are you showing gas movement? Use an arrow.' After 6 minutes: 'Show your model to your partner. Find one thing that is the same and one thing that is different. You have 2 minutes.' [Allow 2 minutes.] Bring the class back together. Cold-call 2 students to briefly hold up and narrate their model: 'Wanjiku, point to one part of your model and tell us what it shows.' Close by saying: 'This model is the beginning. Every lesson from now on, you will have a chance to revise it as you learn more. By Lesson 6, your model will be completely different — and that is what learning looks like in science.'	Initial Model Construction (NGSS SEP 2: Developing and Using Models): Asking students to build a physical representation of their mental model forces them to commit to spatial and mechanistic claims — which gaps in understanding they must then confront. The 'I think / I am not sure' annotation frame builds metacognitive honesty and creates explicit revision targets. Comparing models with a partner surfaces diversity of prior knowledge and models collaborative scientific discourse.	Teacher collects model pages (or photographs them) at the end of the lesson. Assess: (a) whether students represent the leaf as a flat surface only (surface-level thinking) or attempt to show internal structure (deeper prior knowledge), (b) whether gas movement is shown as entering/exiting at specific points or uniformly across the whole leaf, (c) whether temperature is represented at all (few students will — this sets up cognitive need for later lessons). Observable indicator: every student's notebook contains a labeled model with at least three annotations and the 'I think / I am not sure' statement.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) Which student predictions surprised you most — what do they reveal about prior conceptions you need to address in Lesson 2? (2) Did all groups clearly observe a difference between warm and cold water bubble production? If any group did not, what variable may have interfered (leaf age, temperature accuracy, leaf size), and how will you address this in Lesson 2's observation setup? (3) Review the DQB photograph: Are the wonder questions rich enough to sustain six lessons of inquiry? If the questions are too surface-level (e.g., 'What are the bubbles?') rather than mechanistic (e.g., 'How does the leaf control when gas goes in and out?'), plan a 5-minute DQB revisit at the start of Lesson 2 to deepen the questions using a 'But why...?' prompt. (4) Identify 2–3 students whose initial models showed unexpectedly sophisticated thinking (e.g., drew stomata unprompted, labeled guard cells, mentioned ATP or photosynthesis) — plan to use these students as 'science consultants' during peer discussion in Lesson 3. (5) Were all students, including those who may be less confident in English, able to participate in the sticky note activity? If language was a barrier, consider allowing DQB entries in Kiswahili for the next lesson and co-constructing a bilingual DQB key vocabulary strip.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed freshly cut sukuma wiki leaves submerged in warm (~42°C) and cold (~15°C) water, recording the location and rate of bubble emergence on annotated leaf diagrams. They noted that bubbles appeared to emerge from specific point-sources on the leaf surface — more abundantly from warm water than cold — rather than uniformly across the entire leaf.
What did I learn?	Students learned that gases are present inside plant leaves and can be released when the leaf is submerged in water; that the rate of gas release is affected by water temperature; and that bubble production is concentrated at specific sites on the leaf rather than the whole surface. Students also learned that scientists use models, annotated diagrams, and public prediction-tracking to organise their thinking.
How does this explain the phenomenon?	Students constructed initial explanations (revised from their pre-observation predictions) proposing that the bubbles come from openings or spaces inside the leaf, and that warmth increases the rate at which gas escapes. Students could not yet fully explain what the gas is, why only specific spots produce bubbles, or how temperature mechanistically increases bubble production — these gaps are recorded on the DQB as productive open questions for the rest of the unit.

LESSON 2 (40 minutes): Sites of Gaseous Exchange in Plants	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To identify and describe the structural sites through which plants exchange gases with their environment, and to connect those sites to the bubble locations observed on sukuma wiki leaves in Lesson 1.
Knowledge	Students will be able to identify and describe the four main sites of gaseous exchange in plants — stomata, cuticle, lenticels, and pneumatophores — including the role of guard cells in opening and closing stomata.
Skills	Students will practise reading for scientific information, labelling biological diagrams, and constructing a leaf cross-section model that maps gas-exchange sites to observable evidence.
Attitudes	Students appreciate that plants have evolved specialised, finely tuned structures to balance gas exchange and water conservation — reflecting the wonder and complexity of living systems.
Key Inquiry Question	How do plants exchange gases with their environment?
Purpose in Storyline	In Lesson 1 students observed that bubbles emerged from specific spots on submerged sukuma wiki leaves, not uniformly across the entire surface. Lesson 2 resolves the first layer of that mystery: those spots are stomata (and other specialised structures). By naming, locating, and modelling these sites students gather the anatomical evidence needed to later explain why bubbles appear where they do, why temperature matters, and — ultimately — how plant respiration works at the cellular level.
Safety Notes	No chemicals or sharp instruments are used in this lesson. If printed leaf cross-section diagrams are cut out with scissors, supervise safe scissor use. Ensure the classroom projector or screen is positioned so all learners can view the animation without straining their eyes.

B. LESSON OVERVIEW
Learners revisit the sukuma wiki bubble phenomenon from Lesson 1, then read the relevant textbook section on sites of gaseous exchange. They observe a teacher-guided animation showing gas movement through stomata, the cuticle, lenticels, and pneumatophores. Using a structured note-taking frame, groups identify and label each site on a leaf cross-section diagram and begin constructing a physical leaf cross-section model using locally available materials (cardboard, cellophane, clay). The lesson closes with a DQB update that connects newly gathered anatomical evidence back to the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners individually re-examine the class anchor photograph (or sketch) of bubbles rising from sukuma wiki leaves submerged in warm water — displayed on the board from Lesson 1. Each learner writes a silent prediction on a	Display the anchor photograph and say: 'Before we open any books, look carefully at this image from our Lesson 1 investigation. Where exactly are the bubbles coming from — the whole leaf surface, or specific spots? Write	Predict-before-reading (prediction activation): By committing to a prediction before instruction, learners activate prior knowledge and create a cognitive need to resolve the discrepancy. This lowers the barrier to engagement	Teacher scans written sticky notes/exercise books to check whether learners can articulate at least one reason for their prediction (not just a guess). Listen during cold-calls for the use of prior vocabulary (e.g., 'pores,' 'skin

📖 **HTML:** [Evolution of Simple Cells (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	sticky note or in their exercise book, answering: 'Where exactly on the leaf do you think the bubbles are escaping from, and why?' They then share predictions in pairs before two or three pairs are cold-called to share with the class. Predictions are pinned to the Driving Question Board (DQB) under the column 'What we think we know.'	your best prediction now. You have 90 seconds — go.' [Allow 90 seconds silent writing.] After writing, say: 'Turn and share your prediction with your partner. 30 seconds.' [Allow 30 seconds.] Cold-call 3 pairs: 'Amina and Kiprono — what did you predict?' 'Brian and Wanjiru — do you agree or do you have a different idea?' 'Fatuma and Otieno — what was your reasoning?' After each response, respond neutrally: 'Interesting — keep that idea in mind as we investigate today.' Do NOT confirm or deny predictions. Record key prediction phrases on the board in two columns: 'Whole surface' vs 'Specific spots.' Say: 'By the end of today's lesson you will have evidence to decide which column is correct — and why.'	with the text and animation that follow.	of the leaf'). Flag learners who write 'the whole surface' for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** [Photorespiration] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Evolution of Simple Cells (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Guided Reading (15 min): Learners open their student textbook to the section on sites of gaseous exchange. They read silently and independently, using a structured READ-MARK-NOTE frame printed on a half-sheet: (1) CIRCLE any new vocabulary; (2) UNDERLINE the sentence that best explains the function of each structure; (3) NOTE in the margin one connection to the sukuma wiki bubble observation. Structures covered: stomata and guard cells (upper and lower epidermis), the waxy cuticle, lenticels on woody stems, and pneumatophores in mangrove roots (e.g., Mida Creek mangroves, Kilifi). Part B — Teacher-Guided Animation (7 min): Teacher plays a short animation (or annotated diagram sequence) showing gas molecules moving through each of the four sites. Teacher pauses the	Before reading, distribute the READ-MARK-NOTE frame and say: 'You have 15 minutes to read pages [X–Y]. Use the frame — circle, underline, note. For every structure you read about, ask yourself: could this be the spot where I saw bubbles on the sukuma wiki leaf? Go.' [Set a visible timer for 15 minutes. Circulate and kneel beside struggling readers; point to key sentences and whisper: 'What does this sentence say the stomata does?'] After reading, say: 'Pencils down. Let's confirm your labels with the animation.' Play the animation. Pause at stomata: 'Everyone point to where you drew the stomata on your diagram. [Wait 5 seconds.] Good — now watch the guard cells open and close. What do you notice?' [Wait 10 seconds. Cold-call 2 learners.] Pause at lenticels: 'Has anyone	READ-MARK-NOTE with anchored annotation: The structured reading frame keeps learners actively processing text rather than passively skimming. Simultaneous diagram labelling during the animation forces learners to translate between verbal, visual, and spatial representations — deepening encoding of the anatomical sites.	Circulate during reading and check READ-MARK-NOTE frames: every learner should have at least four structures circled and one margin note connecting to the bubble phenomenon. During animation pauses, use show-of-hands to confirm correct label placement ('Raise your hand if your stomata label is on the lower epidermis'). Listen for correct use of 'guard cells' during cold-calls — if absent, re-teach before moving on.

	animation at each structure and uses a laser pointer or finger to trace the path of CO_2 and O_2. Learners annotate a blank leaf cross-section diagram in their books as the animation proceeds, labelling: upper cuticle, upper epidermis, palisade mesophyll, spongy mesophyll, air spaces, stomatal pore, guard cells, lower epidermis, lower cuticle. For woody stem: bark, lenticel. For mangrove root: pneumatophore above water line.	seen rough, slightly raised spots on the bark of a mango tree or a Nandi flame near your home? Those are lenticels — this is how the woody part of the plant breathes.' Pause at pneumatophores: 'Think about the mangrove trees at Mida Creek in Kilifi — their roots grow upward out of the mud. Why might that be an advantage for gas exchange?' [Wait 10 seconds. Cold-call 1 learner.]		
Explain Phase 📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group receives a large printed (or hand-drawn) blank leaf cross-section outline and a set of label cards (stomata, guard cells, cuticle, palisade cells, spongy mesophyll, air spaces, lenticel, epidermis). Groups collaboratively place and glue label cards onto the diagram and then write a one-sentence functional description below each label. Groups then answer three 'Connect-Back' questions in writing: (1) In Lesson 1, bubbles appeared at specific points on the submerged sukuma wiki leaf — which structure do you now believe is responsible? Give evidence. (2) Why did far fewer bubbles appear in cold water? Use what you know about stomata to suggest a reason. (3) The waxy cuticle limits gas exchange — why might this be an advantage for a sukuma wiki plant growing under the hot Nairobi sun? Groups share answers in a whole-class debrief; teacher records agreed explanations on the board.	Distribute label cards and diagram outlines. Say: 'In your group, place every label card where you believe it belongs. You must agree as a group — if there is disagreement, discuss it. You have 8 minutes.' [Circulate. If a group places stomata on the upper epidermis only, say: 'Look at your textbook diagram again — which surface has more stomata in most leaves, and why might that be?'] After 8 minutes, say: 'Now answer the three Connect-Back questions in your exercise books. Every person writes. 5 minutes.' [Circulate, prompt with: 'Use evidence from Lesson 1 in your answer.'] Whole-class debrief: cold-call one group per question. For Question 2 say: 'Jomo's group — read your answer to question 2 aloud.' After the answer: 'Does another group agree? Does anyone want to add or challenge that idea?' [Wait 10 seconds before accepting hands.] Summarise on the board: 'STOMATA are the primary site — that is where the Lesson 1 bubbles came from. Guard cells control opening — temperature affects this, which explains the cold-water result. The cuticle limits water loss	Collaborative labelling with evidence-linked questioning (Claim-Evidence-Reasoning lite): Placing physical label cards forces spatial reasoning. The three Connect-Back questions scaffold CER thinking by requiring learners to link new anatomical knowledge explicitly to the phenomenon observations, building a causal narrative that advances the driving question.	Check that all groups correctly place stomata on the lower epidermis and write a functionally accurate sentence (e.g., 'Stomata are openings controlled by guard cells that allow CO_2 in and O_2 out'). Listen during debrief for causal language: learners should say stomata 'allow' or 'control' gas movement, not just that they 'are holes.' If causal language is absent, prompt: 'What do the guard cells do to cause that?'

		— a vital adaptation in warm climates like Nairobi.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group writes one new 'We now know...' statement and one new 'We still wonder...' question on separate sticky notes and adds them to the class DQB. The teacher facilitates a 3-minute whole-class gallery walk past the DQB. Learners place a small dot sticker next to the 'We still wonder...' question they find most interesting or important. The most-dotted question is read aloud and recorded as the class's priority question to pursue in upcoming lessons.	Say: 'Based on today's reading, animation, and labelling activity, each group will add two sticky notes to our Driving Question Board. On the GREEN note, write one thing you now KNOW that you didn't know at the start of today — connect it to the sukuma wiki bubbles. On the YELLOW note, write one question you still have about how plants breathe or release energy. You have 3 minutes.' [Distribute sticky notes. Circulate and prompt groups that write vague statements: 'Can you be more specific — which structure, and how does it connect to the bubbles?'] After posting, say: 'Stand up, walk to the board, and put a dot sticker next to the yellow question you most want answered.' After dot voting, read the top question aloud: 'Our class most wants to know: [read question]. We will keep this question in mind as we move into Lessons 3 and 4, where we will investigate how and when stomata open and close, and what gases actually move through them.'	Driving Question Board dot-voting: The DQB makes the progression of understanding visible over time. Dot-voting on 'wonder' questions gives every learner a stake in the inquiry direction and helps the teacher identify persistent misconceptions or gaps to address in subsequent lessons.	Read all 'We now know...' sticky notes before the next lesson. Flag any that reveal misconceptions (e.g., 'Plants only breathe through their roots' or 'Stomata are only on the stem'). These become targeted re-teaching points at the start of Lesson 3. The priority 'wonder' question guides lesson framing.
Model Building Phase 📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook)	Groups begin constructing a 3D leaf cross-section model using locally available materials: layers of cardboard for epidermis and mesophyll, green cellophane or tissue paper for palisade and spongy layers, clay or playdough for guard cells, and small holes punched with a pencil tip for stomatal pores. Each group labels their model with small flag labels on toothpicks. Groups do not complete the model today — they store it for revision and extension	Distribute materials. Say: 'We are going to build a model of a leaf cross-section that we will keep adding to across this unit. Today your job is to build the basic layers and mark the sites of gaseous exchange we identified. You have 8 minutes to build as much as you can. This model will grow as our understanding grows — so do not worry about making it perfect today.' [Circulate. If a group omits air spaces in the spongy mesophyll, say: 'Where do the	Progressive model building (incremental modelling): Beginning the model now, with only anatomical evidence, makes thinking visible and creates a physical artefact that will be revised as new evidence accumulates lesson by lesson. Writing the key claim on the model base reinforces that models are evidence-based, not decorative — a core science and engineering practice.	Check that each group's model includes at minimum: two epidermal layers, a cuticle indication, stomatal pores on the lower epidermis, guard cells flanking the pores, and distinct palisade and spongy mesophyll layers. Check that the key claim written on the base card correctly names stomata as the bubble source and mentions guard cells. Groups missing guard cells or writing a claim that does not reference Lesson 1 evidence are

Source: CK-12 🔍 Search ARES for similar readings	in subsequent lessons as new evidence is gathered. On the back of their model's base card, groups write today's key claim: 'Bubbles in Lesson 1 escaped through stomata — small pores in the lower epidermis of the sukuma wiki leaf, controlled by guard cells.'	gases travel once they enter through the stomata? Show me that space in your model.'] With 2 minutes remaining, say: 'Write today's key claim on the back of your base card — start with the words: Bubbles in Lesson 1 escaped through...' [Allow groups to write. Collect models for storage or have learners store them safely in their project folders.] Close the lesson: 'Today we answered part of our driving question — we know WHERE plants exchange gases. In our next lessons we will investigate HOW and WHEN that exchange happens, and what the plant is actually doing with those gases inside its cells.'		prompted before storing their models.

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) Did the READ-MARK-NOTE frame support all readers, including those who struggle with dense scientific text? If not, consider preparing a simplified glossary card for Lesson 3. (2) During cold-calls, did learners spontaneously use causal language ('because,' 'which causes,' 'so that') when explaining stomatal function, or did they describe structures without function? Causal language is a prerequisite for the respiration content in Lessons 5–8. (3) Were the Connect-Back questions (linking today's anatomy to Lesson 1's bubbles) answered with specific evidence, or were answers vague? If vague, plan a brief anchor-phenomenon revisit at the start of Lesson 3. (4) Look at the DQB 'wonder' questions — do they reveal that learners are thinking about gas movement (good) or are they stuck on irrelevant tangents? Use this to calibrate the depth of the Lesson 3 stomatal mechanism lesson. (5) Were all four sites (stomata, cuticle, lenticels, pneumatophores) represented in group diagrams, or did learners focus only on stomata? Lenticels and pneumatophores often get neglected — reinforce with a real mango branch or mangrove photograph at the start of Lesson 3 if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Bubbles escaped from specific spots on the submerged sukuma wiki leaf surface, not uniformly from the whole leaf. The animation showed gas molecules moving in and out through distinct openings on the leaf and stem surfaces. Fewer bubbles appeared in cold water compared to warm water.
What did I learn?	Plants exchange gases through four main sites: stomata (pores on the epidermis controlled by guard cells), the waxy cuticle (limited exchange), lenticels (on woody stems and branches, e.g., mango bark), and pneumatophores (specialised roots in waterlogged areas like Mida Creek mangroves). Stomata are the primary site of gas exchange in leaves and are responsible for the bubble locations observed in Lesson 1.
How does this explain the phenomenon?	The bubbles in Lesson 1 appeared at specific spots because those spots are stomata — microscopic pores surrounded by guard cells that can open and close. Stomata are concentrated on the lower epidermis of most leaves. The reduction in bubbles in cold water suggests that low temperature affects the opening of stomata or the rate of gas release, a question the class will investigate

	further in upcoming lessons.

LESSON 3 (40 minutes): Stomatal Mechanisms: How Guard Cells Control Gas Exchange	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students investigate the mechanisms by which guard cells open and close stomata, evaluating two competing scientific theories to explain stomatal movement and connecting this mechanism to the bubble observations from Lesson 1.
Knowledge	Students will be able to: (1) describe the potassium ion theory of stomatal opening and closing; (2) describe the starch-sugar hypothesis of stomatal movement; (3) explain how changes in guard cell turgor pressure cause stomatal pores to open or close; (4) distinguish between the two theories and identify evidence supporting each.
Skills	Students will practise: evaluating competing scientific explanations using animation evidence; constructing written scientific arguments; collaborative group discussion and structured debate; annotating model diagrams with mechanistic detail.
Attitudes	Students will appreciate that science involves competing explanations that are evaluated against evidence, and will develop confidence in defending a reasoned scientific position during structured debate.
Key Inquiry Question	How do plants exchange gases with their environment?
Purpose in Storyline	Lesson 3 advances the storyline by explaining the mechanism behind the specific bubble locations observed in Lesson 1. Students now move from knowing WHERE gas exchange occurs (stomata, Lesson 2) to understanding HOW stomata are controlled. This mechanistic understanding is essential before students investigate respiration rates in Lesson 4, because they need to understand that gas movement into and out of leaves is actively regulated, not passive.
Safety Notes	No chemical or flame hazards in this lesson. If printed materials are distributed, ensure no student has paper-cut risk from sharp-edged cards during sorting activities. Students with photosensitive conditions should be seated away from direct projection light during animation viewing.

B. LESSON OVERVIEW

Students begin by revisiting their Lesson 1 bubble observations and Lesson 2 guard cell diagrams to pose a new predictive question: what causes the stomata to open or close? They then observe two teacher-curated animations — one illustrating the potassium ion theory and one the starch-sugar hypothesis — pausing to record key mechanistic steps. In the Explain phase, groups engage in structured collaborative inquiry, constructing written explanations and debating which theory better accounts for why a maize seedling wilts in the midday heat near Kisumu but recovers at dusk. The lesson closes with DQB updating and revision of cumulative leaf cross-section model diagrams to include guard cell mechanisms.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Photorespiration Source: Khan	Students individually re-examine their Lesson 1 annotated diagrams showing bubble emergence points on sukuma wiki leaves and their	Teacher displays the Lesson 1 sukuma wiki bubble photograph or sketch on the board alongside a Lesson 2 stomata diagram.	Prediction journaling with annotation connection — students anchor new thinking to prior evidence (Lesson 1 bubbles,	Teacher circulates during the 3-minute writing time and reads over shoulders, noting whether students are drawing on osmosis concepts

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Lesson 2 labelled stomata diagrams. The teacher poses a focusing question on the board: 'The bubbles came from specific spots. We now know those spots are stomata controlled by guard cells. But what makes guard cells open or close — and why might temperature matter?' Students write a 3-sentence prediction in their exercise books, addressing: (1) what they think triggers guard cell movement; (2) whether they think the process requires energy; (3) whether they think temperature would affect stomatal opening. Students share predictions with a shoulder partner before two or three are shared with the class.	Teacher says: 'Last lesson we agreed that bubbles came from stomatal pores controlled by guard cells. Today we ask a deeper question — what is the engine inside the guard cell that drives the pore open or shut?' Teacher writes the focus question prominently. After giving 3 minutes of individual writing time, teacher uses cold-calling to hear two contrasting predictions, then says: 'Hold those ideas — we are going to test them against evidence from animations.' WAIT TIME instruction: after posing the focus question, teacher waits a full 10 seconds in silence before accepting any verbal response, signalling that deep thinking is expected.	Lesson 2 guard cell structure) before encountering new information, activating prior knowledge and creating cognitive hooks for the animation evidence to follow.	from earlier learning or defaulting to vague ideas like 'the plant decides to open.' This informs how much scaffolding to provide during animation pauses. Teacher mentally notes students who invoke water or ion movement — these students can be called on during the Explain phase to bridge to the potassium ion theory.
Observe Phase 📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students watch two sequential animations (each approximately 3 minutes) curated by the teacher. Animation 1 illustrates the potassium ion (K+) theory: under light, K+ ions are actively pumped into guard cells, lowering water potential; water enters by osmosis; guard cells become turgid; the thickened inner wall causes the cell to bow outward, opening the pore. The reverse process closes the pore in darkness or drought. Students pause the animation at three teacher-designated moments to sketch and label diagrams in their exercise books — (a) K+ movement direction; (b) water movement direction; (c) guard cell shape change. Animation 2 illustrates the starch-sugar hypothesis: in light, starch in guard cells is converted to soluble sugars, lowering water potential and drawing water in by osmosis; in darkness, sugars convert back to starch. Students again sketch	Teacher introduces Animation 1: 'This animation shows a theory proposed from research in the 1960s and 1970s involving potassium ions. As you watch, focus on three things: what moves into the guard cell, what follows it, and what happens to the cell shape.' Teacher pauses the animation at the three designated frames and says: 'Pencils up — sketch what you see. Label the direction of movement with an arrow.' After Animation 1, teacher distributes or projects the comparison table template. Teacher introduces Animation 2: 'Now a second, older theory. Watch for similarities and differences with what you just saw.' After Animation 2, teacher says: 'You have 90 seconds to complete the comparison table before we move to group discussion.' WAIT TIME instruction: after asking 'What is the key difference between the two	Structured observation with designated pause-and-sketch moments and a two-column comparison table. This prevents passive video watching and ensures students extract specific mechanistic information that they will need to construct arguments in the Explain phase. The comparison table is a disciplinary literacy scaffold for scientific argumentation.	Teacher checks sketch diagrams during the 90-second table completion window, looking for correct arrow directions for K+ and water movement. Common misconception to probe: students who draw water moving OUT of guard cells when K+ moves IN (confusing high solute concentration with high water potential). Teacher addresses this with a targeted prompt: 'If you dissolve a lot of salt in water, which side of a membrane has lower water potential — the salty side or the plain water side?' This re-activates osmosis understanding without full re-teaching.

	key moments. Between the two animations, students complete a two-column comparison table: 'What drives water into guard cells?' and 'What is the energy source?' for each theory.	theories?', teacher waits 8 seconds before nominating a student, maintaining an expectant, scanning gaze across the room.		
Explain Phase 📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four engage in structured collaborative inquiry using the Claim-Evidence-Reasoning (CER) framework. Each group receives a scenario card: 'A maize seedling growing in a shamba near Kisumu wilts badly by noon on a hot sunny day, but recovers its turgor and stands upright again by dusk without being watered. Using either the potassium ion theory or the starch-sugar hypothesis — or both — explain why stomata close at noon and reopen in the evening.' Groups must: (1) write a group Claim stating which theory (or combination) they find more convincing; (2) cite at least two pieces of Evidence from the animations and comparison table; (3) write a Reasoning paragraph connecting the evidence to the claim. Groups then participate in a structured debate: one group presents their position, a second group offers a counter-argument or extension, and the teacher facilitates a class vote on which theory was better argued — not which is definitively correct. Students individually revise their Lesson 2 guard cell diagrams to annotate the mechanism they found most convincing.	Teacher distributes scenario cards and says: 'This maize plant near Kisumu wilts at noon but recovers at dusk. Your group has five minutes to agree on a Claim and gather Evidence from your notes. Then you will argue your case to the class.' Teacher circulates with prompts: 'What does your comparison table say about the energy source for each theory? Does that fit what happens in bright midday sun versus cool dusk?' After group writing, teacher selects two groups with different positions and facilitates the debate: 'Group A, state your claim. Group B, what evidence do you have that challenges or extends that explanation?' Teacher closes the debate by saying: 'Scientists still debate the relative contributions of these two mechanisms. What matters is that you used evidence to reason — that is exactly what biologists do.' Teacher says: 'Before you revise your diagrams, tell your partner in one sentence what actually moves water into the guard cell.' WAIT TIME instruction: during the debate, after each group speaks, teacher waits 6 seconds before prompting the next group, giving the class time to process and formulate responses.	Claim-Evidence-Reasoning (CER) structured argumentation followed by a whole-class structured debate. CER is a high-leverage disciplinary practice that builds scientific writing skills and forces students to distinguish between assertion and evidence-backed reasoning. The Kisumu maize scenario grounds abstract ion chemistry in a Kenya-relevant agricultural context.	Teacher listens to group CER writing and targets groups whose Reasoning paragraph merely restates the Claim without connecting evidence. Prompt: 'Your claim says stomata close at noon. Your evidence says K+ moves out. Now write the sentence that connects those two — what does K+ leaving DO to the guard cell?' The structured debate also surfaces misconceptions publicly, allowing whole-class correction. The individual diagram revision at the end serves as an exit-level formative product.
Driving Question Board (DQB) Creation	The class returns to the Driving Question Board established in Lesson 1. Students review the 'What we wonder' sticky notes and	Teacher directs attention to the DQB: 'In Lesson 1 you asked why warm water produced more bubbles. You asked why bubbles	DQB revision using connection notes — this is a metacognitive practice that helps students track their own growing understanding	The quality of connection notes reveals whether students have linked the mechanism (ion movement, osmosis, turgor) to the

📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	identify any questions that the stomatal mechanism evidence now partially or fully answers. Each student writes one new sticky note: either a question their learning today has raised (e.g., 'Does the potassium ion pump use ATP? Is it affected by temperature?') or a connection note linking today's learning to the original phenomenon (e.g., 'The warm water in Lesson 1 might have opened more stomata because temperature affects turgor — more open stomata, more bubbles!'). Students place their notes on the DQB in the appropriate column. The teacher reads two or three aloud and highlights how today's mechanistic evidence advances the driving question.	came from specific spots. Today you have evidence about how those spots open and close. Look at the DQB — which questions can we now partially answer?' Teacher gives 2 minutes for students to write their sticky notes. Teacher then reads one connection note aloud and models moving it from the 'Wonder' column to a 'Growing Understanding' column: 'This student connects temperature in Lesson 1 to turgor pressure today — that is a sophisticated link. Does the class agree?' Teacher closes by restating the driving question: 'We are building toward answering how plants breathe AND how they release energy from food. Today we answered a piece of the first part — we know HOW the breathing pores are controlled.' WAIT TIME instruction: after asking 'Which Lesson 1 questions can we now partially answer?', teacher counts silently to 8 before accepting responses.	across lessons. Moving questions from 'Wonder' to 'Growing Understanding' gives students a visible record of intellectual progress and maintains the storyline coherence across the 12-lesson unit.	original observation (bubble location and temperature effect). Teacher notes students who write only surface-level notes (e.g., 'stomata open because of guard cells') versus those who draw causal chains (e.g., 'warm water increases ion pump activity, more K+ enters, more osmosis, turgid guard cells, open pore, more gas escapes, more bubbles'). The latter group can be peer-tutors in Lesson 4.
Model Building Phase 📹 **VIDEO:** Photorespiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their leaf cross-section model diagrams begun in Lesson 2 and add a zoomed-in inset panel specifically for the stomatal guard cell mechanism. The inset must include: (1) guard cells in both open and closed states shown side by side; (2) arrows showing K+ movement direction for each state; (3) arrows showing water movement direction for each state; (4) labels for turgid and flaccid states; (5) a brief annotation (one or two sentences) stating which theory they used and why. Students who finish early add a second annotation connecting the mechanism to the Lesson 1 observation: 'This is why warm water produced more bubbles —	Teacher projects a partially completed model inset on the board with deliberate errors — for example, K+ arrows pointing outward when the pore is shown as open. Teacher says: 'I have started this inset but I think I made a mistake. Take 30 seconds with your partner to find the error and fix it on my model.' After students identify the error, teacher says: 'Good — now build your own correct version in your model diagram. You have 6 minutes.' Teacher circulates and uses targeted questions: 'Your guard cell is shown as turgid — which ion moved in to cause that? Show me with an arrow.' Teacher gives a 2-minute warning and prompts peer	Error-detection task followed by individual model construction and peer review. The deliberate error on the projected model activates critical evaluation skills and primes students to apply correctness criteria to their own diagrams. Peer review with a checklist builds accountability and rehearses the scientific practice of checking others' reasoning — directly foreshadowing the peer review in Lesson 11.	Peer-reviewed model insets are collected or photographed by the teacher at the end of the lesson. Teacher scans for: (a) correct directionality of K+ and water arrows in both open and closed states; (b) correct use of the terms turgid and flaccid; (c) logical consistency between the chosen theory and the annotated explanation. Students whose insets show persistent arrow-direction errors are identified for targeted follow-up at the start of Lesson 4.

	because...' The completed inset is initialled by a peer reviewer who checks that all five elements are present.	review: 'Exchange books with your partner. Check the five elements on the list on the board. Sign your name if all five are present.' WAIT TIME instruction: after projecting the error model and asking students to find the mistake, teacher waits 30 full seconds in silence, scanning the room, before taking any responses.		

D. TEACHER REFLECTION

After the lesson, consider: (1) Did students move beyond describing WHAT guard cells do to explaining WHY (mechanistically) using ion movement and osmosis? Listen back to any debate audio or review CER paragraphs. (2) Were students able to connect the K+ pump or starch-sugar hypothesis to the temperature-bubble observation from Lesson 1, or did these remain disconnected? If disconnected, plan a brief 5-minute bridging activity at the start of Lesson 4. (3) Which students dominated the debate and which were silent? Adjust group composition or assign specific roles (Claim writer, Evidence finder, Reasoner, Speaker) in Lesson 4 to ensure equitable participation. (4) Did the Kisumu maize scenario feel relevant and authentic to students, or did it feel contrived? If students seemed disengaged, consider substituting the scenario with a locally specific example — for instance, sukuma wiki plants in Nairobi city gardens wilting during the dry season versus thriving after the long rains. (5) Are there questions on the DQB that now point naturally toward respiration — for instance, questions about ATP or energy in the K+ pump? Flag these for Lesson 4 introduction.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the animations, we observed that guard cells change shape when ions and water move in or out of them. In Lesson 1, we observed that warm water produced more bubbles from specific spots on sukuma wiki leaves.
What did I learn?	We learned that two theories explain stomatal movement: the potassium ion theory (K+ is actively pumped into guard cells, lowering water potential, causing osmotic water entry and turgidity, opening the pore) and the starch-sugar hypothesis (starch converts to soluble sugars in light, similarly driving water entry by osmosis). Both theories agree that osmosis and turgor pressure drive guard cell shape changes.
How does this explain the phenomenon?	We can now explain why bubbles in Lesson 1 came from specific spots — those spots are stomata whose guard cells were opened by ion and water movement. We can also partially explain why warm water produced more bubbles: higher temperatures may increase the rate of ion pumping and osmotic movement, leading to more open stomata and greater gas release. The wilting maize near Kisumu closes stomata to conserve water by reversing these ion movements, then reopens at cooler dusk.

LESSON 4 (80 minutes): Aerobic & Anaerobic Respiration — Research, Experiment & Data Analysis	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and compare aerobic and anaerobic respiration in living cells, construct word and symbol equations for both pathways, and connect cellular respiration to the gaseous exchange structures studied in previous lessons.
Knowledge	By the end of the lesson, the learner should be able to: (a) explain the process of aerobic respiration in plant cells and write its word and symbol equations; (b) explain the process of anaerobic respiration in plant cells and yeast cells and write its word and symbol equations; (c) compare and contrast aerobic and anaerobic respiration in terms of reactants, products, and energy yield.
Skills	By the end of the lesson, the learner should be able to: (a) design and conduct a simple controlled investigation comparing CO_2 production in aerobic vs. anaerobic conditions using limewater as an indicator; (b) collect, tabulate, and interpret experimental data; (c) construct word and symbol equations for aerobic and anaerobic respiration.
Attitudes	By the end of the lesson, the learner should be able to: (a) appreciate the importance of cellular respiration as the universal energy-releasing process in all living organisms; (b) demonstrate curiosity and precision when designing and conducting fair-test investigations; (c) value collaborative data collection and honest reporting of results.
Key Inquiry Question	How do plants obtain energy from organic compounds through respiration?
Purpose in Storyline	In Lessons 1–3, students identified where gas exchange happens (stomata, lenticels, cuticle, pneumatophores) and gathered evidence that plants take in CO_2 and release O_2 during photosynthesis. The sukuma wiki bubbles raised a deeper question: do plants also release CO_2? Lesson 4 shifts the investigation inward — into the cell — to answer why plants produce CO_2 even in the dark, and why warming the water increased bubble production. By comparing aerobic and anaerobic pathways using yeast (a model organism), students construct the biochemical story of how glucose is broken down to release energy, completing a critical piece of the driving-question puzzle.
Safety Notes	Limewater (calcium hydroxide solution) is mildly irritating; students must avoid contact with eyes and wash hands after use. Boiling water for the anaerobic set-up must be handled by the teacher only. Yeast suspensions should be disposed of responsibly. Ensure adequate ventilation in the laboratory.

B. LESSON OVERVIEW
Students begin by revisiting the sukuma wiki phenomenon and the Driving Question Board to identify the unanswered question about CO_2 production. They read a structured text on aerobic and anaerobic respiration, then design and conduct a paired investigation in which yeast suspensions are set up under aerobic and anaerobic conditions and their CO_2 output is detected using limewater. Groups tabulate turbidity observations at timed intervals, analyse trends, and construct word and symbol equations for both pathways. The lesson closes with a whole-class model revision and DQB update connecting cellular respiration to the gas-exchange structures already mapped.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students are shown two sealed	Display the two bag set-ups on the	Think-Pair-Share with Written	Circulate during individual writing

📹 **VIDEO:** [Cellular respiration introduction](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	transparent bags: one containing fresh sukuma wiki leaves in warm water (aerobic, leaves in light), and one containing the same leaves wrapped tightly in black polythene (no light, simulating dark/anaerobic-like conditions). They are asked to predict in writing: (1) Which bag do you think will accumulate more CO_2 over 10 minutes? (2) Where does the CO_2 in the dark bag come from if photosynthesis cannot occur? (3) A mandazi vendor in Gikomba Market noticed that her dough rises faster on a warm day than a cold day — what do you think is happening inside the dough? Students record individual predictions in their science notebooks, then share with a partner using a Think-Pair-Share protocol.	demonstration bench. Say: 'Look carefully at both set-ups. Do NOT touch them yet. In your notebook, write your three predictions. You have 3 minutes — go.' [Allow 3 minutes individual writing time.] Then say: 'Turn to the person next to you. Share your prediction for question 2. You have 90 seconds.' [Allow 90 seconds.] Cold-call 3 pairs (one from each bench cluster): 'Amina and Brian, what did you predict for bag two — where does the CO_2 come from in the dark?' After each response, press: 'What evidence from our previous lessons makes you say that?' [WAIT TIME: 12 seconds before accepting responses.] Record divergent predictions on the board in two columns: 'CO_2 comes from…' and 'CO_2 does NOT come from…'. Do not confirm or deny. Say: 'Hold these predictions — our experiment today will help us decide who is right.'	Prediction: Writing predictions before discussion forces individual commitment to an idea, reducing groupthink. Sharing predictions publicly creates cognitive dissonance that motivates evidence-seeking. Connecting to the mandazi dough context activates prior knowledge of yeast fermentation in a familiar Kenyan context, bridging everyday experience to cellular biochemistry.	and scan notebooks. Look for: (1) whether students can distinguish photosynthesis (light-dependent, CO_2 consumed) from respiration (always occurring, CO_2 produced); (2) whether students connect the dark-bag CO_2 to cellular respiration. Flag students who write 'no CO_2 in the dark bag' for targeted questioning during Phase 3. Record names on clipboard.
Observe Phase 📹 **VIDEO:** [Cellular respiration introduction](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	PART A — Structured Reading (15 min): Each student reads a two-page ARES text card titled 'Two Roads to Energy: Aerobic and Anaerobic Respiration.' The text covers: (i) the definition and equation for aerobic respiration (glucose + oxygen → carbon dioxide + water + energy); (ii) the definition and equation for anaerobic respiration in plants (glucose → ethanol + carbon dioxide + energy) and in animals/yeast; (iii) comparison table of reactants, products, location, and relative ATP yield; (iv) a real-world link — how yeast respires anaerobically in maize porridge/dough fermentation used in brewing traditional Kenyan beverages like busaa. Students	PART A: Distribute text cards. Say: 'Read carefully and annotate with ✓, ?, and !. You have 12 minutes. At least one ! annotation must connect the reading to the sukuma wiki leaves from Lesson 1.' [Set visible timer.] At 12 minutes, say: 'Find one partner and share your single most important ! annotation. 60 seconds.' Cold-call 2 students: 'Fatuma, read us your ! annotation and explain the connection to sukuma wiki.' [WAIT TIME: 10 seconds.] PART B: Before students begin the investigation, say: 'Before we set up, tell me — what is our independent variable? What is our dependent variable? What are we	Annotation Coding + Controlled Experiment with Quantitative Observation: Annotation coding (✓/!/?) trains students to read scientifically rather than passively, and the ! code explicitly requires them to link new information back to the anchoring phenomenon. The controlled experiment gives students first-hand, quantitative evidence (limewater turbidity as a proxy for CO_2 concentration) that they generated themselves, making the abstract biochemistry of respiration tangible. Using glucose (sukari) and yeast — familiar ingredients in Kenyan baking and brewing — grounds the chemistry in a known cultural context.	During PART A, collect 5 notebooks at random and check for at least one ! annotation linked to the sukuma wiki phenomenon. During PART B, check each group's data table for: (1) correct identification of independent and dependent variables; (2) consistent turbidity recording at all 5 time points; (3) greater turbidity increase in Tube A (aerobic) than Tube B (anaerobic) — if a group records the opposite, probe immediately: 'Walk me through how you set up Tube B — was the stopper on before or after mixing?' Flag anomalous results for class discussion in Phase 3.

	annotate the text using the coding strategy: '✓' for information that confirms their prediction, '?' for surprises, and '!' for connections to the sukuma wiki phenomenon. PART B — Yeast CO_2 Investigation (25 min): Groups of 4 receive a printed investigation design scaffold. Guided by the teacher, they set up two test tubes: – Tube A (Aerobic): 5 mL warm water (35°C) + 1 g dry yeast + 2 g glucose (sukari) → left open to air → connected via delivery tube to a test tube of fresh limewater. – Tube B (Anaerobic): Same mixture → sealed with a rubber stopper immediately after mixing → connected via delivery tube to a separate test tube of fresh limewater. Groups observe both limewater tubes at 0 min, 5 min, 10 min, 15 min, and 20 min, recording the turbidity on a scale of 0 (clear) to 3 (milky-white). They photograph or sketch the limewater appearance at each interval. A parallel class data table is maintained on the board.	keeping the same to make this a fair test?' Cold-call 3 different students for each variable type. [WAIT TIME: 12 seconds each.] Demonstrate how to connect the delivery tube and how to read limewater turbidity. Stress the safety note: 'I will pour the boiling water used to pre-boil the anaerobic mixture — do not attempt this yourselves.' Circulate every 5 minutes. At each observation point, prompt: 'Record your turbidity score NOW — be precise, be honest even if the result surprises you.' At the 20-minute mark, say: 'Complete your data table and calculate the total turbidity score for each tube across all time points.'		
Explain Phase 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration	PART A — Data Analysis & Equation Construction (15 min): Groups plot their turbidity data as a line graph (time on x-axis, turbidity score on y-axis, two lines for Tube A and Tube B) on graph paper provided. They answer three analysis questions: (1) Which condition produced more CO_2? How does your graph support this? (2) Using your reading, explain WHY aerobic respiration produces more CO_2 than anaerobic respiration per unit of glucose. (3)	PART A: Say: 'You have 12 minutes to plot your graph and answer the three questions. Question 3 is the most important — the equations must be balanced. Use your text card if needed.' Circulate and ask probing questions to groups that are struggling: 'Look at the products column in your comparison table — what does aerobic respiration produce that anaerobic does NOT?' [WAIT TIME: 12 seconds.]	Graphing + Mini Whiteboard Gallery Critique: Graphing transforms raw numbers into a visual pattern, making the difference between aerobic and anaerobic CO_2 production immediately visible. The gallery critique with agree/disagree/add is a low-stakes peer-assessment strategy that surfaces misconceptions publicly so the teacher can address them efficiently. The Part C individual writing task is a 'closing the loop'	Collect all mini whiteboard photos (or have students leave boards up) and assess: (1) aerobic equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (+ energy); (2) anaerobic equation: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (+ energy). Mark errors on your clipboard. For Part C, read 6 individual notebooks while circulating — look for: correct use of 'enzyme,' correct direction of the effect of temperature on enzyme activity, and explicit connection to CO_2 bubble production. Students

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Write the word equation and balanced symbol equation for aerobic respiration and for anaerobic respiration (yeast pathway). Groups write equations on mini whiteboards. PART B — Whole-Class Equation Gallery (10 min): Each group holds up their mini whiteboard equations. The class critiques using the protocol: 'Agree, Disagree, or Add.' The teacher scribes the consensus equations on the board. Students copy the verified equations into their notebooks and annotate each term (e.g., '$C_6H_{12}O_6$ — this is glucose, the sugar from sukuma wiki leaves'). PART C — Connecting to Phenomenon (5 min): Students individually answer in their notebooks: 'The sukuma wiki leaves submerged in WARM water produced more bubbles than those in cold water. Using what you now know about respiration, explain this observation. Include the word ENZYME in your answer.'	PART B: For the gallery critique, say: 'When I point to a group's board, everyone look carefully. Thumbs up if you agree, thumbs sideways if you want to add something, thumbs down if you think there is an error. No calling out — I will cold-call.' Cold-call at least 4 students for the critique. When errors arise (e.g., unbalanced equations), say: 'Count the carbon atoms on both sides — are they equal? Fix it.' Do not correct for students; redirect them to fix it themselves. PART C: After students write their individual explanation, cold-call 2 students: 'Wanjiku, read your explanation.' After the first student, ask the class: 'Did she include the word enzyme correctly? Thumbs up or sideways.' [WAIT TIME: 10 seconds.] Reinforce: 'Enzymes are proteins — they speed up chemical reactions. Warming the water increases enzyme activity, which speeds up respiration, which produces more CO_2 — those are the bubbles we saw!'	strategy — it explicitly requires students to use their new mechanistic knowledge (respiration, enzymes, temperature) to re-explain the original phenomenon, deepening the connection between evidence and explanation.	who do not mention enzymes or temperature need targeted follow-up in Lesson 5.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs	The class DQB (established in Lesson 1 around the sukuma wiki phenomenon) is revisited. Each student receives two sticky notes. On the BLUE note, they write one question that today's lesson has ANSWERED from the DQB. On the YELLOW note, they write one NEW question that today's lesson has RAISED. Students post their notes on the DQB in the appropriate columns: 'Now We Know' (blue) and 'Still Wondering' (yellow). The class reads the new yellow notes aloud and votes	Display the DQB prominently. Say: 'Look at the questions we posted in Lesson 1. Take your two sticky notes and spend 3 minutes writing — blue note: one question we can now answer; yellow note: one new question today raised for you.' [Allow 3 minutes.] Have students post notes. Then say: 'Let me read three of the yellow notes aloud.' Read them. 'By show of hands — which of these questions do you most want to investigate next?' Tally votes visibly. Say: 'These top questions will guide our next	Driving Question Board Update — Claim-Evidence-Question Tracking: The DQB functions as a living, visible record of the class's growing understanding. The two-colour sticky note protocol forces students to distinguish between answered and unanswered questions, building metacognitive awareness of what they know and do not know. Voting on priority questions gives students agency over the inquiry direction and sustains motivation across the 12-lesson sequence.	Read all blue sticky notes before the next lesson. Assess whether students have correctly identified a question that today's evidence genuinely answers. A correct blue note demonstrates that the student can connect experimental evidence to a specific claim. Students whose blue note describes a question NOT answered by today's lesson (e.g., 'we learned about stomata') may be conflating lessons — note their names for a brief check-in at the start of Lesson 5.

Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(show of hands) on the top 2 'Still Wondering' questions to prioritise for the next lesson. Expected 'Still Wondering' questions include: 'Where exactly in the cell does respiration happen?' 'If plants respire all the time, why don't they run out of glucose?' 'How does the CO_2 from respiration get OUT of the leaf — through the same stomata?'	lessons. Notice that today we answered WHERE the CO_2 comes from — from INSIDE the cell, from respiration — but we have not yet explained exactly HOW the CO_2 leaves the leaf. That question connects respiration back to gaseous exchange structures. That is where we are headed.' Point to relevant sticky notes from Lessons 1–3 to make the connection explicit.		
Model Building Phase 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative cell/leaf model diagram (started in Lesson 1, revised in Lessons 2 and 3). Working individually for 5 minutes, they add a NEW layer to their model: a zoomed-in view of a single plant cell showing: (1) the mitochondrion labelled as the 'site of aerobic respiration'; (2) arrows showing glucose entering the mitochondrion and CO_2 + H_2O + energy exiting; (3) a dashed arrow showing CO_2 travelling from the cell toward the nearest stoma (drawn from their Lesson 2 model); (4) a colour-coded annotation: RED for anaerobic pathway (ethanol + CO_2 produced in cytoplasm, no oxygen needed), BLUE for aerobic pathway (in mitochondrion, oxygen required). Below the diagram, students write a 2-sentence 'model statement' beginning: 'My model now shows… because the evidence from today's experiment showed…'	Distribute the cumulative models. Say: 'Open your model from Lesson 1. Today we add the inside of the cell. You have 5 minutes to add the mitochondrion, the two pathways, and the dashed arrow to the stoma. Use RED for anaerobic, BLUE for aerobic.' [Set timer.] Circulate and ask: 'Show me on your model — where does aerobic respiration happen? Where does the CO_2 go after it is made?' Cold-call 2 students to share their model with the class under the document camera. After each share, ask the class: 'What did this model show well? What could be added or improved?' [WAIT TIME: 10 seconds.] Collect models at the end of the lesson for review, or photograph them for the ARES platform portfolio.	Cumulative Model Revision with Colour-Coded Pathways: Revising a single evolving model across all 12 lessons makes scientific knowledge-building visible as a cumulative process rather than isolated lessons. The colour-coding of aerobic (blue, oxygen-rich) vs. anaerobic (red, oxygen-absent) pathways creates a visual shorthand that helps students hold both processes in working memory simultaneously. The dashed arrow connecting mitochondrion to stoma explicitly bridges today's cellular biochemistry to the macroscopic gas-exchange structures from Lessons 1–3, preventing compartmentalisation of knowledge.	When collecting/photographing models, check for: (1) mitochondrion correctly identified as site of aerobic respiration; (2) cytoplasm identified as site of anaerobic respiration; (3) CO_2 shown moving toward a stoma; (4) model statement uses 'evidence' language and references limewater data. Models missing the dashed arrow (CO_2 to stoma) indicate a student who has not yet connected cellular respiration to the macroscopic gaseous exchange system — this is a critical gap to address in Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students' limewater turbidity data clearly show a difference between aerobic and anaerobic conditions? If not, what set-up variable (temperature, yeast concentration, glucose amount, seal quality on the anaerobic tube) could be adjusted for next time? (2) During the equation gallery critique, were students able to identify unbalanced equations independently, or did they need significant scaffolding? If the latter, consider pre-teaching balancing equations as a 10-minute warm-up before Lesson 5. (3) How many students correctly used the word 'enzyme' in their Part C explanation (Phase 3)? If fewer than 60% of

students made the enzyme-temperature-rate connection, plan a brief enzyme activity review at the start of Lesson 5. (4) Review the yellow DQB sticky notes: are students asking deeper mechanistic questions (e.g., 'where in the cell does this happen?') or still surface-level questions (e.g., 'what is glucose?')? This reveals the depth of conceptual engagement and informs pacing for Lessons 5–6. (5) Were there equity issues — did the same dominant voices answer cold-call questions? Review your clipboard notes and ensure different students are targeted in Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that limewater connected to the aerobic yeast tube (Tube A, open to air) turned milky-white/turbid faster and more intensely than the limewater connected to the anaerobic yeast tube (Tube B, sealed). They also observed that warming increased the rate of turbidity change, mirroring the warm-water effect on sukuma wiki bubble production from the anchoring phenomenon.
What did I learn?	Students learned that all living cells — including plant cells and yeast — release energy from glucose through respiration. Aerobic respiration ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy) occurs in the mitochondria and produces more energy and more CO_2 per glucose molecule. Anaerobic respiration ($C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ + energy) occurs in the cytoplasm without oxygen and produces less energy. Both pathways produce CO_2, which must exit the plant through the gas-exchange structures (stomata, lenticels) studied in Lessons 1–3.
How does this explain the phenomenon?	The tiny bubbles streaming from the sukuma wiki leaves submerged in warm water were partly CO_2 produced by cellular respiration in the leaf cells (in addition to oxygen from photosynthesis). Warm water increases enzyme activity in the leaf cells, which speeds up the rate of respiration, producing CO_2 more quickly — hence more bubbles in warm water than in cold water. The CO_2 produced inside leaf cells diffuses out through the same stomata that regulate gaseous exchange, connecting the cellular (mitochondrion) level to the tissue (stomata) level of organisation.

LESSON 5 (80 minutes): Importance of Anaerobic Respiration: From Busaa to Biogas	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and evaluate the economic and ecological importance of anaerobic respiration in real-world Kenyan contexts, connecting cellular processes to food production, energy generation, and biotechnology.
Knowledge	By the end of the lesson, the learner should be able to: (a) explain the process and products of anaerobic respiration in yeast and plant cells; (b) describe the role of anaerobic respiration in fermentation, biogas production, and ethanol manufacture; (c) distinguish between aerobic and anaerobic respiration in terms of conditions, products, and energy yield.
Skills	By the end of the lesson, the learner should be able to: (a) analyse evidence from previous lessons to construct explanations for anaerobic respiration products; (b) design a simple investigation linking yeast fermentation to CO_2 production in mandazi dough; (c) communicate findings using annotated diagrams and scientific vocabulary.
Attitudes	By the end of the lesson, the learner should be able to: (a) appreciate the relevance of Biology to everyday Kenyan economic activities such as brewing, baking, and biogas generation; (b) show curiosity and open-mindedness when connecting laboratory findings to community practices.
Key Inquiry Question	How do plants obtain energy from organic compounds through respiration?
Purpose in Storyline	In Lessons 1–4, students identified the sites of gas exchange (stomata, lenticels, pneumatophores), gathered evidence that plants produce CO_2 and consume O_2, and compared aerobic and anaerobic pathways at the equation level. Lesson 5 now deepens understanding by anchoring anaerobic respiration to vivid Kenyan economic contexts — fermentation of uji and busaa, yeast-leavened mandazi, biogas digesters, and industrial ethanol — making the biochemistry tangible and personally meaningful. This positions students to begin planning their unit project, in which they will design a yeast-based CO_2 detection system to answer the driving question quantitatively.
Safety Notes	– If live yeast and warm sugar solution are used in class, ensure no student with known yeast/mould allergies is in direct contact with cultures. • Busaa and ethanol are alcoholic products — discussion is academic; no tasting or handling of fermented beverages in school. • Biogas (methane) is flammable; only discuss and view images/videos — no live biogas demonstrations unless a certified school biogas plant is available and supervised by a technician. • Wash hands after handling any biological materials. • Ensure adequate ventilation in the laboratory.

B. LESSON OVERVIEW

Students open by revisiting the anchoring phenomenon — the mandazi dough rising on Mama Wanjiku's counter — and connect it to the CO_2 bubble evidence gathered in Lessons 1–4. Through a structured gallery walk of Kenyan anaerobic respiration case studies (uji fermentation, busaa brewing, biogas digesters in Kisumu, industrial ethanol at KWAL), students identify patterns: oxygen-limited environments, glucose as substrate, CO_2 and ethanol/lactic acid as products, and varying energy yields. A class fermentation demo with yeast, sugar, and a balloon provides real-time evidence of CO_2 production. Students then use a comparison matrix sensemaking strategy to consolidate aerobic vs anaerobic respiration, update their Driving Question Board, and draft the question they will investigate in their unit project. The lesson closes with a written Exit Ticket connecting anaerobic respiration to both the mandazi phenomenon and the sukuma wiki bubbles from the anchoring phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown two images side by side on the class display: (1) a ball of mandazi dough that has risen and is full of air pockets, and (2) a flat, dense piece of dough that has not risen. They are also shown a photo of a biogas digester at a dairy farm in Kisumu. Working individually for 3 minutes, students write in their science notebooks: (a) What gas do you think is inside the air pockets in the risen mandazi dough? (b) Where did that gas come from — the flour, the water, the yeast, or somewhere else? (c) How might the Kisumu biogas digester be related to what happens inside a plant cell? Students then share predictions with a 'turn-and-talk' partner for 2 minutes before a whole-class cold-call share-out.	Display the two dough images and the Kisumu biogas digester photo. Say: 'Before we explain anything today, I want you to predict. Look at these two images carefully. In your notebook, answer the three questions on the board. You have 3 minutes — work silently and independently.' [SET TIMER — 3 minutes WAIT TIME]. After timer: 'Now turn to your partner. Share your prediction on question (b) — where did the gas come from? You have 2 minutes.' [2 minutes partner talk]. Cold-call 4 students (select 2 who predicted yeast, 1 who predicted flour, 1 who is unsure) using lolly sticks. Say: 'Akello, what did you and your partner predict?' Record key prediction phrases on the board verbatim under the heading: OUR PREDICTIONS — DO NOT ERASE. Ask: 'What evidence from our previous lessons — the sukuma wiki bubbles, the lime water experiment — might help us check these predictions today?' [10 seconds WAIT TIME]. Cold-call 2 students.	Prediction Anchoring — students activate prior knowledge from Lessons 1–4 (CO_2 production in aerobic respiration, the sukuma wiki bubble phenomenon) and generate a testable claim before instruction. Recording predictions verbatim creates cognitive accountability: students will return to these exact words in Phase 3 to evaluate whether their thinking has shifted.	Circulate during the 3-minute individual write. Look for: (a) students who correctly name CO_2 as the gas (consolidating L3–L4 learning); (b) students who link yeast to fermentation (prior knowledge); (c) students who cannot connect the biogas digester to cellular respiration (a gap to address in Phase 2). Use sticky-note flags — green for strong prior knowledge, yellow for partial, red for significant gap — to guide cold-call selection and Phase 2 grouping.
Observe Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration	PART A — Gallery Walk (20 minutes): Six A3 poster stations are set up around the room, each presenting a Kenyan anaerobic respiration case study with images, a short text, and two guiding questions. Stations: (1) Uji Fermentation — sour uji made from fermented millet in western Kenya; (2) Busaa Brewing — traditional maize/sorghum beer and the role of yeast; (3) Mandazi Dough Rising — commercial baker in Mombasa explains yeast leavening; (4) Biogas Digester —	Before gallery walk: 'You have 3 minutes at each station. Use your Evidence Table — every column must be filled before you rotate. Designated recorder rotates each round.' Circulate continuously. At Station 6 (lactic acid in waterlogged roots), prompt: 'How does this connect to the sukuma wiki leaves we submerged in Lesson 1? Could the roots of that plant also be doing something similar?' [10 seconds WAIT TIME at each group]. At Station 4 (biogas), prompt: 'What is the	Evidence Table + Gallery Walk — a structured data-collection protocol that transforms qualitative case studies into comparable evidence rows. By filling identical columns across six different contexts, students begin to see the pattern: anaerobic respiration recurs wherever oxygen is limited, regardless of whether the context is a brewery, a kitchen, a digester, or a flooded root. This pattern recognition is the foundation for the explanation phase.	Collect Evidence Tables at the end of Phase 2 (or photograph them). Check: (a) Are students correctly identifying CO_2 and ethanol as products across stations 1–5, and lactic acid at station 6? (b) Are students marking 'No' in the 'Is oxygen present?' column for all six stations? (c) Are students articulating at least one concrete economic benefit per station? Students who leave 'Is oxygen present?' blank or write 'Yes' need targeted questioning in Phase 3. Balloon demo: observe whether

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Kisumu dairy farm producing methane from cow dung and plant waste for cooking; (5) Industrial Ethanol — Kenya Wine Agencies Limited (KWAL) in Nairobi produces ethanol from molasses using yeast fermentation; (6) Lactic Acid Fermentation in Waterlogged Plant Roots — sukuma wiki roots in flooded soil produce lactic acid. Groups of 4 rotate every 3 minutes. At each station, students record in their Evidence Table: Substrate used | Organism/agent | Products formed | Is oxygen present? | One economic or ecological benefit. PART B — Live Demo (10 minutes): Teacher sets up (or has set up before class) three identical 500 mL bottles each containing 200 mL warm water (37 °C), 1 teaspoon sugar, and labelled amounts of dry yeast: Bottle A = no yeast, Bottle B = 1 g yeast, Bottle C = 2 g yeast. Each bottle has a balloon stretched over its mouth. Students observe balloon inflation over 10 minutes and record: Which bottle inflates fastest? What gas is filling the balloon? How does this connect to the mandazi dough?	substrate here — what is being broken down? Is this aerobic or anaerobic? How do you know?' For the live demo, say: 'While we finish the gallery walk, these bottles are working. When we reconvene, you will observe the balloons and I want you to explain what you see using the word equation for anaerobic respiration from Lesson 3.' After gallery walk, direct all students to face the demo bottles. Ask: 'Mwangi, describe what you observe — be precise, use scientific language.' [10 seconds WAIT TIME]. Cold-call 3 students to narrate observations. Do NOT explain yet — save explanation for Phase 3.		students spontaneously link faster inflation in Bottle C to more yeast → more fermentation → more CO_2 — this predicts readiness for the Explain phase.
Explain Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration	PART A — Comparison Matrix (12 minutes): Each student receives a pre-printed Comparison Matrix with two columns (Aerobic Respiration | Anaerobic Respiration) and eight rows (Location in cell | Oxygen required? | Substrates | Products | Energy released per glucose | Speed | Examples in Kenya | One advantage). Students complete the matrix individually using their Evidence Tables, notebooks from Lessons 3–4, and today's gallery	Distribute Comparison Matrix. Say: 'You have 12 minutes. Use only evidence — your Evidence Table, your notebooks, and the word equations from Lesson 3. If a cell is empty, write a question mark and circle it — those are the gaps we will discuss.' [Set 12-minute timer]. Circulate — target students who struggled in Phase 2. After 12 minutes, cold-call 3 students to read one row each from their matrix. For any disagreement between peers, ask: 'Which piece	Comparison Matrix + Phenomenon Explanation — the matrix forces students to hold aerobic and anaerobic respiration in working memory simultaneously, surfacing contrasts that verbal explanation alone would obscure. The structured 5-sentence Phenomenon Explanation then compels students to apply abstract biochemistry to the concrete mandazi context, closing the loop between classroom science and everyday Kenyan life. The	Read Phenomenon Explanations during writing (circulate and annotate with a quick +/Δ mark). Indicators of proficiency: (a) Sentence 2 includes correct reactants (glucose) and products (ethanol + CO_2) with 'in the absence of oxygen'; (b) Sentence 3 references enzyme activity and connects to the warm-water data from Lesson 1; (c) Sentence 4 correctly uses the term 'anaerobic conditions' for waterlogged soil; (d) Sentence 5 names ATP yield

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	walk data. They then peer-check with a partner, reconciling any differences using evidence — not opinion. PART B — Explain the Phenomenon (8 minutes): Students return to the mandazi prediction from Phase 1. In their notebooks, they write a 5-sentence 'Phenomenon Explanation': Sentence 1 — What gas is in the mandazi air pockets and where did it come from? Sentence 2 — Write the word equation for anaerobic respiration in yeast. Sentence 3 — Why must the dough be warm (connect to enzyme activity and lesson 2 warm-water bubble data). Sentence 4 — Why does the same process happen in sukuma wiki roots when the soil is waterlogged? Sentence 5 — What is the trade-off — what does the plant/yeast lose by respiring anaerobically instead of aerobically? PART C — Real-World Application (5 minutes): Each group is assigned one Kenyan industry (uji, biogas, ethanol, bread) and must produce a 3-sentence 'Scientist's Statement' explaining how understanding anaerobic respiration helps a Kenyan farmer, brewer, or engineer optimise their process.	of evidence from today or from Lessons 3 and 4 helps us decide?' [10 seconds WAIT TIME]. Transition to Phenomenon Explanation: 'Now I want you to use everything in that matrix to explain what happened to Mama Wanjiku's mandazi dough. Five sentences — you have 8 minutes.' [Set timer]. After writing: cold-call 2 students to read their sentence 4 (waterlogged roots). Ask the class: 'Does this connect back to what we saw with sukuma wiki in Lesson 1? Raise your hand if your explanation includes a connection to the anchoring phenomenon.' For Part C: 'Your Scientist's Statement must include: the substrate, the product that matters economically, and one way knowledge of anaerobic respiration improves the process. You have 5 minutes.' Cold-call one group per industry to share. Respond with: 'Strong scientific reasoning — can you add the specific ATP trade-off to make that even more precise?'	Scientist's Statement extends this to economic application, embedding Science and Engineering Practice 6 (Constructing Explanations) and SEP 8 (Obtaining, Evaluating, and Communicating Information).	difference (2 ATP anaerobic vs 36–38 ATP aerobic). Flag any student whose Sentence 5 is missing or incorrect — this is a key misconception to address before the unit project.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Students return to the class Driving Question Board — a large display that has been accumulating sticky notes since Lesson 1. Each student writes TWO sticky notes: (1) a GREEN note — one question from our Driving Question Board that they can now answer with confidence using today's evidence (they write the answer in one sentence on the back); (2) an ORANGE note —	Hand out green and orange sticky notes. Say: 'Two minutes — write your green note first. Which question from our board can you now answer? Write the answer on the back.' [2 minutes WAIT TIME]. 'Now your orange note — what does today's lesson make you want to investigate further?' [90 seconds]. After posting, say: 'Three-minute silent gallery read — walk slowly, read every new note,	Driving Question Board Curation — a metacognitive sensemaking strategy in which students publicly track their own epistemic progress. The green/orange binary forces students to discriminate between consolidated knowledge and productive uncertainty. Clustering unanswered questions around the unit project question gives students agency: they see that their own curiosity is driving the	Photograph the DQB after posting. Analyse green notes: are students correctly attributing CO_2 production to yeast fermentation (anaerobic respiration) rather than aerobic respiration? Are they distinguishing ethanol from lactic acid as products in different organisms? Orange notes reveal conceptual gaps — if many students ask 'Does the plant suffer when it respires anaerobically?',

📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	one new question that today's lesson raised for them that is NOT yet answered. Students post their notes in the correct columns: QUESTIONS WE CAN NOW ANSWER | QUESTIONS WE STILL HAVE. The class then does a 3-minute silent gallery read of newly posted notes. One student volunteer reads aloud the three most-posted unanswered questions. The teacher frames these as the investigable questions for the unit project.	put a small tick on any note that matches something you also wonder.' After gallery read, ask a volunteer: 'Fatuma, read the three notes with the most ticks.' Record these on a separate 'UNIT PROJECT CANDIDATES' strip of paper on the board. Say: 'These unanswered questions are exactly what scientists work on. In our next two lessons, you will design an investigation to answer one of them — specifically: How can we measure and compare CO_2 production in aerobic vs anaerobic conditions using yeast and a simple indicator system?' Connect explicitly: 'This is the same CO_2 that made the sukuma wiki bubbles in Lesson 1 — now we are going to measure it.'	scientific inquiry, mirroring how real scientists identify gaps in the literature.	this signals a need to address ATP yield trade-offs more explicitly in the unit project framing.
Model Building Phase 📹 **VIDEO:** [Biodiversity flourishes in Phanerozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative plant respiration model from Lesson 4 — a diagram they have been building across the unit showing glucose breakdown pathways, reactants, products, and sites within the cell. Today they make three mandatory additions: (1) Add an ANAEROBIC BRANCH to the pathway diagram — showing glucose → pyruvate → (in absence of O_2) → ethanol + CO_2 (in yeast) OR lactic acid (in plant roots/bacteria), with '2 ATP' labelled; (2) Annotate with a KENYAN CONTEXT ICON — a small drawing of mandazi dough, a biogas digester, or a busaa pot next to the anaerobic branch to anchor the biochemistry in lived experience; (3) Draw an arrow from the anaerobic branch back to the ANCHORING PHENOMENON box at the top of their model (labelled since Lesson 1) with the sentence: 'CO_2 from anaerobic	Return Lesson 4 models. Say: 'Open your model to where you left it in Lesson 4. You are adding three things — they are on the board. You have 10 minutes. Your model must be able to explain both the mandazi dough AND the sukuma wiki bubbles by the time you finish.' [Set 10-minute timer]. Circulate — check that students are labelling '2 ATP' on the anaerobic branch and '36–38 ATP' on the aerobic branch. Prompt: 'Odhiambo, your model shows ethanol and CO_2 — excellent. Now show me the arrow that connects this back to the Lesson 1 phenomenon. Where does CO_2 go after it is produced?' [10 seconds WAIT TIME]. After 10 minutes: 'Swap with the group behind you. Peer review — one praise, one suggestion using the stem on the board. You have 2 minutes.' After peer review: 'One volunteer — Chebet, what was the strongest	Cumulative Model Revision — a Progressive Modelling strategy (NGSS SEP 2: Developing and Using Models) in which students maintain a single artefact across all 12 lessons, revising it as new evidence accrues. Adding the anaerobic branch today makes the model's two-pathway structure visible and comparable. The mandatory connection arrow back to the anchoring phenomenon prevents the model from becoming an abstract diagram disconnected from observable reality — every addition must explain something students have seen or will see.	Collect or photograph all models. Evaluate against a 3-point rubric: (1) Anaerobic branch present with correct reactants, products, and 2 ATP label; (2) Aerobic branch still correctly shows 36–38 ATP, CO_2, H_2O, and mitochondrion label from Lesson 4; (3) Connection arrow to anchoring phenomenon is present and the annotated sentence correctly names CO_2 as the linking product. Students scoring 1/3 require a 5-minute targeted conversation at the start of Lesson 6. Students scoring 3/3 are invited to act as 'model consultants' during Lesson 6 peer review.

	yeast respiration inflates mandazi dough just as CO_2 from aerobic leaf respiration produced the sukuma wiki bubbles.' Groups share models with an adjacent group for a 2-minute 'Scientist Peer Review' — one specific praise, one specific suggestion using the stem: 'Your model would be more accurate if...'	suggestion your reviewers gave you?' Cold-call 2 more. Close by saying: 'Your model is now showing TWO pathways — aerobic and anaerobic. In Lesson 6, we will investigate the rate of each pathway under different conditions and gather the final evidence we need to fully answer our driving question: How do plants breathe, and what happens inside their cells when they release energy from food?'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the gallery walk, were students able to independently identify 'no oxygen present' as the common condition across all six Kenyan case studies, or did they require significant prompting? If many students needed prompting at Station 4 (biogas) or Station 6 (waterlogged roots), consider adding a 'What is missing from this environment?' guiding question card to those stations in future iterations. (2) Did the live yeast-balloon demo produce visible balloon inflation within the 10-minute observation window? If not (e.g., due to cold classroom temperature), note the ambient temperature and plan to pre-warm the yeast mixture for 5 minutes before class, or use the demo as a take-home overnight observation with a report-back in Lesson 6. (3) Review the Phenomenon Explanation notebooks — specifically Sentence 5 (ATP trade-off). If fewer than 70% of students correctly stated that anaerobic respiration yields less ATP, this concept needs explicit re-teaching at the start of Lesson 6 before the unit project planning begins. (4) Were the DQB orange notes generating genuinely investigable questions, or were students writing vague statements? If vague, model a 'testable question' format ('How does X affect Y?') explicitly at the start of Lesson 6 before project planning. (5) Equity check: Did female students and students from pastoralist or rural backgrounds have equitable opportunity to contribute examples from their own community contexts (e.g., fermented milk — mursik — from Kalenjin communities, or fermented cassava)? If not, plan to add a 'community fermentation' station in future runs of this lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed: (1) six Kenyan case studies (uji, busaa, mandazi, biogas, ethanol, waterlogged roots) during the gallery walk, identifying substrate, agent, products, and oxygen conditions at each station; (2) balloon inflation on yeast-sugar-water bottles, providing real-time evidence of CO_2 gas production in the absence of oxygen; (3) comparison of balloon inflation rates across different yeast concentrations (Bottles A, B, C).
What did I learn?	Students learned: (1) anaerobic respiration occurs in the absence of oxygen, producing ethanol and CO_2 in yeast (fermentation) or lactic acid in plant roots and bacteria; (2) anaerobic respiration yields only 2 ATP per glucose molecule compared to 36–38 ATP in aerobic respiration; (3) fermentation by yeast underpins economically important Kenyan processes including uji souring, busaa brewing, mandazi leavening, biogas production, and industrial ethanol manufacture; (4) plants shift to anaerobic respiration in their roots when soil is waterlogged and oxygen is depleted — connecting plant cell biochemistry to agricultural and ecological contexts.
How does this explain the phenomenon?	Students explained: (1) the mandazi dough phenomenon — CO_2 produced by yeast during anaerobic fermentation of sugar creates gas pockets that cause the dough to rise, exactly as CO_2 produced during aerobic respiration in sukuma wiki leaves caused the bubbles observed in the anchoring phenomenon; (2) the word equation for anaerobic respiration in yeast: Glucose → Ethanol + Carbon dioxide (+ 2 ATP); (3) why the same glucose molecule can follow two different pathways depending on oxygen

	availability, and why the anaerobic pathway is less energetically efficient but essential under oxygen-limited conditions.

LESSON 6 (40 minutes): Factors Affecting Respiration Rate: Temperature, Glucose, and Oxygen

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students investigate how temperature, glucose concentration, and oxygen availability affect the rate of cellular respiration in yeast, using locally available materials such as sugar-cane juice and maize flour suspension, then explain their findings using enzyme activity principles.
Knowledge	Students will be able to: (1) identify temperature, glucose concentration, and oxygen availability as key factors that affect respiration rate; (2) explain how each factor influences enzyme activity and therefore the rate of ATP production; (3) distinguish between the effects of these factors on aerobic versus anaerobic respiration pathways.
Skills	Students will be able to: (1) design a fair-test investigation controlling one variable at a time; (2) collect and tabulate quantitative data on CO2 bubble production rate as a proxy for respiration rate; (3) construct line graphs from experimental data; (4) interpret graph trends to draw evidence-based conclusions.
Attitudes	Students will demonstrate curiosity when unexpected results arise, precision and honesty in recording data, and collaborative responsibility when sharing roles within investigative groups.
Key Inquiry Question	How do plants obtain energy from organic compounds through respiration? — specifically, what environmental and biochemical conditions speed up or slow down that energy-release process?
Purpose in Storyline	Lesson 6 builds directly on Lessons 4 and 5 by moving from what respiration produces to how fast it operates. Students now gather quantitative evidence about the conditions that control respiration rate, which will later (Lesson 8) help them explain why the sukuma wiki leaves in the anchoring phenomenon produced more bubbles in warm water than in cold water — connecting enzyme kinetics to the observable phenomenon for the first time with numerical data.
Safety Notes	Hot water (above 60 degrees C) must be handled by the teacher only. Students use warm water at 35 to 40 degrees C and cold water at approximately 10 degrees C. Yeast suspensions and sugar-cane juice are non-toxic but students should avoid ingesting them. Glassware should be inspected for cracks before use. Spills should be wiped immediately to prevent slipping. Students with known yeast or sugar allergies should inform the teacher before the lesson.

B. LESSON OVERVIEW

Students begin by revisiting the anchoring phenomenon — more bubbles in warm water than in cold — and predict which factors might control how fast yeast respires. They then carry out a structured three-part investigation using sugar-cane juice and dried yeast: one group varies temperature (cold, warm, hot), a second varies glucose concentration (dilute maize flour suspension, standard sugar solution, concentrated sugar-cane juice), and a third varies oxygen availability (open beaker versus sealed bottle). Groups count CO2 bubbles per minute, tabulate results, and plot line graphs. In the Explain phase the class synthesises all three data sets using enzyme activity concepts. The lesson closes with a DQB update and a revised model arrow showing that respiration rate is not constant but is regulated by environmental conditions — directly advancing the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Worked example: Calculating molar mass and number of moles Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(0–6 minutes) The teacher projects or draws the anchoring phenomenon scene: sukuma wiki leaves releasing more bubbles in warm water than in cold. The teacher asks students to open their notebooks and write answers to two prompts: (1) List THREE things you think could make yeast respire faster or slower. (2) For each factor, predict whether increasing it will speed up, slow down, or have no effect on respiration, and write one sentence explaining your reasoning. Students write individually for 2 minutes, then share with a shoulder partner for 1 minute. Three or four pairs share out. The teacher records predictions on the board in a two-column table: Factor | Predicted Effect. This prediction table remains visible throughout the lesson as an accountability anchor.	Teacher opens by saying: 'Cast your minds back to Lesson 1. Mwalimu Njoroge's class saw more bubbles in warm water than in cold. We now know those bubbles contain CO2 from respiration. But today I want to push further: what controls HOW FAST respiration happens?' After individual writing, teacher says: 'Share your top factor with your partner. You have 60 seconds — go.' [WAIT TIME: 60 seconds of partner talk.] Teacher then cold-calls three pairs using a random name stick, recording each factor on the board without evaluating. Teacher asks the class: 'Do you notice any patterns in our predictions? What factors keep coming up?' [WAIT TIME: 10 seconds.] Teacher says: 'Today we are going to test three of these factors with real yeast and materials you could find in any Kenyan kitchen or market — sugar-cane juice from Thika Road market, maize flour from home, and ordinary dry yeast from the duka. Let us see whose predictions are right.'	Prediction table on the board with student-generated factors creates cognitive dissonance targets — students have skin in the game and will compare final data against their own written predictions.	Teacher scans written predictions during partner talk. Look for: (1) whether students mention temperature, substrate concentration, and oxygen — if oxygen is missing, prompt: 'Think back to Lesson 4 and 5 — what was different about aerobic and anaerobic conditions?'; (2) whether explanations reference enzymes — if not, note this as a gap to address in the Explain phase.
Observe Phase 📹 **VIDEO:** Worked example: Calculating molar mass and number of moles Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular	(6–26 minutes) Groups of four are pre-assigned to one of three investigation stations. Each station has a printed half-page instruction card. STATION A — Temperature Effect: Three 250 mL beakers filled with (i) cold water at 10 degrees C with ice, (ii) warm water at 37 degrees C, and (iii) hot water at 55 degrees C. Each beaker holds a small glass bottle containing 5 g dried yeast dissolved in 100 mL of standard 10% sugar-cane juice solution. Groups count CO2	Teacher circulates with a checklist of four probing questions, asking each group at least one: (1) 'What is your independent variable and how are you keeping everything else the same?' (2) 'You have been counting bubbles — what does each bubble represent in terms of chemistry?' (3) 'Your cold water bottle is producing almost no bubbles. Does that mean no respiration is happening, or very slow respiration? How would you know?' (4) 'Your Station C sealed	Jigsaw data sharing: after graphing, each group nominates one spokesperson. The class reconvenes briefly (2 minutes) and each spokesperson reads their key result aloud while the teacher adds the three data sets to a class results table on the board. This creates a shared evidence base for the Explain phase.	Teacher uses the circulation checklist to note which groups correctly identify their independent and dependent variables and which cannot. Groups who cannot articulate what CO2 bubbles represent are flagged for targeted questioning in the Explain phase. Completed tables and graphs are checked for labelled axes, appropriate scales, and data plotted correctly — teacher provides immediate verbal feedback during circulation.

[Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	bubbles emerging from a short rubber tube inserted in the bottle cap for 1 minute at each temperature, recording results in a table (Temperature | Bubbles per minute). STATION B — Glucose Concentration Effect: Three identical bottles at room temperature containing (i) 100 mL of thin maize flour suspension at 2% concentration, (ii) 100 mL of 10% sugar solution, and (iii) 100 mL of undiluted sugar-cane juice at approximately 20% sugar. Each bottle has 5 g yeast. Groups count CO2 bubbles per minute for each concentration. STATION C — Oxygen Availability Effect: Two setups at 37 degrees C with 10% sugar-cane juice and 5 g yeast. Setup 1 is an open beaker stirred gently (aerobic). Setup 2 is a sealed bottle with the tube submerged in water to create anaerobic conditions and prevent oxygen entry. Groups count CO2 bubbles per minute for both. All groups record data in pre-drawn tables in their notebooks. After 15 minutes of data collection, each group spends 2 minutes plotting a simple line graph (Station A and B) or bar chart (Station C) in their notebooks.	bottle — is this aerobic or anaerobic? How do you know?' Teacher also watches for groups not controlling variables (e.g., different amounts of yeast at different temperatures) and intervenes with: 'Is there a variable you have accidentally changed here? What could that do to your results?' If a group finishes early, teacher directs them to write one sentence predicting what they expect to find at another station. Teacher signals the end of data collection at 20 minutes: 'Pause your counting. Make sure your table is complete and your graph axes are labelled with units. You have 2 minutes to finish your graph — go.' [WAIT TIME: 2 minutes silent work.]		
Explain Phase 📹 **VIDEO:** Worked example: Calculating molar mass and number of moles Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular	(26–34 minutes) The class uses the shared results table on the board. Teacher guides a whole-class discussion structured in three rounds, one per factor. In each round students are asked: (1) Describe the trend in your graph. (2) Explain the trend using what you know about enzymes and respiration. (3) Connect the trend back to the anchoring phenomenon or a Kenyan real-world context. Students write a	Teacher opens the Explain phase with: 'Station A group, read me your trend from your graph in one sentence — just describe what you see in the data, no explanation yet.' [WAIT TIME: 5 seconds.] After the description, teacher asks the whole class: 'Who can explain WHY that trend occurs? Think about enzymes. You have 15 seconds to think before I ask someone.' [WAIT TIME: 15 seconds — critical thinking pause.]	Claim-Evidence-Reasoning (CER) writing frames scaffold scientific explanation. Prediction table annotation creates metacognitive closure — students see their thinking evolve within a single lesson. The Kenyan jiko and uji analogy anchors abstract enzyme kinetics in lived experience.	Teacher collects one CER card per group (an index card on which groups write their best CER for the temperature factor) to review after class. During discussion, teacher listens for three common misconceptions: (1) 'Heat makes yeast stronger' — redirect to enzyme language; (2) 'More CO2 always means more ATP' — correct using aerobic vs. anaerobic yield comparison from Lesson 4; (3) 'Yeast is a plant so these

Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	three-sentence explanation for each factor in their notebooks: a Claim sentence, an Evidence sentence citing their own data, and a Reasoning sentence linking evidence to enzyme activity. For example, for temperature: Claim — increasing temperature from 10 to 37 degrees C increased the respiration rate in yeast. Evidence — our group recorded 3 bubbles per minute at 10 degrees C but 24 bubbles per minute at 37 degrees C. Reasoning — higher temperatures increase the kinetic energy of molecules, so enzyme-substrate collisions in respiration are more frequent, increasing the rate of glucose breakdown until temperatures exceed the enzyme optimal point and cause denaturation. Students also compare their original predictions on the board against the actual data and annotate their prediction table with a tick or cross and one correction sentence.	Teacher then cold-calls a student who has not yet spoken. If the enzyme connection is incomplete, teacher says: 'You mentioned faster reactions. What is doing the reacting? What molecules are involved in breaking down glucose inside the cell?' Teacher uses the board to sketch a quick enzyme-substrate diagram if needed, labelling it with the Kenyan context: 'This is exactly why uji ferments faster when left near a warm jiko than in a cold pantry.' For Station C, teacher asks: 'The sealed bottle produced MORE or FEWER bubbles than the open beaker. What does that tell us about which pathway — aerobic or anaerobic — produces more CO2 per unit time? Does more CO2 mean more ATP? Think before answering.' [WAIT TIME: 20 seconds.] Teacher guides students to the nuance: anaerobic fermentation in yeast produces CO2 at a comparable early rate but yields far less ATP, connecting back to Lesson 5 findings. Teacher closes the Explain phase: 'Now annotate your prediction table. Put a tick next to each prediction that was supported by your data and a cross next to those that were not. Write one sentence correcting any wrong prediction using evidence.'		results do not apply to sukuma wiki' — address by connecting to the anchoring phenomenon directly.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Worked example: Calculating molar mass and number of moles Source: Khan Academy (English - US curriculum)	(34–37 minutes) Students return to the class Driving Question Board displayed at the front of the room. Each student writes one new sticky note using the sentence starter: 'Today my data showed that _____, which helps explain the anchoring phenomenon because _____.' Students place their sticky notes in the EVIDENCE column of the DQB.	Teacher says: 'You have 90 seconds to write your sticky note. It must include your own data — a number or a trend — and it must connect to sukuma wiki. Go.' [WAIT TIME: 90 seconds silent writing.] Teacher reads two sticky notes aloud and models upgrading a vague note: 'This one says just temperature matters — let us make it stronger by adding the	DQB update makes the cumulative storyline visible and gives students agency in marking their own knowledge growth. The evidence column creates a wall of student-generated scientific claims that will be drawn on in Lesson 8 synthesis.	Teacher checks that sticky notes contain quantitative evidence references, not just vague statements. Any note lacking a data reference is returned with the prompt: 'Add a number or trend from your table to make this scientific claim.'

🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher reads two or three aloud and checks whether they specifically reference the sukuma wiki bubble observation. The teacher also directs students to check whether any questions from earlier lessons on the DQB can now be answered or partially answered. For example, the question 'Why did warm water produce more bubbles?' can now be answered with enzyme kinetics evidence. Students draw a line through resolved questions and write a one-phrase answer beside them.	evidence and the connection.' Teacher then points to the DQB: 'Scan the questions we posted in Lessons 1 through 5. If today gave you an answer, go up and mark it resolved. You have 30 seconds.' [WAIT TIME: 30 seconds.]		
Model Building Phase 📹 **VIDEO:** Worked example: Calculating molar mass and number of moles Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(37–40 minutes) Students open their cumulative unit models (started in Lesson 2 and revised in Lessons 3, 4, and 5). They add a new annotation layer titled FACTORS AFFECTING RESPIRATION RATE. Using a coloured pencil, they draw three small arrow-and-label additions to their model: (1) a temperature arrow pointing to the respiration pathway arrow with a note — optimal temperature increases enzyme activity and respiration rate; (2) a glucose concentration arrow pointing to the substrate entry of the pathway with a note — more substrate increases reaction rate until enzymes are saturated; (3) an oxygen availability fork — a label showing that high O2 favours aerobic respiration producing 36 to 38 ATP, while low O2 shifts the pathway to anaerobic fermentation producing only 2 ATP. Students write one sentence at the bottom of their model page: 'The rate of respiration in plant cells — including those in the sukuma wiki leaves in Lesson 1 — depends on temperature, substrate availability,	Teacher says: 'In the last 3 minutes, add the three factors to your model. Use a new colour so we can see what we learned today versus what was already there. Make sure your model shows the aerobic and anaerobic fork we first drew in Lesson 4 — today we are adding WHY the rate on each fork changes.' Teacher circulates and checks that students are updating existing models rather than drawing new ones — continuity of the model across lessons is essential. Teacher closes the lesson: 'Next lesson we will look at plants in very different environments — the Kenyan coast, the Sahel, and Lake Victoria — and ask how their structures are adapted for gaseous exchange. Everything you learned today about what controls respiration rate will help you understand why those adaptations matter.'	Cumulative model annotation with colour-coded layers makes conceptual growth visible across the unit. The closing sentence requirement forces students to linguistically connect today's evidence back to the anchoring phenomenon, reinforcing the storyline thread.	Teacher spot-checks five models during the 3-minute update, verifying that: (1) three factors are labelled; (2) the aerobic-anaerobic fork is present and annotated with ATP yield; (3) the closing sentence references sukuma wiki. Students who have not yet drawn the aerobic-anaerobic fork are reminded to revisit their Lesson 4 notes before Lesson 7.

	and oxygen, because these factors control enzyme activity in the respiratory pathway.'			

D. TEACHER REFLECTION
After the lesson, consider: (1) Did all three investigation stations produce reliable enough data for students to identify clear trends? If Station B maize flour suspension was too cloudy to count bubbles accurately, plan to use clear sugar solutions of known concentrations next time. (2) Were students able to move from describing trends to explaining them using enzyme language without excessive prompting? If enzyme vocabulary was weak, plan a 5-minute warm-up review of enzyme activity at the start of Lesson 7. (3) Did the CER cards collected reveal genuine understanding of the aerobic versus anaerobic CO_2 production distinction, or did students conflate CO_2 output with ATP yield? This misconception must be corrected before Lesson 8 synthesis. (4) Did the DQB update produce specific, evidence-rich sticky notes, or were they vague? Modelling stronger DQB language at the start of Lesson 7 may be needed. (5) Reflect on which students were silent during whole-class discussion — use their names for cold-calling in Lesson 7 to ensure equitable participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Yeast in warm sugar-cane juice produced significantly more CO_2 bubbles per minute than yeast in cold water; increasing glucose concentration increased bubble rate up to a point; sealed anaerobic bottles produced rapid early bubble bursts but the rate declined quickly compared to open aerobic setups.
What did I learn?	Respiration rate in yeast — and by extension in plant cells — is controlled by temperature (through its effect on enzyme kinetic energy and collision frequency), glucose concentration (through substrate availability for respiratory enzymes), and oxygen availability (which determines whether the aerobic or anaerobic pathway is used and therefore how much ATP is produced).
How does this explain the phenomenon?	The anchoring phenomenon of more bubbles from sukuma wiki in warm water than in cold water can now be explained quantitatively: warmer temperatures increase the kinetic energy of respiratory enzyme molecules and their substrates, increasing the frequency of successful enzyme-substrate collisions in the glycolysis and Krebs cycle pathways, thereby increasing the rate of CO_2 production and the number of bubbles observed — directly linking cellular enzyme kinetics to the visible macroscopic phenomenon first observed in Lesson 1.

LESSON 7 (80 minutes): Adaptations for Gaseous Exchange in Different Environments	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to compare structural adaptations for gaseous exchange in xerophytes, hydrophytes, and mesophytes, and relate these adaptations to environmental conditions across Kenya's diverse ecosystems.
Knowledge	Learners identify and describe the structural features of stomata, cuticle, lenticels, and pneumatophores in xerophytes (e.g., cactus, acacia), hydrophytes (e.g., water hyacinth from Lake Victoria), and mesophytes (e.g., sukuma wiki), explaining how each feature is suited to its environment.
Skills	Learners observe and compare leaf cross-sections and whole-plant specimens, construct comparative anatomy tables, and build evidence-based arguments linking structure to function across contrasting Kenyan ecosystems (Rift Valley, Lake Victoria basin, Turkana/arid north).
Attitudes	Learners appreciate Kenya's ecological diversity as a living laboratory for understanding plant adaptation, and develop curiosity about how organisms solve the shared challenge of gas exchange under radically different environmental constraints.
Key Inquiry Question	How do plants exchange gases with their environment? How do plants obtain energy from organic compounds through respiration?
Purpose in Storyline	In Lessons 1–6, learners established that the bubbles streaming from submerged sukuma wiki come from stomata and identified the roles of guard cells, the cuticle, and lenticels. Lesson 7 now zooms out: if sukuma wiki (a mesophyte) has stomata arranged the way we observed, how do plants that live in Kenya's scorching Turkana desert or in the waterlogged shores of Lake Victoria solve the same gas-exchange problem? By comparing xerophytes, hydrophytes, and mesophytes, learners deepen their model of plant gaseous exchange and add new explanatory power to answer the driving question.
Safety Notes	Handle cactus and acacia specimens with care — thorns and spines can cause puncture wounds; use forceps or thick gloves. If using razor blades or scalpels for cross-sections, follow laboratory safety protocols: cut away from the body, use cutting mats, and ensure the teacher demonstrates technique first. Water hyacinth from Lake Victoria may carry pathogens; rinse specimens in dilute disinfectant solution before handling and wash hands thoroughly after the activity. Dispose of plant material in the designated bin, not the drain.

B. LESSON OVERVIEW

Learners engage in a structured comparative analysis of three plant types — xerophytes, hydrophytes, and mesophytes — drawn from Kenya's contrasting ecosystems. Using prepared microscope slides, hand specimens, diagrams, and a guided reading passage, they identify key gaseous-exchange adaptations (sunken stomata, thick cuticle, reduced leaf area, aerenchyma, pneumatophores, floating leaves, and typical stomata distribution) and record findings in a three-way comparison table. The lesson culminates in a class gallery walk, a structured argumentation discussion, and a model revision that adds environmental-adaptation annotations to each learner's cumulative plant gaseous-exchange model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are presented with three	Display the three photographs and	Think-Pair-Share Prediction	Observable indicator: Every

📹 **VIDEO:** [Watch Me] Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	unlabelled photographs displayed on the board or printed cards: (A) a cactus from Turkana County, (B) water hyacinth floating on Lake Victoria near Kisumu, and (C) sukuma wiki growing in a Nairobi school garden. Each learner silently writes in their science notebook: 'Which plant do you think loses the MOST water through its leaves each day? Which loses the LEAST? Predict one structural feature on each plant that controls how much gas — and water — moves in and out.' Learners then share predictions with a shoulder partner (Think-Pair) before three cold-call shares to the whole class.	ask: 'Before we look closely at these plants, I want you to think about where each one lives in Kenya. Picture Turkana — hot, dry, almost no rain. Picture Lake Victoria near Kisumu — water everywhere. Picture our school garden here in Nairobi — moderate rain, moderate temperature.' Pause. 'Now, in your notebooks, write your prediction: which plant loses the most water through its surfaces each day, and which loses the least? And predict ONE feature you think each plant might have to control gas and water movement.' Allow 3 minutes of silent individual writing. Then say: 'Turn to your shoulder partner and compare your predictions for 2 minutes.' After pair discussion, cold-call three learners: 'Amina, tell us your prediction for the cactus.' — 'Ochieng, what did you predict for the water hyacinth?' — 'Wanjiku, what did you and your partner agree on for the sukuma wiki?' Record key prediction words on the board under columns: CACTUS / WATER HYACINTH / SUKUMA WIKI. Do NOT confirm or correct yet. Say: 'Hold onto these predictions — we will test them with real evidence in the next phase.'	Mapping. By articulating and publicly recording predictions before instruction, learners activate prior knowledge from Lessons 1–6 (stomata, guard cells, cuticle) and create cognitive anchors. The public prediction board creates accountability — learners are motivated to check whether their predictions hold up against evidence, driving engagement in the Observe phase.	learner has written at least one structural feature per plant type in their notebook before the pair discussion begins. The teacher circulates during individual writing time and looks for: (1) reference to stomata or cuticle from prior lessons — confirms lesson-to-lesson knowledge retention; (2) environmental reasoning (e.g., 'cactus is dry so it must have fewer stomata') — confirms transfer of prior learning. Note any learners who cannot connect environment to structure; these learners will be targeted for questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells	Learners work in groups of four at lab stations. Each station has: (i) a hand specimen or high-quality printed photograph of the assigned plant, (ii) a prepared microscope slide (or printed photomicrograph) of a leaf cross-section, (iii) a laminated 'Observation Scaffold Card' prompting them to record: location of stomata (upper/lower/both surfaces), depth of stomata	Before stations begin, demonstrate microscope or photomicrograph use: 'Look at this cross-section of sukuma wiki — you already know this leaf from Lessons 1 through 6. Find the stomata. Notice where they sit on the surface.' Give 30 seconds for learners to locate them. 'Now when you rotate to the cactus station, your job is to ask: are the stomata in the same position as the sukuma wiki? Are	Structured Station Rotation with Observation Scaffold Cards. The scaffold card ensures learners collect parallel data across all three plant types, making comparison in Phase 3 meaningful and systematic. By rotating through all stations rather than receiving teacher-told information, learners construct their own evidence base — aligned with NGSS Science and Engineering	Observable indicator: Each group's comparison table has at least four of the six observation categories filled in for all three plant types after the rotations. The teacher checks tables during circulation — any group with fewer than four categories completed receives a prompt: 'You have three minutes left at this station — which rows are still empty on your table? Focus there.' Listen for

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(flush/sunken/elevated), thickness of cuticle (thin/moderate/thick), presence/absence of aerenchyma (air spaces), presence/absence of specialised roots (pneumatophores), and leaf area/shape. Station A focuses on a xerophyte (cactus or flat-spined acacia from the Rift Valley). Station B focuses on a hydrophyte (water hyacinth, Eichhornia crassipes, from Lake Victoria). Station C focuses on a mesophyte (sukuma wiki from the school garden — connecting directly to the anchoring phenomenon). After 15 minutes of station observation, groups rotate so each group observes all three plant types. Groups record findings in their comparison table in their notebooks.	they higher? Lower? Buried?' Circulate to each station every 4–5 minutes. Use targeted questions: At Station A (xerophyte) — 'Why might having sunken stomata help the cactus survive in Turkana? What does sunken mean for air movement above the stoma?' (WAIT 10 seconds before cold-calling.) At Station B (hydrophyte) — 'Water hyacinth floats on Lake Victoria. Which surface of the leaf faces the air? Where would you predict the stomata to be, and what do you actually see?' (WAIT 12 seconds.) At Station C (mesophyte) — 'Compare the cuticle thickness here to what you saw at the cactus station. What does that difference tell us?' (WAIT 10 seconds.) After all rotations, instruct groups to complete their three-column comparison tables before moving to Phase 3.	Practice 3 (Planning and Carrying Out Investigations) and Practice 4 (Analysing and Interpreting Data).	misconceptions during station discussions, particularly: 'The cactus has NO stomata' (incorrect — it has sunken stomata, usually on the stem) or 'Water hyacinth stomata are on the bottom because it's a water plant' (incorrect — they are on the upper/abaxial surface because that faces air). Address these in Phase 3 whole-class discussion.
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their home groups with completed comparison tables. The class engages in a Gallery Walk: each group posts one large paper (A2 size or equivalent) on the wall summarising their comparison table with annotated diagrams of the three plant types. Learners do a silent 5-minute gallery walk, adding sticky-note questions or comments to other groups' posters. The class then reconvenes for a structured whole-class discussion in which the teacher uses the Claim–Evidence–Reasoning (CER) framework to guide learners to explain each adaptation. Learners then complete a written CER paragraph individually: 'Claim: [plant type] is adapted to exchange gases in [environment] because…	Open the whole-class discussion by pointing to a sunken stoma diagram: 'I am going to give you a claim. Your job is to support or challenge it with evidence from your stations.' State the claim: 'Sunken stomata in acacia trees from the Rift Valley reduce water loss without preventing gas exchange.' Ask: 'Who has evidence from their observation that supports this claim? Raise your hand.' Cold-call two learners to share evidence. Then ask: 'Now, the reasoning — WHY does being sunken reduce water loss? Think about what is in the pit around the stoma.' WAIT 12 seconds. Cold-call: 'Kipchoge, explain the reasoning to us.' After the sunken stoma discussion, move to water hyacinth: 'Now let's talk about aerenchyma. Fatuma, at	Claim–Evidence–Reasoning (CER) Framework with Gallery Walk. The gallery walk externalises group thinking and allows cross-group learning. The CER framework then structures the transition from observation to explanation — learners are explicitly required to distinguish what they observed (evidence) from why it matters (reasoning), building the scientific argumentation skills demanded by NGSS Science and Engineering Practice 7 (Engaging in Argument from Evidence). The written CER paragraph provides an individual accountability check after the collaborative discussion.	Observable indicator: Each learner's written CER paragraph includes (1) a named structural adaptation specific to their assigned plant type, (2) at least one piece of observational evidence from the station activity (e.g., 'the cross-section showed stomata buried below the leaf surface'), and (3) a reasoning statement that explicitly connects the structure to the environment (e.g., 'this reduces the water vapour gradient above the stoma, slowing evaporation in the dry Rift Valley'). Collect CER paragraphs at the end of Phase 3. Learners who cannot produce a reasoning statement are flagged for targeted support during Phase 4.

	Evidence: … Reasoning: …'	the Lake Victoria station, did you observe large air spaces inside the water hyacinth leaf or stem? What do you think those air spaces do for gas exchange in a waterlogged plant?' WAIT 10 seconds. Draw out the explanation that aerenchyma provide internal gas channels so that oxygen can diffuse to submerged tissues. Connect to the phenomenon: 'Remember the bubbles from the sukuma wiki in warm water in Lesson 1. If sukuma wiki were a water hyacinth and its roots were under water, how might it still get oxygen to those roots?' WAIT 15 seconds before cold-calling. Reinforce with the phrase: 'Every adaptation we are describing today is a solution to the same problem our sukuma wiki had in that beaker — moving gases in and out efficiently.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. They revisit the two anchor questions: 'How do plants exchange gases with their environment?' and 'What happens inside their cells when they release the energy stored in food?' Each learner writes one new sticky note to add to the DQB: either (a) a CLAIM card (green) — something they can now answer with evidence from today, or (b) a NEW QUESTION card (yellow) — a new question today's lesson has raised. The teacher facilitates a 5-minute whole-class share of selected cards, grouping them under the emerging themes: STRUCTURE, ENVIRONMENT, PROCESS.	Point to the DQB and say: 'Six lessons ago, Mwalimu Njoroge's students watched bubbles stream from sukuma wiki in warm water and asked: where exactly do those bubbles come from, and why does temperature matter? We have been building answers. Today we added a new layer — the same gas-exchange problem is solved differently depending on the ecosystem.' Hand out green and yellow sticky notes. Say: 'On your green card, write one specific thing you can now CLAIM about how plants exchange gases — with evidence. On your yellow card, write one NEW question today's lesson has raised for you.' Allow 3 minutes of individual writing. Cold-call three green cards and two yellow cards to read aloud. Cluster them on the DQB. Highlight any	Driving Question Board — Claim and New Question Revision. The DQB is a visible, cumulative record of the class's growing understanding. Distinguishing 'claims we can now make with evidence' (green) from 'new questions raised' (yellow) teaches learners that scientific understanding is always partial and generative — answering one question opens new ones. This metacognitive move also helps the teacher gauge the class's collective readiness to move to respiration content in Lessons 8–12.	Observable indicator: Green cards use lesson-specific vocabulary (xerophyte, hydrophyte, aerenchyma, sunken stomata, cuticle, pneumatophores) and reference specific Kenyan ecosystems. Yellow cards show genuine scientific curiosity extending beyond today's content. A green card that only restates the question without evidence (e.g., 'plants exchange gases through stomata' with no environmental qualification) is insufficient — the teacher notes these learners and will revisit in the Model Building phase.

		yellow cards that preview upcoming lessons (e.g., questions about mangrove pneumatophores, or about whether xerophytes do photosynthesis at night — a preview of CAM metabolism). Say: 'These yellow questions are exactly what scientists ask. We will come back to some of them in Lessons 8 and 9.'		
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative plant gaseous-exchange model from their science notebooks (first drawn in Lesson 1 and revised in each subsequent lesson). Using a new colour (designated for Lesson 7), they add three annotated panels to their model — one for each plant type — showing: the structural adaptation for gaseous exchange, the Kenyan ecosystem it lives in, an arrow indicating the direction of dominant gas movement (in or out) under typical daytime conditions, and a brief label explaining HOW the adaptation helps balance gas exchange and water loss. Learners then write a one-sentence 'model update statement' summarising what their model can now explain that it could not explain before Lesson 7.	Say: 'Open your models to the last page of revisions. Look at what you drew in Lesson 1 — just the sukuma wiki leaf with bubbles. Look at how much it has grown. Today you are adding three new panels.' Display a model template on the board showing the three blank panels labelled TURKANA XEROPHYTE / LAKE VICTORIA HYDROPHYTE / NAIROBI MESOPHYTE. 'In each panel, draw the key structural feature for gaseous exchange in that plant, label the ecosystem, draw an arrow for gas movement, and write one sentence explaining how the structure helps.' Allow 10 minutes for individual model revision. Circulate and prompt: 'Amina, you drew sunken stomata for the acacia — now write a sentence: HOW does this structure help in Turkana's climate?' (WAIT 10 seconds.) 'Ochieng, your water hyacinth panel — where are the stomata on a floating leaf, and why?' (WAIT 12 seconds.) After individual work, conduct a 3-minute pair share: 'Show your partner your three panels and read your model-update sentence aloud. Your partner gives you one piece of feedback: is the connection to the environment clear?' Close by asking the whole class: 'Who would like to share	Cumulative Model Revision with Ecosystem Panels. The three-panel structure forces learners to hold all three plant types in mind simultaneously and represent them in a single unified model — a higher-order synthesis task. Adding a new colour for each lesson makes the cumulative nature of the model visually explicit, reinforcing the idea that scientific models are revised as evidence grows, aligned with NGSS Science and Engineering Practice 2 (Developing and Using Models). The model-update sentence is a metacognitive prompt that requires learners to articulate growth in their own understanding.	Observable indicator: Each learner's model has three new panels, each containing (1) a labelled structural diagram with the correct adaptation for that plant type, (2) the correct Kenyan ecosystem name, (3) a directional arrow for gas movement, and (4) a written sentence linking structure to environmental function. Incomplete or incorrect panels (e.g., labelling aerenchyma as a xerophyte feature) are flagged during circulation for immediate correction. The model-update sentence is collected as an exit artefact — sentences that only describe structure without explaining environmental function indicate learners who have not yet achieved the lesson's core SLO and will need targeted scaffolding at the start of Lesson 8.

		their model-update sentence?' Cold-call two learners. End with the phenomenon connection: 'Next lesson, we will ask — now that we understand HOW gases move in and out of plants, what happens to those gases INSIDE the cell? That is where respiration begins.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the station rotations, were learners able to distinguish between the three plant types on their own, or did they rely heavily on the Observation Scaffold Card? If most groups could not fill in more than three categories without prompting, consider whether the scaffold needs more visual cues (labelled diagrams) for Lesson 8. (2) How many green DQB cards used specific vocabulary AND named a Kenyan ecosystem? If fewer than 70% of cards met both criteria, revisit the vocabulary in the Lesson 8 warm-up using a matching activity. (3) Did any learners confuse aerenchyma (hydrophyte adaptation) with stomata (universal structure)? If so, create a visual comparison chart for the Lesson 8 board display. (4) Were the CER paragraphs able to articulate reasoning — not just evidence? Reasoning is the hardest part of scientific argumentation for Grade 10 learners; if more than one-third of paragraphs lacked a reasoning statement, build an explicit 'reasoning sentence starter' scaffold into Lesson 8's explain phase. (5) Review the model-update sentences: do learners describe plants as PASSIVE structures, or do they use agency language ('the guard cells RESPOND to…', 'the sunken stoma TRAPS humid air…')? Agency language indicates deeper mechanistic understanding and readiness for cellular respiration content in Lessons 8–12.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed prepared slides and hand specimens of three Kenyan plant types — an acacia/cactus xerophyte (Rift Valley/Turkana), water hyacinth hydrophyte (Lake Victoria, Kisumu), and sukuma wiki mesophyte (Nairobi school garden) — and recorded structural features including stomata position, cuticle thickness, aerenchyma presence, and specialised root structures in a three-column comparison table.
What did I learn?	Learners learned that gaseous-exchange adaptations vary systematically with environment: xerophytes have sunken stomata and thick cuticles to minimise water loss in arid conditions; hydrophytes have stomata on the upper (air-facing) leaf surface and internal aerenchyma to channel gases to submerged tissues; mesophytes like sukuma wiki have a moderate arrangement suited to Kenya's temperate highland conditions — and that all three solutions address the same fundamental challenge of moving gases in and out efficiently while managing water loss.
How does this explain the phenomenon?	Learners used the Claim–Evidence–Reasoning framework to explain how each structural adaptation is functionally suited to its ecosystem, revised their cumulative plant gaseous-exchange model by adding three new ecosystem panels (Turkana xerophyte, Lake Victoria hydrophyte, Nairobi mesophyte), and updated the Driving Question Board with evidence-based claims and new questions — advancing their collective answer to 'How do plants exchange gases with their environment?' and setting up the transition to cellular respiration in Lessons 8–12.

LESSON 8 (80 minutes): Photosynthesis, Gaseous Exchange & Respiration Linkages: Systems Synthesis and Model Building	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to synthesise their accumulated evidence about plant gaseous exchange and respiration by mapping the relationships between photosynthesis, gaseous exchange, and cellular respiration as an interconnected system, and to revise their cumulative unit models accordingly.
Knowledge	Learners explain how CO_2, O_2, water, and glucose flow between photosynthesis, gaseous exchange structures (stomata, lenticels, cuticle, pneumatophores), and cellular respiration as an integrated plant system.
Skills	Learners construct and annotate a systems diagram that tracks the cycling of gases and organic molecules across all three processes, and critically evaluate and revise their cumulative unit model using evidence gathered across Lessons 1–7.
Attitudes	Learners appreciate that living systems are interconnected and that understanding one process requires understanding its relationships with others, reflecting the value of holistic scientific thinking.
Key Inquiry Question	How do photosynthesis, gaseous exchange, and respiration work together as a system to sustain plant life?
Purpose in Storyline	By Lesson 8, students have gathered evidence about stomatal structure and function, gas exchange pathways, lenticels, cuticle, pneumatophores, aerobic respiration, and anaerobic respiration. This synthesis lesson draws all those threads together: the bubbles streaming from Mwalimu Njoroge's sukuma wiki leaves are now fully explainable as an output of a coordinated system — photosynthesis producing O_2 and glucose, gaseous exchange structures regulating movement of CO_2 and O_2 in and out of leaves, and respiration consuming glucose and O_2 to release ATP, CO_2, and water. Students revise their unit models and update the Driving Question Board to capture what is now understood versus what still needs investigation.
Safety Notes	No hazardous materials are used in this lesson. If warm water baths are used for a brief re-demonstration, supervise handling of hot water. Ensure scissors used for cutting paper systems-diagram arrows are handled safely.

B. LESSON OVERVIEW
In this 80-minute synthesis lesson, learners work in small groups to construct a colour-coded systems diagram that maps the flow of CO_2, O_2, glucose, and water across photosynthesis, gaseous exchange structures, and cellular respiration in a sukuma wiki plant. They begin by individually predicting how the three processes connect before reviewing their evidence cards from Lessons 1–7. Groups then build their systems diagrams using labelled arrows, negotiate the direction of each molecular flow, and present one key insight to the class. The teacher facilitates whole-class discussion using targeted cold-calling and Socratic questioning to surface and correct remaining misconceptions (e.g., 'plants only photosynthesise, not respire'). Learners then individually revise their cumulative unit models to reflect the integrated system, and the class updates the Driving Question Board, explicitly marking which driving questions are now answered by today's synthesis and flagging any residual questions for Lessons 9–12.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Each learner receives a blank A4	Distribute the blank prediction	Individual Prediction Mapping —	Teacher observes whether

📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	sheet with three labelled boxes: PHOTOSYNTHESIS, GASEOUS EXCHANGE STRUCTURES, and CELLULAR RESPIRATION. Working individually and silently, learners draw arrows between the boxes and label each arrow with the molecule they think moves along that pathway (CO_2, O_2, glucose, water, ATP). They also write a one-sentence prediction answering: 'How do these three processes depend on each other to keep the sukuma wiki plant alive?' Learners retain their prediction sheets throughout the lesson for comparison at the end.	sheet and say: 'Before we connect everything we have learned across the past seven lessons, I want you to work alone — no talking, no sharing — and draw what you think the connections are between these three processes. Use arrows and label them with molecules. You have 6 minutes.' Start a visible timer. Circulate silently; do NOT correct errors — note 3–4 specific misconceptions on a sticky note for targeted follow-up (e.g., arrows showing glucose moving out through stomata; respiration drawn as only occurring at night). After 6 minutes, say: 'Hold up your sheet. Look at your neighbour's. You will have a chance to revise this at the end of the lesson.' Cold-call 3 learners: 'Amina, tell me one arrow you drew and why.' Allow 10–12 seconds WAIT TIME after each question before accepting a response. Do not affirm or correct yet — simply say 'Interesting — keep that idea in mind.'	by externalising prior knowledge as a diagram before instruction, learners activate their mental models and create a personal baseline. This makes conceptual change visible and motivates the synthesis activity that follows.	learners can identify at least two correct molecular flows between processes (e.g., O_2 from photosynthesis used in respiration; CO_2 from respiration used in photosynthesis). Misconceptions noted on sticky note are used to target Phases 2 and 3 questioning. Observable indicator: every learner has at least three labelled arrows on their sheet before Phase 2 begins.
Observe Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive an Evidence Card Set — a set of 12 cards (prepared by the teacher or printed from the ARES platform) summarising key evidence gathered in Lessons 1–7: e.g., Card 1: 'Bubbles from sukuma wiki stomata in warm water = O_2 released during photosynthesis'; Card 4: 'Lenticels in mango tree stems allow CO_2 out and O_2 in for respiration in woody tissues'; Card 7: 'Aerobic respiration equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O +$ ATP'; Card 10: 'Guard cells open stomata in light, allowing CO_2 in for photosynthesis and O_2 out.' Groups sort the cards into three piles — PHOTOSYNTHESIS, GASEOUS EXCHANGE,	Distribute Evidence Card Sets and say: 'These cards summarise the evidence you have gathered across our last seven lessons. Your task is to sort them — but pay special attention to any cards that could belong in more than one category. Those overlap cards are your most important clues about how the system works.' Allow 8 minutes for sorting. Circulate and probe with: 'Why did your group put Card 7 in the overlap zone?' and 'Which molecule appears on the most cards — what does that tell you?' After sorting, cold-call 2 groups to share their overlap zones. Say: 'Group 3, tell me one card you placed in the overlap and explain why in one sentence.'	Evidence Card Sorting with Overlap Detection — this strategy (adapted from claim-evidence-reasoning scaffolding) makes the interdependencies between processes visible before learners attempt to draw them. Placing cards in an overlap zone is a concrete, low-stakes way to identify shared molecular currencies across processes.	Observable indicator: each group correctly places at least 3 cards in the overlap zone and can verbally justify at least one placement using molecular flow language (e.g., 'CO_2 is produced in respiration and used in photosynthesis, so it connects both'). Teacher listens for groups that still separate photosynthesis and respiration entirely — these groups receive a probing card: 'At night, the stomata close. Does respiration stop? How does the plant get O_2?'

	RESPIRATION — and then identify cards that belong in more than one pile, placing them in an OVERLAP zone in the centre. They record which molecules appear in the overlap zone and why.	Allow 12 seconds WAIT TIME. Listen for correct identification that CO_2 and O_2 are shared across all three processes. Write the molecules CO_2, O_2, GLUCOSE, WATER, ATP on the board as groups name them.		
Explain Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Using their sorted evidence cards as reference, each group constructs a large-format (A2 or flipchart paper) colour-coded Systems Diagram of the sukuma wiki plant. The diagram must include: (1) a cross-section of the leaf showing stomata and mesophyll cells, (2) a chloroplast (photosynthesis site) and a mitochondrion (respiration site) inside a cell, (3) colour-coded arrows — GREEN for CO_2 flows, BLUE for O_2 flows, ORANGE for glucose/ATP flows, and RED for water flows — with direction and molecular labels, (4) labels showing which gaseous exchange structure is involved at each entry/exit point (stomata, cuticle, lenticels), and (5) a brief annotation explaining what happens at night versus during the day. Groups then do a Gallery Walk — each group leaves one member as a 'museum guide' while others visit two other groups' diagrams, using sticky notes to add one 'I notice…' and one 'I wonder…' comment to each visited diagram.	Before groups begin, model the starting scaffold on the board: draw a simple leaf outline, place a chloroplast and mitochondrion inside, and say: 'Your diagram must show that these two organelles are in a conversation — molecules are constantly moving between them. Your job is to show exactly what is moving, in which direction, and through which door in the plant.' Allow 15 minutes for diagram construction. Circulate and ask targeted questions to groups showing misconceptions noted in Phase 1: 'You have drawn glucose moving out through the stomata — is that what really happens? Which lesson gave us evidence about where glucose goes?' and 'Your diagram only shows gas exchange during the day — what changes at night when stomata close and photosynthesis stops?' During Gallery Walk (7 minutes), prompt guides: 'If a visitor asks why the bubbles in Mwalimu Njoroge's warm water experiment increased, can your diagram answer that?' After Gallery Walk, cold-call 4 different learners (not group presenters) to share one 'I wonder…' comment they left — allow 10 seconds WAIT TIME each. Record emerging questions on the board.	Colour-Coded Systems Diagram with Gallery Walk — the colour-coding requirement forces learners to track individual molecules across process boundaries, preventing the common misconception that each process operates in isolation. The Gallery Walk introduces peer critique, building metacognitive awareness of gaps in their own models.	Observable indicators: (1) Each group's diagram contains correctly directed arrows for at least CO_2 and O_2 cycling between photosynthesis and respiration. (2) At least one gaseous exchange structure is correctly labelled at a gas entry/exit point. (3) The day/night annotation correctly states that photosynthesis stops at night but respiration continues, and that stomata close, reducing CO_2 entry. Teacher uses sticky-note 'I wonder…' comments collected during Gallery Walk to identify class-wide residual misconceptions for DQB update.
Driving Question	The class returns to the physical or digital Driving Question Board that	Point to the DQB and say: 'Cast your mind back to Lesson 1.	Dual Sticky-Note DQB Protocol — the deliberate separation of	Observable indicators: (1) Every learner posts at least one green

Board (DQB) Creation 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	has been building since Lesson 1. Each learner receives two sticky notes: one GREEN ('Now I can answer…') and one YELLOW ('I still wonder…'). Individually, learners write one statement on each note referencing the original driving questions and phenomenon. Green notes are placed on the DQB next to the question they answer; yellow notes are placed in a 'Still Investigating' column. The class then reads out the green notes as a collective 'What we now know' statement, and the teacher synthesises the yellow notes to preview Lessons 9–12. Learners also return to their original prediction sheets from Phase 1 and annotate them in red pen — circling what was correct, crossing out misconceptions, and adding what changed.	Amara noticed bubbles coming from specific spots on the sukuma wiki leaf in warm water. We have now spent seven lessons gathering evidence. Let's use what we built today to see how many of our original questions we can mark as answered.' Allow 4 minutes for learners to write their sticky notes. Then cold-call 5 learners to read their green notes aloud — after each, ask the class: 'Do we agree this question is answered? Thumbs up, thumbs sideways, thumbs down.' For any thumbs sideways or down, say: 'What evidence from our lessons would you need to feel confident? Add that to a yellow note.' After posting all notes, read the class's green notes together as a chorus statement: 'We now know that…' and then say: 'Look at your yellow notes. These are the questions that Lessons 9 through 12 will help us answer. Which of these is most urgent for you personally?' Cold-call 3 learners. Allow 12 seconds WAIT TIME each.	'answered' from 'still wondering' questions makes epistemic progress visible to learners. Returning to original prediction sheets with red pen annotation makes personal conceptual change concrete and builds metacognitive awareness of learning as an evidence-driven process.	note correctly answering one of the two KICD Key Inquiry Questions using molecular or structural evidence (e.g., 'I can now answer how plants exchange gases — through stomata, lenticels, and the cuticle, which regulate CO_2 and O_2 movement depending on light and temperature'). (2) Every learner posts at least one yellow note that is genuinely investigable (not trivial), providing the teacher with data on residual gaps. (3) On revised prediction sheets, learners have made at least two red-pen corrections relative to their Phase 1 predictions.
Model Building Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar	Each learner retrieves their personal Cumulative Unit Model — a diagram they have been building and revising since Lesson 1 to explain the sukuma wiki bubble phenomenon. Using a different-coloured pen, learners add the new layer from today's lesson: the systems-level connections between photosynthesis, gaseous exchange, and respiration. They must add at least three new labelled elements not present in their Lesson 7 model version. Learners then write a 3-sentence 'Model Narrative' beneath their diagram: Sentence 1 explains what is happening at the stomata.	Say: 'Your cumulative model is the story of your thinking across this entire unit. Today, you are writing a new chapter. Use a different colour — this is important, because it lets you see how your understanding has grown since Lesson 1. You have 10 minutes.' Circulate and prompt with: 'Your model shows stomata but not what is happening inside the mesophyll cell — can you add that?' and 'Your narrative says plants breathe like animals — how is it similar and how is it different? Be precise.' With 2 minutes remaining, cold-call 2 learners to read their Sentence 3 aloud. After each, ask: 'Class, did	Colour-Layered Cumulative Model Revision with Model Narrative — using a new pen colour for each lesson revision creates a visible 'growth ring' structure in the model, making epistemic progress tangible. The three-sentence narrative forces translation from diagram to verbal explanation, consolidating the systems-level understanding built today and directly linking back to the anchoring phenomenon.	Observable indicators: (1) Each learner's revised model contains at least three new labelled elements in the new colour (e.g., mitochondrion with respiration equation, bidirectional CO_2/O_2 arrows between chloroplast and mitochondrion, labels on stomata showing both entry of CO_2 and exit of O_2 and water vapour). (2) Sentence 3 of the Model Narrative correctly references increased enzyme/reaction rate at higher temperatures as the reason for increased bubble production in warm water — connecting the phenomenon to the systems-level understanding. (3) Models that

readings	Sentence 2 explains how photosynthesis and respiration exchange molecules. Sentence 3 explains why warm water in Mwalimu Njoroge's experiment produced more bubbles than cold water, using systems-level language. Learners date and initial this revision and store models in their ARES portfolio.	they connect warm temperature to enzyme activity and reaction rate? If not, what would you add?' Allow 10 seconds WAIT TIME for class responses. Close by saying: 'In our next lesson, we begin examining how animal gaseous exchange compares to what we have built here — your model will be the foundation for that comparison.'		lack this temperature-rate connection are flagged for targeted follow-up at the start of Lesson 9.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Gallery Walk, which molecular flows were most frequently missing or incorrectly directed on groups' systems diagrams — was it the cycling of CO_2 between respiration and photosynthesis, the role of water, or the internal movement of glucose? Use this to adjust the opening of Lesson 9. (2) How many learners' green DQB sticky notes correctly used molecular and structural evidence to answer the Key Inquiry Questions, versus vague or incomplete answers? If fewer than 70% of learners posted a satisfactory green note, consider building a brief 5-minute whole-class synthesis at the start of Lesson 9 before moving to animal gaseous exchange. (3) Did the day/night distinction (photosynthesis stops; respiration continues) surface clearly in learners' diagrams and model narratives? This is a persistent misconception in the KICD curriculum — if it remains unresolved, address it explicitly before Strand 3.0 begins. (4) Were learners able to connect the warm-water bubble increase to reaction rate and enzyme activity in their Sentence 3 narratives? If not, a brief revisit of temperature effects on enzyme activity (from earlier in the Biology course) may be needed. (5) Review the yellow 'I still wonder…' sticky notes — do any point to questions that should be addressed in Lessons 9–12 of this sub-strand, or do they signal gaps from earlier lessons that need re-teaching?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed their own Evidence Card Sets summarising data from Lessons 1–7, sorted cards into process categories and an overlap zone, and reviewed peer Systems Diagrams during the Gallery Walk — all anchored to the original sukuma wiki bubble phenomenon from Mwalimu Njoroge's class.
What did I learn?	Learners synthesised that photosynthesis, gaseous exchange, and cellular respiration are not isolated processes but an interdependent system: CO_2 and O_2 cycle between photosynthesis and respiration; gaseous exchange structures (stomata, lenticels, cuticle) regulate molecular entry and exit; glucose produced in photosynthesis fuels respiration; and the rate of the system — evidenced by bubble production in warm water — is influenced by temperature acting on enzyme-driven reactions.
How does this explain the phenomenon?	Learners explained the sukuma wiki anchoring phenomenon at a systems level: bubbles stream from stomata because O_2 produced in photosynthesis exits through these structures; warm water increases reaction rates in enzyme-driven photosynthesis and respiration pathways, producing more O_2 and CO_2 and thus more bubbles; and the entire gas exchange apparatus of the leaf — guard cells, mesophyll air spaces, chloroplasts, and mitochondria — functions as a coordinated system to sustain the plant's energy and gas balance.

LESSON 9 (80 minutes): Project Launch & Planning: Investigating Anaerobic Respiration with Local Kenyan Materials

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To design and plan a guided inquiry investigation into anaerobic respiration using locally available Kenyan materials, connecting evidence gathered across the unit to the anchoring phenomenon.
Knowledge	Students will know that anaerobic respiration occurs in the absence of oxygen and produces different end-products depending on the organism (e.g., ethanol and CO_2 in yeast fermentation; lactic acid in muscle cells); that locally available Kenyan materials such as unga wa mahindi (maize flour), sukuma wiki, githeri, and mandazi dough can serve as substrates for fermentation investigations; and that a valid scientific investigation requires a clearly stated hypothesis, identified variables, controlled conditions, and a structured data collection plan.
Skills	Students will be able to formulate a testable hypothesis about anaerobic respiration; identify independent, dependent, and controlled variables in their chosen investigation; select appropriate locally available materials and apparatus; design a step-by-step procedure for their investigation; and create a data collection table with appropriate headings and units.
Attitudes	Students will appreciate the relevance of Biology in everyday Kenyan life (e.g., bread-making, busaa brewing, biogas production on farms); demonstrate collaborative responsibility by sharing planning roles equitably within their groups; and show scientific curiosity by connecting their investigation to the broader unit driving question.
Key Inquiry Question	How do plants obtain energy from organic compounds through respiration?
Purpose in Storyline	In Lessons 1–8, students gathered evidence about where and how plants exchange gases (stomata, lenticels, cuticle, pneumatophores) and explored aerobic respiration in plant cells. Lesson 9 pivots students toward the unit project: they now apply everything they know to design an original investigation into anaerobic respiration. By choosing a focus — yeast fermentation, biogas simulation, or food fermentation — students connect the cellular energy story directly to phenomena they encounter in Kenyan daily life (rising mandazi dough, fermenting uji, biogas digesters on Rift Valley farms). This lesson advances the driving question by asking: What happens inside cells when oxygen is NOT available?
Safety Notes	Students should handle yeast and fermentation materials hygienically — wash hands before and after handling food substrates. If using sugar solutions with yeast, remind students that these are not for consumption. Biogas simulation groups must NOT seal containers completely unless supervised — pressure build-up is a hazard. No open flames near any fermentation setup. Dispose of all organic waste in the compost pit or school garden bin. Students with allergies to yeast or gluten should be noted before materials are distributed.

B. LESSON OVERVIEW

In this 80-minute project launch lesson, students transition from evidence-gatherers to inquiry designers. Using all prior knowledge from Lessons 1–8 about plant gaseous exchange and aerobic respiration, students are introduced to the unit culminating project: a guided inquiry investigation into anaerobic respiration. Groups of 3–4 students self-select or are assigned one of three investigation tracks — (A) Yeast Fermentation (e.g., sugar + yeast producing CO_2 measured via balloon inflation), (B) Biogas Simulation (organic kitchen waste in a sealed bottle releasing gas), or (C) Food Fermentation (observing uji or githeri fermentation over time). The lesson moves through a project brief introduction, a gallery walk of sample experimental setups, collaborative hypothesis and variable identification, and ends with each group presenting a 90-second planning pitch. The anchoring phenomenon thread is maintained throughout: students are asked to connect their anaerobic investigation back to the sukuma wiki bubbles from Lesson 1 — if aerobic respiration produces CO_2 bubbles in leaves, what happens to gas production when oxygen is removed?

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown two images on the board (or printed cards): (1) a close-up of sukuma wiki leaves releasing bubbles in warm water (the anchoring phenomenon from Lesson 1), and (2) a photo of a mandazi vendor in Nairobi Market, with dough visibly rising in a bowl beside her. Students write independently for 3 minutes in their science notebooks, responding to the prompt: 'In both images, gas is being produced. In Image 1, the plant is using oxygen from aerobic respiration. What do you think is happening in Image 2 — where is the gas coming from, and is oxygen involved?' Students then share predictions with a partner (Think-Pair-Share) before 3–4 cold-calls to the whole class. Students record their initial predictions in a two-column table: 'What I think is happening' | 'Evidence from previous lessons that supports this.'	Display both images prominently. Say verbatim: 'Look carefully at both images. You have spent eight lessons gathering evidence about how plants breathe and release energy. Now I want you to think like a scientist — what is producing the gas in that mandazi dough, and is it the same process as in the sukuma wiki leaf?' Allow 3 minutes of silent independent writing — enforce quiet. After Think-Pair-Share (2 minutes), cold-call exactly 4 students using a random name method (sticks, cards, or roster). For each response, probe with: 'What evidence from our previous lessons supports that idea?' Apply 12-second wait time after each cold-call before accepting or redirecting answers. Record key student ideas on the board under two columns: 'With Oxygen' and 'Without Oxygen.' Do NOT confirm or correct yet — hold tension for the observe phase.	Think-Pair-Share with Evidence Anchoring: Students first retrieve prior knowledge independently (activating long-term memory), then test their ideas against a partner's, then publicly commit to a claim with evidence. Recording predictions in a two-column table externalises thinking and creates a reference point for revision later in the lesson — a key metacognitive practice in inquiry learning.	Observable indicator: Each student has a completed two-column prediction table in their notebook within 3 minutes. Teacher circulates and scans for (a) correct identification of yeast/microorganisms as the gas-producing agent in the mandazi, (b) any student who connects this to 'no oxygen' — flag these students as potential cold-call anchors for Phase 3. Red flag: students who write 'the dough breathes like the leaf' without differentiating aerobic vs. anaerobic — note for targeted questioning during Phase 3.
Observe Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook)	The teacher distributes the Project Brief card (one per student) which outlines the three investigation tracks in simple Kenyan-context language. Students then participate in a 15-minute Gallery Walk: three stations are set up around the classroom, each representing one investigation track. Station A (Yeast Fermentation): a small bottle of sugar solution with a sachet of baker's yeast (unga wa chachu, available at any Kenyan duka), a balloon stretched over the bottle neck, and a sample data table	Before the gallery walk, say verbatim: 'Each station shows a different way that cells — plant cells, yeast cells, or bacteria — release energy when oxygen is limited or absent. As you walk, you are gathering evidence to help you choose your group's investigation. Ask yourself: which of these connects most directly to our driving question about how plants release energy from food?' Set a timer for 5 minutes per station (use a visible countdown). Circulate actively — spend at least 90 seconds at each station. At Station	Gallery Walk with Guiding Questions (Structured Observation Protocol): By physically moving between stations and responding to anchored questions at each, students build comparative understanding of three manifestations of anaerobic respiration. Connecting each station back to the anchoring phenomenon (sukuma wiki bubbles) prevents isolated fact-gathering and keeps the sensemaking tethered to the unit's explanatory thread.	Observable indicator: Students have written responses to all three questions at each of the three stations (nine total responses) in their notebooks. Teacher checks for (a) correct identification of CO_2 as the gas produced at Stations A and B, (b) student use of the word 'fermentation' or 'without oxygen' at least once, and (c) any student who explicitly links Station A or B back to the sukuma wiki phenomenon without prompting — celebrate these connections publicly.

Source: CK-12 🔍 Search ARES for similar readings	showing balloon diameter over time. Station B (Biogas Simulation): a sealed plastic bottle containing kitchen vegetable scraps (sukuma wiki offcuts, ugali remnants) with a rubber tube leading to an inverted water-filled bottle — gas displacement visible. Station C (Food Fermentation): two jars of uji (fermented porridge base) at different stages — one fresh, one 24-hour fermented — with pH strips, a hand lens, and a smell observation card. At each station, students read a 'What you are looking at' card and answer three questions in their notebooks: (1) What evidence of gas production do you see or predict? (2) What do you think the organism or substrate is doing without oxygen? (3) How does this connect back to the sukuma wiki bubbles from Lesson 1?	A, prompt: 'What is inside the balloon? Where did that gas come from? Is this aerobic or anaerobic?' At Station B, prompt: 'Farmers in the Rift Valley use setups like this to produce cooking gas from cow dung and kitchen waste — what does that tell you about what is happening inside?' At Station C, prompt: 'Uji is fermented all over Kenya every morning — what organism do you think is making the uji sour, and is it using oxygen?' After the gallery walk, allow 2 minutes for students to annotate their notebooks before moving to group formation.		
Explain Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Before groups begin planning, the teacher leads a 12-minute whole-class explanation session using the 'Two Pathways' diagram on the board. Students copy and annotate a simplified concept map in their notebooks showing: Glucose (from food/organic matter) → [With O_2: Aerobic Respiration → CO_2 + H_2O + ATP (lots)] vs. [Without O_2: Anaerobic Respiration → in yeast: Ethanol + CO_2 + ATP (little) | in muscle cells / bacteria: Lactic Acid + ATP (little)]. The teacher uses the sukuma wiki phenomenon as the explicit bridge: 'In Lessons 2–5, we saw that sukuma wiki leaves produce CO_2 bubbles through aerobic respiration. Today we ask: what happens in the yeast cells inside that mandazi dough, or inside bacteria in fermenting uji, when	Draw the 'Two Pathways' diagram live on the board (do not use a pre-drawn poster — drawing it step by step models scientific thinking). As you draw, narrate: 'Glucose is the starting point — whether you are a yeast cell in mandazi dough in Nairobi, a plant cell in a waterlogged rice paddy in Mwea, or a muscle cell in a Kipchoge marathon runner's leg, the question is always: is oxygen available?' After drawing, cold-call exactly 3 students to read back one branch of the diagram each. Apply 10-second wait time. Then say verbatim: 'Now I want you to write one sentence that connects today's new pathway to what you already know about aerobic respiration in plants. You have exactly 5 minutes — this is individual thinking time, no talking.'	Concept Mapping with Comparative Annotation: Students construct meaning by placing new information (anaerobic pathways) in direct structural relationship to prior knowledge (aerobic respiration from Lessons 6–8). The 'Connection Sentence' frame ('similar because... different because...') is a disciplinary literacy strategy that requires students to identify both continuity and contrast — the hallmark of deep conceptual understanding rather than surface recall.	Observable indicator: Each student's Connection Sentence explicitly names at least one similarity (e.g., 'both start with glucose,' 'both release energy/ATP') AND at least one difference (e.g., 'anaerobic does not use oxygen,' 'anaerobic produces ethanol or lactic acid instead of water,' 'anaerobic produces less ATP'). Red flag: students who write only 'both produce CO_2' without acknowledging that lactic acid fermentation does NOT produce CO_2 — address this during the DQB phase.

	oxygen is NOT available? The end products change — but glucose is still being broken down to release energy.' Students are then given 5 minutes to write a 'Connection Sentence' in their notebooks: 'Anaerobic respiration is similar to what we observed in sukuma wiki leaves because _______, but it is different because _______.'	Circulate and read over shoulders. Identify 2 students with strong connection sentences and 1 student with a misconception-revealing sentence — plan to use all three in the DQB phase.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board (established in Lesson 1 around the sukuma wiki phenomenon). In groups, they receive three sticky note colours: GREEN for 'Evidence we now have,' ORANGE for 'New questions this raises,' and PINK for 'Connections to our project.' Each group posts at least one sticky note of each colour. The teacher then facilitates a 7-minute whole-class discussion, reading selected sticky notes aloud and asking students to sort them into two columns on the board: 'Answers we are building' and 'Questions we still need to investigate.' The two original driving questions ('How do plants exchange gases?' and 'How do plants obtain energy from organic compounds?') are re-read aloud, and students vote (hands or mini-whiteboards) on a confidence scale of 1–4: 'How confident are you that you can now answer each driving question?' Results are recorded on the board for comparison at Lesson 12.	Distribute sticky notes before this phase (pre-count: 3 per student). Say verbatim: 'Look back at the questions we posted on this board in Lesson 1 when we first watched the sukuma wiki bubbles. Which of those questions can you now answer with evidence? Which new questions has today's project launch opened up?' Allow 4 minutes of group sticky-note writing. Then facilitate posting — ask each group to briefly (15 seconds) explain their ORANGE sticky note (new question) only — do not process all notes, keep pace. After posting, cold-call the 2 students you identified in Phase 3 (one strong connection sentence, one misconception sentence) and ask them to read their sentences aloud, then place them as sticky notes. For the misconception student, do NOT correct publicly — instead ask the class: 'Does everyone agree with this? What evidence do we have?' Apply 12-second wait time. Close by saying: 'Your project investigations will help us answer the ORANGE questions. That is what scientists do — they design experiments to chase the questions they cannot yet answer.'	Driving Question Board Curation with Confidence Voting: The DQB is a living public record of the class's epistemic journey. Colour-coded sticky notes make the structure of scientific knowledge visible — distinguishing evidence from questions from connections. Confidence voting creates metacognitive awareness: students must evaluate not just what they know, but how well they know it. Returning to the original phenomenon (sukuma wiki) anchors new knowledge in the shared sensory memory of the unit.	Observable indicator: Each group has posted at least 3 sticky notes (one per colour). The teacher scans for (a) GREEN notes that correctly cite aerobic vs. anaerobic respiration differences, (b) ORANGE notes that reveal genuine inquiry (e.g., 'Does the sukuma wiki leaf itself ever do anaerobic respiration?', 'Does temperature affect fermentation like it affects stomata?'), and (c) PINK notes that name a specific Kenyan context (mandazi, uji, biogas, busaa). Confidence vote data: record the class average for both driving questions — expected range at Lesson 9 is 2.5–3.5 out of 4 for DQ1, and 2–3 out of 4 for DQ2.
Model Building	Groups of 3–4 students (pre-assigned or self-selected, with	Distribute the Project Planning Template at the start of this phase.	Project Planning Template with Public Pitch (Disciplinary Practice:	Observable indicator: Each student's Project Planning

Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	teacher guidance for balance) finalise their investigation track choice and complete a Project Planning Template in their notebooks. The template includes: (1) Investigation Title, (2) Kenyan context connection (one sentence), (3) Hypothesis (in 'If... then... because...' format), (4) Independent variable, (5) Dependent variable, (6) At least 3 controlled variables, (7) Materials list (using locally available Kenyan items where possible — unga, sukari, maji ya bomba, plastic bottles from the school kitchen, rubber tubing), (8) Step-by-step procedure (minimum 5 steps), (9) Data collection table (hand-drawn with headings and units). With 12 minutes remaining, each group delivers a 90-second planning pitch to the class. The teacher provides a sentence starter on the board: 'We are investigating _____ because we want to find out _____. We predict that _____ because _____. We will measure _____ by _____.' Peers use a simple Peer Feedback card (one star: something strong; one step: one suggestion for improvement).	Say verbatim: 'You have 20 minutes to complete your planning template as a group — every member must write the full plan in their own notebook. I will be circulating and asking each group: why did you choose this track, and how does your investigation connect back to our sukuma wiki phenomenon from Lesson 1? If you cannot answer that question, your plan is not ready.' Circulate with purpose — spend at least 3 minutes with each group. Use targeted prompts: For hypothesis: 'Is your hypothesis testable? How will you know if it is true or false?' For variables: 'What is the ONE thing you are changing? Everything else must stay the same — name three things you will keep constant.' For materials: 'Can you find everything on this list here in Nairobi or at your home? If not, substitute.' For the pitch session, use a visible 90-second timer. After each pitch, give exactly 60 seconds for peer feedback cards. Do NOT give extended teacher feedback during pitches — hold detailed feedback for written comments returned before Lesson 10. Close the lesson by saying: 'Your plans are the beginning of new evidence. In Lessons 10 and 11, you will carry out your investigations and gather data that will help us finally and fully answer our driving question: How do plants — and other living cells — release the energy stored in food?'	Planning and Carrying Out Investigations — NGSS SEP 3): Completing a structured planning template before beginning an investigation embeds the scientific practice of experimental design as a cognitive tool, not just a procedural step. The 90-second pitch forces concision and public accountability — students must distil their entire plan into its essential logic. Peer feedback using a 'star and step' protocol teaches students to evaluate scientific reasoning, not just procedure.	Template is complete with all 9 sections filled in before the pitch. During pitches, teacher uses a quick checklist (on clipboard) to mark: ✓ Hypothesis is in If-Then-Because format | ✓ Independent and dependent variables are correctly identified | ✓ At least 3 controlled variables named | ✓ Materials list uses at least 2 locally available Kenyan items | ✓ Kenyan context connection named. Groups missing 2 or more checkmarks receive a written prompt slip before Lesson 10: 'Your plan needs revision before you begin your investigation — see the teacher at the start of Lesson 10.' Groups with strong plans are flagged as potential model-sharing groups for Lesson 12.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions in your professional journal: (1) Which investigation track did most groups choose, and what does this reveal about student interest and prior knowledge — did groups gravitating toward yeast fermentation show stronger understanding of cellular respiration than those

choosing biogas? (2) During the gallery walk, were students able to spontaneously connect the fermentation stations back to the sukuma wiki anchoring phenomenon, or did they need explicit prompting? If prompting was needed, what does this suggest about how well the phenomenon has been embedded across Lessons 1–8? (3) Review the confidence voting data from the DQB phase: if most students rated their confidence at 2 or below for Driving Question 2 ('How do plants obtain energy from organic compounds?'), consider opening Lesson 10 with a brief 5-minute recap of the aerobic respiration equation before students begin their investigations. (4) Were the Project Planning Templates detailed enough for students to begin independently in Lesson 10, or are multiple groups likely to need significant revision? If more than 3 groups have incomplete or flawed plans, schedule a 10-minute 'plan clinic' at the start of Lesson 10 rather than proceeding directly to investigation setup. (5) Note any student who explicitly connected anaerobic respiration to a personal Kenyan context (e.g., mentioning a parent who brews busaa, or a family that uses a biogas digester) — these students can serve as local knowledge anchors in the whole-class discussion in Lesson 11.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed three set-up stations representing anaerobic respiration in local Kenyan contexts: (A) a yeast and sugar solution with a balloon inflating over the bottle neck, (B) a biogas simulation bottle with vegetable scraps (sukuma wiki offcuts, ugali remnants) producing gas displacement, and (C) two jars of uji at different fermentation stages with pH strips and hand lenses. Students also observed the anchoring phenomenon image (sukuma wiki bubbles) placed alongside a mandazi dough image to prompt comparison.
What did I learn?	Students learned that anaerobic respiration is the process by which cells release energy from glucose in the absence of oxygen; that in yeast cells this produces ethanol and carbon dioxide (as seen in rising mandazi dough and balloon inflation), while in bacteria and some plant cells it may produce lactic acid; that anaerobic respiration produces less ATP than aerobic respiration; that locally available Kenyan materials (unga wa chachu, sukari, sukuma wiki, uji base) can be used to investigate fermentation; and that a valid scientific investigation requires a testable hypothesis, clearly identified variables, and a structured data collection plan.
How does this explain the phenomenon?	Students explained that the gas bubbles seen in the sukuma wiki phenomenon (Lesson 1) are produced by aerobic respiration in plant cells — a process requiring oxygen. In contrast, the gas produced in yeast fermentation (mandazi dough rising, balloon inflation) comes from anaerobic respiration — the same starting molecule (glucose) is broken down, but without oxygen, producing ethanol and CO_2 instead of water and CO_2. Students also explained the design of their group investigations by formulating hypotheses in 'If-Then-Because' format and identifying variables, connecting their experimental design to the broader unit question of how living cells extract energy from organic food molecules.

LESSON 10 (80 minutes): Project Execution: Carrying Out Investigations on Plant Gaseous Exchange and Respiration	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to carry out their planned investigations on plant gaseous exchange and respiration, collect and record data accurately, control variables, and begin constructing evidence-based explanations that connect their findings to the anchoring phenomenon.
Knowledge	Learners demonstrate understanding that plants exchange gases through stomata, lenticels, the cuticle, and pneumatophores; that stomata are regulated by guard cells; that temperature affects the rate of gas exchange and respiration; and that cellular respiration releases energy from organic compounds such as glucose.
Skills	Learners carry out fair-test investigations by controlling variables; collect, record, and organise quantitative and qualitative data; identify patterns and anomalies in results; and construct preliminary evidence-based explanations linking data to biological mechanisms.
Attitudes	Learners demonstrate scientific rigour, intellectual honesty in recording data (including unexpected results), collaborative responsibility within their investigation groups, and curiosity in pursuing answers to the driving question.
Key Inquiry Question	How do plants exchange gases with their environment, and how do they obtain energy from organic compounds through respiration?
Purpose in Storyline	In Lessons 8–9, student groups designed and planned their investigations. In this lesson (Lesson 10 of 12), groups execute those plans — handling materials, collecting real data, and encountering the messiness of authentic science. Their findings become the evidence base for the explanations and model revisions in Lessons 11–12. Every data point gathered here ties directly back to the sukuma wiki bubbling phenomenon: learners are now generating their own empirical answers to the questions that phenomenon raised.
Safety Notes	Learners handling hot water must use heat-resistant gloves or cloths and place beakers on stable surfaces away from the edge of the bench. Iodine solution (if used for starch testing) must not be ingested; learners must wash hands after contact. Scalpel or razor blade use (for leaf cross-sections) must follow teacher demonstration; blades must be collected and stored by the teacher after use. Learners with plant allergies should wear gloves when handling sukuma wiki or other plant material. All spills must be reported and cleaned immediately.

B. LESSON OVERVIEW

Lesson 10 is the hands-on project execution lesson at the heart of the evidence-gathering sequence. Student groups (formed in Lesson 8) carry out their independently designed investigations into one or more aspects of plant gaseous exchange and respiration — for example: the effect of temperature on bubble production from sukuma wiki leaves, the distribution and density of stomata on upper vs. lower leaf surfaces, the effect of leaf coating (petroleum jelly blocking stomata) on gas exchange, or the detection of CO_2 or heat produced during respiration. The teacher circulates continuously with structured prompts, acting as a formative coach rather than an instructor. Groups record data in pre-prepared tables, begin identifying patterns, and draft preliminary claims supported by evidence. By the end of the lesson, every group has a complete raw data set and at least one written evidence-based explanation ready for peer critique in Lesson 11.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Before touching any equipment, each group opens their	Open the lesson by projecting the anchoring phenomenon image	Hypothesis Sharpening — Revisiting predictions after prior	Observable indicator: Every group's sticky note correctly

📹 **VIDEO:** [Elements and atoms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	investigation plan from Lesson 9 and reads aloud their written hypothesis. Groups then spend 3 minutes discussing: 'Given everything we have learned in Lessons 1–9 about stomata, guard cells, lenticels, and cellular respiration, would you revise your prediction at all? Why or why not?' Each group records any revision (or a written statement confirming no revision) in their science notebook. Groups also identify their independent variable, dependent variable, and at least two controlled variables by writing them on a sticky note and posting it at their workstation for the teacher to check during circulation.	(sukuma wiki leaves bubbling in warm water in Mwalimu Njoroge's class) and the driving question: 'How do plants breathe, and what happens inside their cells when they release energy stored in food?' Say: 'Today is the day we generate our own evidence. Before you touch a single piece of equipment, your group must confirm your hypothesis and variable identification — because a fair test requires you to know exactly what you are changing and what you are measuring.' Circulate to each group's sticky-note variables list. At any group missing a controlled variable, ask: 'What else in your setup could change between trials that you are not intending to change? How will you keep it the same?' Allow 10 seconds of WAIT TIME before prompting. Cold-call one student per group (aim for 5–6 groups) to state their hypothesis aloud in the format: 'We predict that [independent variable] will affect [dependent variable] because [biological reason].' Provide corrective feedback only on scientific accuracy — do not evaluate experimental design at this point.	instruction (Lessons 1–9) activates accumulated conceptual knowledge and forces learners to articulate the biological mechanism behind their prediction, not just a guess. This metacognitive step builds the habit of connecting claims to reasons before data collection, which is central to scientific practice (NGSS SEP 1: Asking Questions and Defining Problems; SEP 6: Constructing Explanations).	identifies at least one independent variable, one dependent variable, and two controlled variables. Teacher checks all sticky notes during circulation; groups with errors are redirected before investigations begin. Red flag: any group whose hypothesis contains no biological mechanism (e.g., 'warm water will make more bubbles' with no reference to stomata, enzymes, respiration rate, or gas solubility) receives a targeted prompt: 'What biological process produces those bubbles, and how does your variable affect that process?'
Observe Phase 📹 **VIDEO:** [Elements and atoms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures](#)	Groups carry out their planned investigations simultaneously. Possible investigation types include: (A) Effect of temperature on bubble rate — sukuma wiki leaf discs submerged in water at 10°C, 25°C, and 40°C; bubbles counted per minute at each temperature using a stopwatch; (B) Stomatal distribution — nail varnish peel method on upper and lower surfaces of sukuma wiki, bean, or maize leaves; stomata counted	Signal the start of investigations: 'You have 40 minutes. Your job is to collect honest data — that includes results that do not match your prediction. Anomalies are valuable evidence, not mistakes. Begin.' Circulate continuously using a structured rotation: spend approximately 4–5 minutes at each group, then move on. Use the following specific prompts during circulation — (1) Variable control check: 'Show me what you are	Structured Data Pause (NGSS SEP 4: Analysing and Interpreting Data) — The mid-investigation pause forces learners to interpret data in real time rather than treating data collection as a mechanical task. Writing one pattern sentence activates analytical thinking while evidence is still being gathered, improving the quality of explanations constructed afterward. This also mirrors authentic scientific	Observable indicators: (1) All data tables contain column headers with units (e.g., 'Number of bubbles per minute', 'Temperature / °C', 'Number of stomata per field of view'). (2) At least two trials are recorded, or one trial with a written reason if time did not permit a second. (3) The 'pattern sentence' written at the Data Pause is specific (e.g., 'Bubble rate increases as temperature increases') rather

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	under a hand lens or microscope in a defined field of view; (C) Effect of blocking stomata — petroleum jelly applied to upper surface only, lower surface only, both surfaces, or neither; leaf discs submerged; bubble rate compared; (D) Detection of CO_2 in respiration — germinating beans (githeri seeds soaked overnight) placed in a sealed flask; hydrogencarbonate indicator or lime water used to detect CO_2 after 15 minutes; (E) Heat production in respiration — temperature probe or thermometer placed in a flask of germinating githeri seeds vs. boiled (dead) seeds. Each learner records raw data in a pre-drawn table in their notebook, including units, time stamps, and any anomalous readings. Groups conduct at least two trials where time permits. The designated 'quality controller' in each group checks that the same person counts bubbles each time, that the stopwatch is reset between trials, and that readings are recorded immediately.	keeping constant. How are you making sure the temperature/light/leaf size is the same across your trials?' (2) Data recording check: 'Is every reading going straight into your table? I should see units in the column headers.' (3) Anomaly probe: 'That reading looks different from the others. Before you cross it out, ask yourselves — could that be real? What might have caused it?' Allow 10 seconds of WAIT TIME after each probe. (4) Mechanism prompt: 'You are seeing [observation]. Which structure or process from our lessons explains why that might be happening?' Cold-call a non-recording group member to answer (aim to cold-call at least 3 different students per group visit across the lesson). At the 20-minute mark, call a 2-minute 'Data Pause': all groups stop, look at their data so far, and write one sentence in their notebook: 'The pattern I am seeing so far is…' This mid-point sensemaking prevents groups from reaching the end with raw numbers and no thinking. At the 35-minute mark, prompt groups to begin their second trial or begin calculating means if two trials are complete.	practice, where researchers analyse data iteratively, not only at the end.	than vague (e.g., 'something changed'). Teacher flags: any group with identical results across all trials (possible error in variable manipulation); any group counting bubbles inconsistently (different group members alternating); any group where the control condition is absent.
Explain Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures	Investigation materials are set aside (not yet cleaned up — teacher will signal clean-up). Each group uses the CER (Claim–Evidence–Reasoning) framework to write a preliminary explanation of their findings. On a large sheet of paper or in notebooks: CLAIM — one sentence stating what their data shows (e.g., 'Increasing water temperature from 25°C to 40°C increased the rate of bubble	Introduce the CER framework by projecting a partially completed example (use a neutral topic so as not to give away results): 'Claim: Githeri seeds soaked in water for 24 hours are heavier than dry seeds. Evidence: Dry seeds averaged 1.2 g; soaked seeds averaged 2.7 g after 24 hours. Reasoning: Seeds absorb water by osmosis through the seed coat, causing the cells to swell and	Claim–Evidence–Reasoning (CER) Framework (NGSS SEP 6: Constructing Explanations) — CER is an explicit structure that separates what learners observed (evidence) from what they conclude (claim) and why the evidence supports the conclusion (reasoning). It builds scientific argumentation skills and prevents the common error of circular reasoning ('more bubbles because	Observable indicators: (1) The claim contains both the independent and dependent variable and states a direction or relationship. (2) Evidence includes at least two specific numerical data points with units. (3) Reasoning explicitly names at least one biological structure (stomata, guard cells, lenticels, mitochondria, enzymes) or process (cellular respiration, gas

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	production from sukuma wiki leaf discs.'); EVIDENCE — two or three specific data points from their table (include numbers and units); REASONING — two or three sentences explaining the biological mechanism that connects the evidence to the claim, referencing stomata, guard cells, gas exchange structures, cellular respiration, enzyme activity, or gas solubility as appropriate. Groups are reminded: 'Your reasoning must use vocabulary from this unit. It must connect your data to at least one biological structure or process we have studied.' Groups then spend 5 minutes reviewing their CER and preparing one 'lingering question' — something their data did not fully explain — to post on the Driving Question Board in Phase 4.	increasing total mass.' Then say: 'Now write your own CER for your investigation. Your reasoning must name at least one structure or process from Sub-Strand 2.3.' Circulate and use targeted prompts: — For weak claims: 'Your claim should state a relationship between your independent and dependent variable. Can you rewrite it to include both?' — For evidence-free claims: 'I see a claim. Where is the data from your table that supports it? Give me numbers.' — For reasoning with no mechanism: 'You have said what happened. Now explain why, using what you know about stomata / respiration / guard cells.' Allow 10 seconds of WAIT TIME after each prompt. Cold-call one group member (not the usual spokesperson) to read their claim aloud. Count: target at least 4 different groups. After 10 minutes of writing, call 2 minutes for groups to review and strengthen their reasoning before the DQB phase.	there were more bubbles'). By requiring biological vocabulary in the reasoning, CER also consolidates conceptual knowledge from Lessons 1–9 within the context of real data the learners themselves generated.	diffusion, opening/closing of stomata). Teacher flags: CER where the reasoning is a restatement of the claim ('the temperature made more bubbles because higher temperature increased the number of bubbles') — redirect with: 'What is happening inside the leaf or cell that explains this?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar	Each group writes their one-sentence claim on a green sticky note and their one 'lingering question' on a yellow sticky note. Groups post both on the class Driving Question Board under the relevant driving question column ('How do plants exchange gases?' or 'How do plants obtain energy through respiration?' or both). The class briefly scans all posted claims. Each learner individually writes in their notebook: (1) one claim from another group that surprised them and why, and (2) one connection between their own group's finding and the bubbling sukuma wiki phenomenon.	Direct groups to the DQB: 'Your green sticky note carries your claim. Your yellow sticky note carries the question your data raised but did not answer. Post them now.' While groups post, circulate and read claims silently. After posting, stand at the DQB and narrate two or three connections between different groups' claims: e.g., 'Group 3 found that blocking lower-surface stomata almost stopped bubble production. Group 1 found that most stomata are on the lower surface. Together, what do those two findings tell us about where gas exchange is happening?' Allow 10 seconds of WAIT TIME.	Driving Question Board Synthesis (NGSS SEP 7: Engaging in Argument from Evidence) — The DQB makes the class's collective evidence visible in one space, allowing learners to see how different investigations address complementary aspects of the same driving question. Narrating cross-group connections by the teacher models the scientific practice of synthesising evidence from multiple sources. The individual notebook connection to the phenomenon ensures personal sensemaking, not just group-level thinking.	Observable indicators: (1) All groups have posted a green (claim) and yellow (lingering question) sticky note. (2) Green sticky notes state a relationship or finding, not a procedure ('We found that…', not 'We tested…'). (3) Notebook entries link the group's finding to a specific feature of the sukuma wiki phenomenon (bubbles, bubble sites, temperature effect). Spot-check of three random notebooks for quality of phenomenon connection.

readings		Cold-call two students from different groups to respond. Then say: 'In your notebook, write one connection between your group's finding and the bubbling leaves we saw at the start of this unit. Specifically — which structure or process does your data help explain?' Allow 3 minutes of silent writing. Collect three notebooks at random as a spot-check formative assessment.		
Model Building Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups retrieve their running explanation model (begun in earlier lessons) — a labelled diagram of a leaf cross-section showing gas exchange structures and the pathway of gases. Using a different colour pen to mark today's additions, groups annotate their model with: (1) an arrow or label showing what their investigation measured or manipulated; (2) one sentence in a speech-bubble style box explaining what their data adds to the model; (3) a star (*) next to any part of the model that their data supports most strongly, and a question mark (?) next to any part that remains uncertain. Groups also update their model's title to include today's lesson number and date. Models are stored in group portfolios for use in Lesson 11 (peer critique and explanation sharing).	Distribute group models and say: 'Add to your model in a new colour — this shows us how your understanding has grown. Annotate what your investigation found and where it fits on the model.' Circulate and prompt: 'Your data showed [X]. Where on the model does that happen? Add an arrow.' For groups whose investigation focused on respiration (germinating githeri seeds), prompt: 'Cellular respiration happens inside cells, not just at the leaf surface. Where would you add mitochondria and the respiration equation to your model?' Allow 10 seconds of WAIT TIME. Cold-call one student from two groups to describe one new annotation they added and what it means. Close the lesson: 'In Lesson 11, every group will share their CER with the class. Others will ask questions and challenge your reasoning. Tonight, review your data and make sure your reasoning is as strong as possible — because your peers will test it.' Signal clean-up: all equipment returned, beakers of water disposed of safely, blades collected by teacher.	Cumulative Model Revision with Colour-Coding (NGSS SEP 2: Developing and Using Models) — Using a new colour to annotate the model makes conceptual growth literally visible: learners and the teacher can see at a glance what was understood before this lesson versus what new evidence has been added today. Anchoring the annotation to a specific location on the leaf cross-section diagram reinforces spatial understanding of where gaseous exchange and respiration occur, directly connecting investigative data to structural biology.	Observable indicators: (1) New annotations are in a different colour from prior lessons. (2) At least one annotation references the specific gas exchange structure or cellular process relevant to the group's investigation. (3) At least one star (*) and one question mark (?) are present, showing honest assessment of the model's completeness. Teacher flag: models with no new annotations or identical colour throughout (group has not engaged with revision) — these groups are prioritised for targeted support in Lesson 11.

D. TEACHER REFLECTION

After Lesson 10, reflect on the following questions before planning Lesson 11:

1. DATA QUALITY: Did all groups collect sufficient data for a meaningful CER? Which groups have incomplete tables, missing units, or only one trial? These groups will need prioritised support at the start of Lesson 11 before peer sharing begins.

2. VARIABLE CONTROL: During circulation, which groups struggled most to control variables? Was the most common error in the independent variable (e.g., not standardising leaf size) or the controlled variables (e.g., inconsistent timing)? Note these groups for targeted questioning in Lesson 11.

3. CER REASONING: Review the three notebooks collected as spot-checks. Is the reasoning biological and mechanistic, or is it descriptive and circular? If more than half the class is writing descriptive reasoning, begin Lesson 11 with a whole-class worked example of strong vs. weak reasoning before peer critique begins.

4. DQB PATTERNS: Look at the yellow sticky notes (lingering questions). What are the most common unanswered questions? These should guide your selection of which groups present first in Lesson 11 — start with the finding that addresses the most yellow questions from other groups.

5. PHENOMENON CONNECTION: How many students independently connected their group's finding to the sukuma wiki bubbling phenomenon in their notebook entries? If fewer than 60% made an explicit connection, open Lesson 11 by returning to the phenomenon image and asking the class collectively: 'Which investigation today most directly explains why bubbles appeared from specific spots on the leaf?'

6. ENGAGEMENT AND SAFETY: Were all learners actively participating, or were some group members passive observers? In Lesson 11, assign specific presentation roles to learners who were less active today. Were all safety protocols (hot water, blades, iodine) followed? Note any incidents in the safety log.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the outcomes of their own investigations — for example: bubble production rates from sukuma wiki leaf discs at different temperatures; stomatal counts on upper vs. lower leaf surfaces using nail varnish peels; colour changes in hydrogencarbonate indicator or lime water when exposed to germinating githeri seeds; or temperature differences between germinating and boiled seeds. They recorded raw data in tables, noted anomalous readings, and documented the behaviour of leaf material, seeds, and indicator solutions over time.
What did I learn?	Students learned how to execute a fair-test investigation by actively controlling variables and recording data with appropriate units and precision. They applied unit knowledge — stomata, guard cells, lenticels, cellular respiration, enzyme activity, gas diffusion — to explain patterns in their own data. They learned that real scientific data is sometimes messy or unexpected, and that anomalies are evidence to be explained, not discarded. They also learned to structure their thinking using the CER framework, distinguishing a claim from evidence and from reasoning.
How does this explain the phenomenon?	Students explained how the biological structures and processes they studied in Lessons 1–9 (stomatal regulation by guard cells, gas exchange through lenticels and stomata, cellular respiration releasing CO_2 and energy from glucose) directly account for the patterns in their data. They connected their group findings to the anchoring phenomenon — specifically explaining why the sukuma wiki leaves in Mwalimu Njoroge's class produced bubbles from specific points (stomata), why warm water increased bubble production (higher temperature increases respiration and enzyme activity rates), and why blocking the lower surface had a greater effect than blocking the upper surface (more stomata on the abaxial surface).

LESSON 11 (40 minutes): Project Presentations and Peer Review: Communicating Evidence on Plant Gaseous Exchange and Respiration	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will communicate their investigation findings using the CER framework, evaluate peer arguments using a shared rubric, and connect their evidence back to the anchoring phenomenon and driving question.
Knowledge	Students consolidate understanding of plant gaseous exchange sites, aerobic and anaerobic respiration pathways, factors affecting respiration rate, and the roles of stomata, guard cells, and lenticels in gas movement.
Skills	Students practise scientific argumentation, oral and visual communication of evidence, structured peer feedback, and critical evaluation of claims against evidence.
Attitudes	Students demonstrate intellectual honesty when reporting anomalous data, respectful critique of peers, and confidence in presenting scientific reasoning to an audience.
Key Inquiry Question	How do plants exchange gases with their environment, and how do plants obtain energy from organic compounds through respiration?
Purpose in Storyline	This lesson is the public evidence-sharing moment of the unit. After designing and executing their investigations in Lessons 9 and 10, groups now present findings to the whole class. The presentations pool evidence from multiple investigations — yeast fermentation, biogas simulation, food fermentation — so the class builds a richer collective answer to the driving question than any single group could achieve alone. The DQB is updated with confirmed answers and outstanding tensions to be resolved in Lesson 12.
Safety Notes	No new hazardous materials are introduced. If any fermentation or biogas apparatus is brought to the front for display, lids must be secured to prevent spills. Students should not taste any fermentation samples. Groups handling glassware during display should place items on non-slip mats on the demonstration bench.

B. LESSON OVERVIEW
Lesson 11 opens with a brief structured prediction task in which each group writes one key claim they expect to defend and one question a peer might challenge. Groups then present their investigation findings in a timed carousel format (approximately 5 minutes per group), using posters, data tables, or simple displays prepared in Lesson 10. After each presentation, peers provide structured written feedback using the co-created rubric. The class then convenes for a whole-class argumentation discussion, debating how collective findings confirm or complicate the original Lesson 1 predictions on the DQB. Students revise their cumulative unit models to incorporate new evidence, and the teacher bridges to Lesson 12 by identifying questions still unresolved on the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Biodiversity]	Before presentations begin, each group receives a half-sheet prompt. They write: (1) The	The teacher distributes the half-sheet prompts and says: 'Before anyone presents, I want every	Anticipation writing activates prior knowledge and creates cognitive readiness to listen critically. Cross-	The teacher scans written CER claims during circulation. Red flags: claims without quantitative

flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evidence of Evolution - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	strongest claim they can defend from their own data, expressed as a full CER statement. (2) One question they predict an audience member might raise to challenge their claim. (3) Their prediction of what another group investigating a different phenomenon (for example, a biogas group predicting what a fermentation group found, or vice versa) will have discovered. Students work in groups for 3 minutes. This activates prior knowledge, prepares them to listen critically to peers, and surfaces predictions that the Explain phase will confirm or complicate. Individually, students also briefly revisit their Lesson 1 DQB entry and write one sentence: 'I originally thought the bubbles from the sukuma wiki were because... I now think the evidence from our investigation shows...' This reconnects the entire lesson arc to the anchoring phenomenon before a single presentation begins.	group to commit to their strongest claim in writing. Science is not about hoping your results look good — it is about being honest about what your data actually shows. Take 3 minutes. Write your CER claim, write the toughest question a skeptic might ask you, and write one prediction about what a different group found.' The teacher circulates, scanning claims for vague language and prompting with: 'Your claim says the rate increased — can you be more precise? By how much, under what conditions?' After 3 minutes the teacher says: 'Hold your predictions. We will come back to them after all groups present. You will mark your own prediction as confirmed, partially confirmed, or not confirmed.' WAIT TIME: After issuing the prediction prompt, the teacher waits a full 60 seconds in silence before circulating, allowing groups to write without interruption.	group prediction builds the habit of seeing science as a community of inquirers rather than isolated experiments. Reconnecting to the Lesson 1 DQB entry maintains the phenomenon thread throughout the lesson.	evidence, reasoning that merely restates the claim, or groups that cannot identify a potential challenge to their argument. The teacher makes a mental note of which groups need a probing question during the presentation carousel.
Observe Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evidence of Evolution - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups present their investigations in a timed carousel. Each group has 4 minutes to present and 1 minute for one clarifying question from the audience. The order is: (1) Yeast fermentation groups, (2) Biogas simulation groups, (3) Food fermentation groups (uji, mandazi, or similar). Each group stands beside their display — poster, data table on manila paper, or simple apparatus — and delivers their CER argument aloud. Audience members are not passive: each student has a peer-review card divided into three columns — 'Evidence I found convincing,' 'A question I still have,' and 'A connection I can make to the	The teacher stands to the side, not at the front, reinforcing that the presenting group owns the space. Before the first presentation the teacher says: 'Your job as an audience is not to be polite — it is to be a fair scientific reviewer. Use your peer-review card. If you cannot find one piece of convincing evidence in a presentation, write why. If you can connect their findings to the sukuma wiki bubbles from Day 1, write that connection. That is the work of a scientist.' The teacher uses a visible timer. After each presentation, the teacher selects one audience member to ask their 'question I still have' — not a	Structured observation through the three-column peer-review card prevents passive reception of information. The card scaffolds the argumentation skills that the Explain phase demands. Requiring the audience to make connections to the anchoring phenomenon keeps the lesson coherent with the unit storyline rather than letting presentations become isolated show-and-tell events.	The teacher collects peer-review cards at the end of the carousel phase. These serve as a formative snapshot of how well students can identify evidence, generate scientific questions, and make cross-phenomenon connections. The teacher specifically checks the 'connection to the anchoring phenomenon' column to assess whether students are integrating unit learning.

	anchoring phenomenon or another lesson.' Students fill in the card for each presentation they observe. This structures observation and prevents passive listening. Groups are reminded that anomalous or unexpected results must be reported honestly and explained, not hidden — consistent with Lesson 10 learning about real scientific data.	volunteer, a chosen student. If the presenting group cannot answer, the teacher says: 'Park that question. We will return to it in our class discussion.' The teacher does NOT evaluate presentations publicly during the carousel — evaluation is reserved for the Explain phase. WAIT TIME: After each presentation ends and before the clarifying question, the teacher waits 15 seconds in silence so all students can finish writing on their peer-review cards before speaking begins.		
Explain Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evidence of Evolution - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	After all presentations, the class convenes for a 10-minute whole-class argumentation discussion structured around three questions written on the board: (1) 'Across all our investigations, what is the strongest collective evidence that respiration produces gases?' (2) 'Where do our findings agree with our Lesson 1 predictions about the sukuma wiki bubbles — and where do they disagree or add something new?' (3) 'If someone challenged us and said plants do not really breathe — they only make food through photosynthesis — how would we use our investigation data to argue back?' Students refer to their peer-review cards and their own data. The teacher facilitates a structured turn-taking discussion. Students are expected to cite specific evidence: 'In our yeast fermentation experiment, when we sealed the tube, the limewater turned cloudy after 8 minutes, which shows CO2 was produced anaerobically.' Groups that predicted what another group would find now publicly compare prediction to outcome and explain any discrepancy. The teacher	The teacher writes the three questions on the board and says: 'We have just heard from groups who investigated fermentation, biogas, and food respiration. Now we think together as a whole class. I am not going to tell you the answers — you are going to build them from your collective evidence. Who can give me a claim supported by data from at least two different groups?' The teacher uses cold-calling to ensure broad participation. When a student makes a claim, the teacher asks the class: 'Does anyone have evidence that supports or complicates that claim?' When disagreement arises, the teacher says: 'Good — this is exactly what scientists do. Do not back down just because someone disagrees. What does your data say?' If a group's results contradicted expectations, the teacher elevates this: 'This group found something unexpected. Instead of ignoring it, let us try to explain it. What variables might have affected their result?' WAIT TIME: After posing each of the three board questions, the teacher waits a full 30 seconds	The three structured questions move students from reporting (what we found) to arguing (what it means) to defending (why it matters). The cross-group comparison simulates the practice of scientific peer review. Connecting to the anchoring phenomenon in question 2 ensures that the Explain phase does not become a decontextualised debate but remains anchored in the observable reality of the sukuma wiki leaves producing bubbles from specific sites.	The teacher listens for: quality of evidence citation (specific data versus vague assertions), ability to distinguish claim from reasoning, and willingness to acknowledge anomalies. The teacher makes anecdotal notes on a class roster, marking students who are not participating so they can be individually prompted in the DQB phase.

	introduces the term 'scientific consensus' — explaining that when multiple independent investigations converge on the same finding, confidence in the conclusion increases, just as in real science.	before accepting any response, giving all students time to formulate evidence-based answers rather than reacting impulsively.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evidence of Evolution - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the DQB — the living board that has been updated since Lesson 1. Each group takes two sticky notes of different colours. On the first colour, they write one question from the original DQB that their investigation has now answered, stating the answer in one sentence and citing their evidence. On the second colour, they write one question that their investigation raised or left unanswered. Groups post both notes on the DQB in the designated columns: 'Now answered by evidence' and 'Still investigating.' The teacher then reads aloud three questions from the original Lesson 1 DQB — specifically those about where the bubbles came from, what gas was in the bubbles, and why warm water increased bubble production — and asks the class to vote with thumbs: fully answered, partially answered, or still unclear. The class discusses each briefly, and the teacher annotates the DQB accordingly. Students individually revisit their own Lesson 1 'What we wonder' entries and write a brief sentence in their science notebooks: 'My original question was... The evidence from this unit now suggests...' This closes the loop on the unit-long prediction thread.	The teacher distributes the two sticky note colours and says: 'Every question on this board was real — it came from your own curiosity on Day 1. Your job now is to be honest: which of these can you answer with evidence from our investigations? And which ones do we still not fully understand? Both are valuable.' After groups post their notes, the teacher reads the three specific Lesson 1 questions aloud and says: 'Raise your hand if you think we have fully answered this question with evidence from our investigations — not from the textbook, from our own data.' The teacher probes partial answers: 'You say partially answered — what exactly is still missing? What experiment would close that gap?' The teacher then bridges to Lesson 12: 'Tomorrow we finalise our models and resolve every remaining question on this board. I want you to come with your notebook and your best thinking.' WAIT TIME: After each of the three Lesson 1 questions is read aloud, the teacher waits 20 seconds before asking for the thumbs vote, giving students time to genuinely reflect rather than follow the majority.	The DQB update externalises the class's collective epistemic progress — students can literally see which questions they have answered with evidence and which remain open. This reinforces the nature of science: inquiry does not always close all questions; it often opens new ones. The individual notebook entry personalises the collective discussion and ensures every student, not just vocal participants, processes the change in understanding.	The teacher reads the posted sticky notes after class to assess whether students can correctly identify answered questions with appropriate evidence, and whether the remaining questions are genuine scientific uncertainties or misunderstandings that need to be addressed in Lesson 12 consolidation.
Model Building	In the final 6 minutes, students open their cumulative unit models	The teacher says: 'Your model is the story of everything we have	Progressive model revision throughout a unit is a core	The teacher listens to individual model narrations during

Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evidence of Evolution - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	— the annotated leaf cross-section diagrams and systems diagrams built progressively since Lesson 2. They add one new layer to their model based specifically on evidence from the investigation presentations they observed today. The addition must: (1) Label or annotate a site of gaseous exchange (stomata, lenticels, cuticle, or pneumatophores in hydrophytes if relevant to their group). (2) Show the pathway of at least one gas (CO_2 or O_2) through that site. (3) Include a note connecting their group's investigation result to the model — for example: 'Our fermentation experiment showed CO_2 production in anaerobic conditions. In the leaf, CO_2 produced by cellular respiration diffuses out through stomata when they are open.' (4) Add an arrow or annotation showing how today's class evidence updates or confirms any prediction made in Lesson 1. Students who investigated different phenomena will annotate different parts of the model, so the class collectively covers the full range of gaseous exchange and respiration concepts. The teacher circulates and asks individuals to narrate their model annotation aloud in one sentence, checking for accurate use of terminology.	learned in this unit. It started with a leaf in a beaker of warm water. It should now be able to explain not just where the bubbles came from but what gases are involved, what cellular process produces them, what structures they pass through, and how conditions like temperature change the rate. Add today's evidence to that story. You have 5 minutes.' The teacher circulates and uses targeted prompts: 'Where on your model does anaerobic respiration happen inside the plant cell? Show me the gas that is produced and the structure it exits through.' For students annotating accurately, the teacher asks an extension prompt: 'If the stomata were closed — say at midday during drought — what would happen to the CO_2 produced by respiration inside the mesophyll cells?' WAIT TIME: After issuing the model-building instruction, the teacher waits 45 seconds in silence before beginning circulation, allowing students to make their own first decision about where to annotate rather than waiting for teacher direction.	scientific practice. By requiring students to integrate today's presentation evidence into their personal model, the lesson ensures that the carousel presentations were not passive events but active learning inputs. The model serves as both a formative assessment tool and a consolidation artefact that will be finalised in Lesson 12. Narrating the model aloud activates verbal-linguistic processing alongside visual-spatial representation, deepening encoding.	circulation. Key indicators of understanding: correct directionality of gas movement (CO_2 out, O_2 in during respiration; O_2 out, CO_2 in during photosynthesis), accurate naming of exit structures, and the ability to connect their own investigation result to the model annotation. Misconceptions flagged here are noted for explicit address in Lesson 12 consolidation.

D. TEACHER REFLECTION

After the lesson, consider: Did all groups present evidence rather than just procedures? Were students citing specific data during the argumentation discussion, or were they making vague claims? Did the peer-review cards show genuine cross-phenomenon thinking, or did students struggle to connect presentations to the anchoring phenomenon? Which questions remain on the DQB, and are they conceptual gaps or language gaps? Are there students whose models are still incomplete or inaccurate in ways that will need targeted support in Lesson 12? Note which groups handled anomalous data honestly and which groups glossed over it — this informs feedback for the consolidation lesson. Also reflect on timing: if the carousel ran long, consider whether the argumentation discussion was truncated, and plan to open

Lesson 12 with a brief revisit of the three DQB questions before moving to model finalisation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Groups presented their investigation findings on anaerobic respiration using yeast fermentation, biogas simulation, and food fermentation with locally available Kenyan materials. Peers observed that CO2 production was confirmed across multiple independent investigations, and that conditions such as temperature, glucose availability, and oxygen exclusion affected gas production rates in consistent ways.
What did I learn?	Scientific confidence increases when multiple independent investigations converge on the same finding. Plants and yeast release CO2 during cellular respiration through both aerobic and anaerobic pathways. This CO2 exits plant cells through stomata, lenticels, and other gaseous exchange sites — the same sites from which bubbles emerged from the sukuma wiki leaves in Lesson 1. Argumentation requires citing specific evidence, not just asserting claims.
How does this explain the phenomenon?	The bubbles observed streaming from specific points on the sukuma wiki leaves in warm water were CO2 produced by cellular respiration in mesophyll cells, exiting through stomata and lenticels. Warm water increased metabolic and diffusion rates, producing more bubbles. Collective investigation evidence confirms that plants do exchange gases with their environment continuously — through respiration at all times, and through photosynthesis when light is available — making the answer to the driving question a synthesis of all unit learning.

LESSON 12 (80 minutes): Consolidation, Model Finalisation & Unit Review: How Do Plants "Breathe" and Release Energy?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all learning across the unit by finalising annotated models, resolving all Driving Question Board entries, and articulating a complete, evidence-based explanation of plant gaseous exchange and respiration.
Knowledge	By the end of this lesson, the learner should be able to: (a) explain how plants exchange gases with their environment through stomata, cuticle, lenticels, and pneumatophores; (b) describe how plants obtain energy from organic compounds through aerobic and anaerobic respiration; (c) connect the roles of photosynthesis and respiration in the overall carbon and energy cycle of a plant.
Skills	By the end of this lesson, the learner should be able to: (a) construct and annotate a finalised, evidence-based cumulative model of plant gaseous exchange and respiration; (b) evaluate the accuracy of their Lesson 1 predictions against their final explanations; (c) communicate scientific explanations clearly using correct biological terminology.
Attitudes	The learner appreciates that science is an iterative process of questioning, gathering evidence, revising ideas, and constructing better explanations — and values how collaborative inquiry has deepened their understanding of the living world around them, including everyday observations such as the sukuma wiki bubbles from Mwalimu Njoroge's class.
Key Inquiry Question	1. How do plants exchange gases with their environment? 2. How do plants obtain energy from organic compounds through respiration?
Purpose in Storyline	This is the final lesson of the unit. Students have spent eleven lessons gathering evidence — from stomatal observations under microscopes, to lenticel dissections on woody stems, to fermentation experiments using yeast and maize flour, to data analysis of ATP production in aerobic vs. anaerobic respiration. In this lesson, all threads converge. Students finalise their cumulative unit models, resolve every question posted on the Driving Question Board since Lesson 1, compare their naïve Lesson 1 models with their final scientific models, and deliver a whole-class Final Explanation of the anchoring phenomenon. The sukuma wiki bubbles that puzzled the class in Lesson 1 are now fully explained.
Safety Notes	No hazardous materials are used in this lesson. If printed or physical models involve scissors, remind learners to handle them carefully. Ensure the classroom is well ventilated as students may be working in close groups for extended periods.

B. LESSON OVERVIEW

Lesson 12 is the capstone lesson of Sub-Strand 2.3. Students work in home groups to finalise and fully annotate their cumulative unit models, incorporating every concept and piece of evidence gathered across all 12 lessons — including stomatal structure and guard cell mechanism, cuticle, lenticels, pneumatophores, aerobic respiration (glycolysis, Krebs cycle, oxidative phosphorylation), anaerobic respiration, the relationship between photosynthesis and respiration, and environmental factors affecting both processes. The class then conducts a structured Driving Question Board (DQB) resolution ceremony, working through every posted question and confirming which have been fully answered, partially answered, or which have generated new questions for further study. A paired Lesson 1 vs. Lesson 12 model comparison surfaces the depth of conceptual change. The lesson culminates in a whole-class co-constructed Final Explanation of the anchoring phenomenon — the sukuma wiki bubbles in Mwalimu Njoroge's Nairobi classroom — delivered as a formal scientific narrative. The teacher closes with a Forward Connection linking plant respiration to the next strand (Animal Gaseous Exchange and Respiration, Strand 3.3), using nyama choma and githeri as bridging contexts.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a printed copy (or view on-screen) of their own Lesson 1 initial model — the sketch they drew in the very first lesson showing where they thought the sukuma wiki bubbles came from and what process they thought was happening. They spend 4 minutes silently reviewing their old model and writing three 'I used to think… now I know…' statements in their science notebooks. They then share with a partner, identifying the biggest conceptual shift they personally experienced during this unit. Selected pairs share aloud with the whole class. This phase reactivates the full unit narrative and emotionally anchors students back to the original phenomenon before the final consolidation work begins.	Distribute or display Lesson 1 model snapshots before learners enter or at the start of class. Say: 'Before we do anything else today, I want you to travel back in time to Lesson 1. Look at the model you drew when you first saw those sukuma wiki bubbles. Don't judge it — just read it carefully.' Allow 4 minutes of silent individual reflection. Then say: 'In your notebook, complete three sentences beginning with: I used to think… but now I know…' Allow 3 minutes of writing. Then cold-call 4 students: 'Mwangi, what was one thing you used to think? Akinyi, what was your biggest surprise? Otieno, what concept changed the most for you? Wanjiku, what question do you still have?' After each response, affirm and bridge: 'That shift — from thinking bubbles came randomly to knowing they emerge specifically from stomata controlled by guard cells — that is exactly what scientists experience when evidence changes their models. That is science.' WAIT TIME: 10–12 seconds after each cold-call question.	I Used to Think / Now I Know — This metacognitive strategy makes conceptual change visible and concrete. By writing and verbalising their prior vs. current understanding, students consolidate the learning arc of the entire unit and develop awareness of themselves as learners whose ideas evolved through evidence. It also reminds the class of the shared anchoring phenomenon, creating narrative coherence before the final model work.	Teacher circulates during the writing phase and scans notebooks. Observable indicator: each student has written at least three complete 'I used to think… now I know…' statements that reference specific biological concepts (e.g., stomata, guard cells, ATP, aerobic respiration) rather than vague generalities. Red flag: students who write only surface-level statements ('I used to think plants don't breathe; now I know they do') without referencing mechanisms — these students need targeted scaffolding during Phase 2 model work.
Observe Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook)	Students return to their established home groups of 4–5. Each group retrieves their cumulative unit model — the large annotated diagram they have been building and revising since Lesson 1. Groups are given a structured Model Finalisation Checklist (displayed on the board or printed) listing every concept that must appear in the final model: (1) Leaf cross-section showing stomata, guard cells, mesophyll cells, cuticle; (2) Mechanism of stomatal	Display the Model Finalisation Checklist prominently. Say: 'Your model has been growing since Lesson 1. Today we complete it. Every item on this checklist must appear — correctly labelled, with arrows showing direction of gas movement and energy flow. You have 20 minutes. Each person in your group must personally add at least two annotations — sign your initials next to what you add so I can see individual contributions.' Circulate to all groups in the first 5	Structured Model Annotation with a Checklist — The checklist transforms an open-ended task into a systematic evidence audit. By requiring students to locate and correctly place every concept from the unit onto a single coherent diagram, this strategy forces integration across all 11 previous lessons. The requirement for individual initials ensures every student engages with the model rather than delegating to dominant group members. Circulating	Teacher uses the checklist as a walk-through rubric during circulation. Observable indicators: (1) All 9 checklist items are present and correctly labelled; (2) Gas movement arrows are directionally accurate (e.g., CO_2 exits stomata during respiration in the dark, O_2 exits during net photosynthesis in bright light); (3) ATP yield numbers are correctly stated for aerobic vs. anaerobic respiration; (4) Each student's initials appear on at least two

Source: CK-12 🔍 Search ARES for similar readings	opening/closing with CO_2, O_2, and H_2O labelled with arrows; (3) Lenticel on a woody stem with gas exchange arrows; (4) Pneumatophore in a waterlogged environment (e.g., Kenyan mangrove coast near Gazi Bay, Kwale County); (5) Aerobic respiration equation and summary of glycolysis → Krebs cycle → oxidative phosphorylation with ATP yield; (6) Anaerobic respiration pathway and products; (7) The relationship between photosynthesis and respiration (inputs/outputs cycling); (8) Effect of temperature on both stomatal aperture and respiration rate (connecting back to the warm vs. cold water observation from Lesson 1); (9) At least one real-world application (e.g., post-harvest storage of sukuma wiki, waterlogged soils in Kano Plains, Kisumu). Groups spend 20 minutes completing any missing annotations, adding arrows, correcting errors identified in previous lessons, and writing a 2–3 sentence scientific caption below the model. Each student must personally annotate at least two components to ensure individual accountability.	minutes, checking that groups have retrieved their models and are actively working. Pose probing questions to prompt deeper annotation: 'Why does your guard cell diagram show K^+ ions moving in? What does that tell us about the energy cost of opening stomata?' / 'Your aerobic respiration pathway — where exactly in the cell does glycolysis happen? Where does the Krebs cycle happen? Label the organelle.' / 'You've drawn a pneumatophore — can you add a label showing what environmental condition makes this structure necessary? Think of Gazi Bay.' / 'Your temperature arrow — which direction does it point for reaction rate in respiration? What molecule explains that?' WAIT TIME: after each probing question, wait 10 seconds before offering a hint. If a group is stuck on the photosynthesis-respiration relationship, ask: 'Think about the sukuma wiki on a bright, sunny day in Nairobi. Which process is faster? How does that change the net gas exchange through the stomata?' Cold-call check at minute 15: ask 3 groups to share one annotation they just added and explain the evidence behind it.	teacher probes push annotation beyond labelling toward causal explanation.	annotations. Red flag: groups whose models show photosynthesis and respiration as separate, unconnected processes — intervene immediately with the bridging question about net gas exchange.
Explain Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures	The class gathers around the Driving Question Board — the large display that has collected every question students posted since Lesson 1. The teacher leads a whole-class 'resolution ceremony' in which each posted question is read aloud and the class collectively determines its status: (A) Fully Answered — we can explain this completely using evidence; (B) Partially Answered	Stand beside the DQB and say: 'This board has held our questions since Lesson 1. Some of you were curious, some of you were confused, some of you were sceptical. Every question you posted here was valid. Now, let's see how far our evidence has taken us.' Read each question aloud. For each, say: 'Hands up if you think this is Fully Answered. Hands up for Partially Answered.	Driving Question Board Resolution Ceremony — This public, structured review of all posted questions makes the entire unit's learning arc visible to every student simultaneously. Categorising questions as Fully/Partially/New communicates that science always generates new questions, normalising intellectual humility. Requiring student volunteers to deliver evidence-	Observable indicators: (a) Students correctly categorise at least 80% of DQB questions by consensus; (b) Student volunteers who deliver answers reference specific biological mechanisms (e.g., 'The bubbles come from stomata because guard cells respond to warm water by increasing membrane permeability and turgor, opening the pore and allowing dissolved respiratory CO_2

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	— we have some evidence but more investigation is needed; (C) New Question Generated — answering this question raised a deeper question for further study. For Fully Answered questions, a student volunteer stands and delivers a brief (30–60 second) evidence-based answer. For Partially Answered and New Questions, students suggest what further investigation could resolve them, connecting to real Kenyan research contexts (e.g., KEFRI — Kenya Forestry Research Institute — studies on mangrove respiration along the coast). The two KICD Key Inquiry Questions ('How do plants exchange gases with their environment?' and 'How do plants obtain energy from organic compounds through respiration?') are addressed last and answered in full by the class collectively. This ceremony takes approximately 15 minutes.	Hands up for New Question Generated.' Use the majority response to categorise it, then cold-call a student to justify: 'Kamau, you voted Fully Answered for this one — give us the evidence in 30 seconds.' For the two Key Inquiry Questions, do not cold-call individuals — instead say: 'I want to hear this answered as a class. I'll start the sentence and I want you to complete it in chorus: Plants exchange gases through…' then pause. For questions that remain partially answered or generate new questions, write them on a 'Questions for Future Study' section of the board and say: 'These are exactly the questions that Kenyan scientists at KEFRI and the University of Nairobi are still investigating. You are thinking like researchers.' WAIT TIME: 12 seconds after reading each DQB question before calling for hands.	based answers provides authentic formative assessment of explanatory depth. Connecting unresolved questions to real Kenyan research institutions (KEFRI, University of Nairobi) validates students' curiosity as scientifically meaningful.	and O_2 to escape'); (c) Students generate at least one new, deeper question during the ceremony (e.g., 'If stomata close at night to conserve water, how do plants get enough CO_2 for night-time processes?'). Red flag: if students cannot answer the two Key Inquiry Questions fully, the teacher should pause and facilitate a targeted 5-minute recap before proceeding to Phase 4.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	This is the centrepiece of the entire unit. Each group collaborates for 8 minutes to write a Final Explanation paragraph — a complete, evidence-based scientific narrative that answers the original anchoring phenomenon question: 'Why did bubbles stream from specific spots on the sukuma wiki leaves when submerged in warm water, and what does this tell us about how plants breathe and release energy?' The paragraph must include: (1) the specific structural identity of the bubble-emitting spots (stomata); (2) the guard cell mechanism; (3) the identity of the gases in the bubbles (CO_2 from respiration, O_2 from photosynthesis); (4) why warm	Signal the transition: 'We have been building toward this moment since Lesson 1. Now I want each group to write the definitive scientific explanation of what Mwalimu Njoroge's students saw in that Nairobi classroom. You have 8 minutes. Your explanation must be complete enough that a student who missed this entire unit could read it and understand exactly what is happening and why.' Display the 7 required components on the board as a writing scaffold. Circulate during writing and whisper targeted nudges: 'You've described the stomata — now explain the guard cell mechanism specifically. What ion moves? What happens to osmotic pressure?' / 'Good start on	Final Explanation Writing with Structured Peer Feedback — The Final Explanation task is the highest-order sense-making activity of the unit. Writing a complete causal narrative requires students to integrate structural knowledge (stomata, guard cells), process knowledge (respiration, photosynthesis), quantitative knowledge (temperature effects, Q10), and contextual knowledge (the specific phenomenon) into a single coherent account. Structured peer feedback using sentence starters ensures feedback is evidence-specific rather than vague ('it was good'), and models the scientific practice of peer review.	Teacher collects all written Final Explanations at the end of the lesson as a summative assessment artefact. Observable indicators for proficiency: (1) All 7 required components are present; (2) Causal language is used throughout ('because,' 'therefore,' 'as a result of,' 'this causes'); (3) Correct biological terminology is used consistently (stomata, guard cells, mesophyll, ATP, aerobic respiration, enzyme activity, Q10); (4) The explanation is internally consistent — no contradictions between the photosynthesis and respiration sections. Indicator of exceeding expectations: student explains the net gas exchange scenario (in bright light, O_2 output from photosynthesis exceeds CO_2

	water increases bubble production (temperature effect on enzyme-driven respiration rate — Q10 principle); (5) why cold water decreases bubble production; (6) the connection to cellular respiration and ATP production in the mesophyll cells; (7) how this connects to the broader carbon cycle. Two groups are selected to read their Final Explanation aloud to the class. The class provides structured peer feedback using the prompt: 'I heard evidence about… / I noticed they explained the mechanism of… / One thing that could make this explanation even stronger is…'	temperature — now name the biological reason. What kind of molecules speed up when temperature rises?' After 8 minutes, say: 'Group 3 and Group 5 — please read your explanations to the class.' After each reading, facilitate peer feedback: 'Using our sentence starters on the board — what did you hear? What mechanism did they explain well? What would make it stronger?' After both readings, synthesise: 'What I'm hearing across both explanations is a complete scientific story. The bubbles were not random — they were the visible signature of plant metabolism. The stomata were the doors. The guard cells were the gatekeepers. The enzymes were the engines. And the temperature of the water was the throttle.' WAIT TIME: 10 seconds after posing each writing-nudge question.		output from respiration, so O_2 bubbles dominate; in low light or dark, only respiratory CO_2 is released).
Model Building Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students place their Lesson 1 initial model and their completed Lesson 12 final model side by side (physically or digitally). They spend 5 minutes completing a structured 'Model Evolution Analysis' in their notebooks, answering: (1) What was correct in my Lesson 1 model? (2) What was missing from my Lesson 1 model? (3) What was incorrect in my Lesson 1 model and why? (4) What is the most important piece of evidence that changed my model? (5) What question does my final model still not fully answer? They share responses with their partner, then the teacher facilitates a 3-minute whole-class reflection. The lesson closes with a Forward Connection: the teacher draws a large arrow on the board from the	Say: 'Side by side — your Lesson 1 model and your Lesson 12 model. These are not the same person's understanding. Look at the distance you have travelled.' Allow 5 minutes for the Model Evolution Analysis writing. Circulate and read over shoulders. Then cold-call 3 students for specific questions: 'Fatuma, what was the most important piece of evidence that changed your model?' / 'Kipchoge, what is one thing your Lesson 1 model got right — give yourself credit for that.' / 'Zawadi, what question does even your final model not fully answer?' After responses, affirm the iterative nature of science: 'Every scientist in history — including Kenyan botanist Professor Hamisi Miraji who	Model Evolution Analysis + Forward Connection Bridging — The side-by-side model comparison is a powerful metacognitive tool that makes intellectual growth tangible and visible. Students who can articulate what was wrong, what was missing, and what evidence drove the change have achieved deep conceptual understanding rather than surface memorisation. The Forward Connection activates analogical reasoning — students who understand plant gaseous exchange and respiration are primed to recognise both homologies and key differences in animal systems, reducing cognitive load in Strand 3.3.	Exit ticket: one written prediction about similarities and differences between plant and animal gaseous exchange and energy release. Observable indicators of readiness for Strand 3.3: predictions that reference specific mechanisms (e.g., 'I predict animals will also use ATP from aerobic respiration, but will not have stomata — they will have a different gas exchange surface') rather than only surface features (e.g., 'animals have lungs and plants don't'). Teacher reviews exit tickets before the next lesson to identify misconceptions to address at the start of Strand 3.3. Final model collection: teacher collects or photographs all final group models as a unit portfolio artefact.

	sukuma wiki leaf model toward a diagram of a cow (animal cell), asking: 'In our next unit, Strand 3.3, we will investigate how animals — including us — exchange gases and release energy from nyama choma and githeri. What do you predict will be similar to plants? What do you predict will be different?' Students write one prediction in their notebooks. This bridges the unit to the next strand and activates prior knowledge for Strand 3.3.	studied mangrove respiration along our coast — started with incomplete models. The willingness to revise your model when evidence demands it is the single most important scientific habit you can develop.' Then draw the Forward Connection arrow and pose the bridging prediction question. Collect one written prediction per student as an exit ticket. WAIT TIME: 12 seconds after the bridging prediction question before accepting responses.		

D. TEACHER REFLECTION

After this final lesson, reflect on the following questions before beginning Strand 3.3:

1. PHENOMENON RESOLUTION: Did students' Final Explanations fully account for all five original puzzles from Lesson 1 — the specific location of bubbles (stomata), the identity of the gases (CO_2 / O_2), the temperature effect (enzyme kinetics / Q10), the specificity of bubble sites (stomatal density and distribution), and the analogy to animal breathing (cellular respiration as the common thread)? If any of these five elements were consistently missing from Final Explanations, plan a 5-minute bridge at the start of the first Strand 3.3 lesson to resolve them.

2. MODEL QUALITY: Were final group models genuinely more sophisticated than Lesson 1 models, or did some groups simply add more labels without demonstrating causal understanding? Groups whose models lack directional arrows and causal connectors ('because,' 'causes,' 'results in') need explicit instruction in scientific modelling conventions at the start of the next unit.

3. DQB UNRESOLVED QUESTIONS: Which questions remained on the DQB as Partially Answered or New Questions? These are valuable diagnostic data. Questions about night-time gas exchange, CAM and C4 plants, or the effect of altitude (relevant to tea growing in Kericho) indicate high-achieving students who are ready for extension tasks.

4. EQUITY CHECK: Did all students contribute personally annotated components to the final model (evidenced by initials)? Were there students who consistently deferred during the DQB ceremony and Final Explanation writing? Identify these students by name for targeted engagement strategies in Strand 3.3.

5. FORWARD READINESS: Review the exit ticket predictions. Students who can already articulate 'animals will use aerobic respiration for ATP but will have specialised gas exchange surfaces instead of stomata' are ready for advanced Strand 3.3 content. Students who predict only surface anatomical differences (lungs vs. leaves) will need the conceptual bridge between cellular respiration in plants and animals made explicit in Lesson 1 of Strand 3.3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Tiny bubbles streaming from specific spots on sukuma wiki leaves submerged in warm water — not from the entire leaf surface, and far fewer bubbles appearing when the water was cold.
What did I learn?	The bubbles emerge specifically from stomata — microscopic pores on the leaf surface controlled by guard cells. The gases in the bubbles are CO_2 (produced by cellular respiration in mesophyll cells) and O_2 (produced by photosynthesis in chloroplasts). Warm water increases bubble production because higher temperature increases enzyme activity, accelerating both aerobic respiration (producing more CO_2) and, where light is available, photosynthesis (producing more O_2). Cold water slows enzyme-catalysed reactions, reducing gas production. Plants exchange gases not only through stomata but also through the cuticle, lenticels on woody stems, and pneumatophores in waterlogged environments. Inside mesophyll cells, aerobic respiration proceeds through glycolysis (cytoplasm), the Krebs cycle (mitochondrial matrix), and oxidative phosphorylation (inner mitochondrial membrane), yielding approximately 36–38 ATP molecules per glucose. When oxygen is absent, anaerobic respiration produces ethanol and CO_2 (or lactate), yielding only 2 ATP per glucose. Photosynthesis and respiration are complementary processes: the glucose and O_2 produced by photosynthesis serve as substrates for respiration, and the CO_2 and H_2O produced by respiration are raw materials for photosynthesis.
How does this explain the phenomenon?	The sukuma wiki leaves were not 'breathing' randomly — they were performing precisely regulated, enzyme-driven metabolic processes at the cellular level. The specific spots from which bubbles emerged are stomata: openings whose size is actively regulated by guard cells through changes in turgor pressure driven by potassium ion (K^+) transport. The bubbles themselves are the visible, macroscopic signature of microscopic chemistry: aerobic cellular respiration in the mitochondria of mesophyll cells releasing CO_2, and photosynthesis in chloroplasts releasing O_2. The temperature of the water acted as a throttle on enzyme activity — warm water accelerated the enzymes of both respiration (e.g., pyruvate dehydrogenase, enzymes of the Krebs cycle) and photosynthesis (e.g., RuBisCO), producing more gas and therefore more bubbles. This observation — puzzling in Lesson 1, fully explained by Lesson 12 — demonstrates that plants do 'breathe' in a deep biochemical sense: they continuously exchange gases with their environment and convert stored chemical energy into usable ATP through the same fundamental respiration chemistry that powers all living cells, from sukuma wiki leaves in a Nairobi school garden to the cells of a Rift Valley marathon runner.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.